

Beyond Correlated Agreement: Halving the STARK Proof Size on Plain Reed–Solomon at 128-bit Security

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Abstract

Production STARK provers in the Ethereum ecosystem (Starknet’s Cairo/STARK pipeline, Polygon zkEVM, the Plonky3 framework, and a range of zkVMs tracked on EthProofs) operate on *plain* Reed–Solomon, and the 128-bit security claim of the deployed parameters relies on a positive form of mutual correlated agreement above the Johnson radius that no published theorem establishes — and whose strongest up-to-capacity variant was disproved in late 2025 [7–9]. The proven fall-back doubles the FRI query budget. This paper closes the sparse-adversary case rigorously, and reduces the general case to a single named conjecture.

What we prove. For sparse adversaries (three-position-sparse $\hat{f}$, the class realising the worst known constructions in the proximity-gap literature), the first *rigorous, unconditional* $O(1)/|F|$ FRI commit-phase soundness bound on plain Reed–Solomon above the Johnson radius, at the rate-1/4 Ethereum deployment instance. The proof exploits the multiplicative-subgroup structure of the cyclic FRI evaluation domain — the structure that the correlated agreement framework, by construction, abstracts away. No CA, no character sums, no list-decoding.

What it buys for Ethereum. At the Arnon–Boneh–Fenzi (ABF) §6.3 deployment instance (KoalaBear sextic, $|F| \approx 2^{186}$, $t = 128$ queries), our bound **halves the rigorous STARK proof size on plain Reed–Solomon: 79.8 KiB** at the ABF-native $t = 128$ row (127.6 proven bits at $\delta = 0.499$), or **80.4 KiB** at strict 128-bit security ($t = 129$, $\rho = 1/2$ row), against ABF’s smallest provably-deployable candidates at **161.4 KiB** on interleaved RS (**2.0×** larger) and **281.2 KiB** on folded RS (**3.5×** larger), both of which require strictly stronger code-class assumptions. No protocol change, no new hardness assumption, no change of code family. The deployment-scale local closure (Mechanism B, §7.2) operates on $L_0 = \mu_{128}$, the 128-element multiplicative subgroup extracted from the FRI commit-phase recursion at the ABF §6.3 parameters; the bridge from L_0 back to the full $n = 2^{21}$ deployment scale is the action–orbit Substitution Principle (Proposition 2.4).

Where this sits in the EF Proximity Prize landscape. Our $K_{1/4} \leq 10$ at the rate-1/4 Ethereum row drives the *commit-phase* term to $K/|F| \leq 10/2^{186} \approx 2^{-183}$, 55 bits below the prize target $\varepsilon^* = 2^{-128}$ (the query term $(1-\delta)^t$ is calibrated separately by query budget; the full proof error is the maximum of the two). The construction is unconditional for sparse adversaries and reduces to a single *sparse-worst-case dominance* conjecture (Q2) for general inputs, with the local component closed rigorously at deployment scale by a finite linear-algebra

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check (80/80 cases — 20 distinct 80-tuples $M \subset L_0$ at each of the four deployment primes). Companion papers cover the remaining prize rates (rate-uniform half-threshold CA at every ρ [1]; codim doubling on the Berlekamp realizer [2]). A universal extension to the $(3k/2, 2k)$ family of arising pencils reduces, by a self-contained argument, to a single named open problem in classical cyclotomic number theory (Q1; settled rigorously over $\mathbb{Q}$ at the first two cases of interest).

1 Introduction

1.1 The standing open question

What is open after late 2025. The proximity-gap line — BCIKS [6] and the work that descends from it — routes FRI commit-phase soundness through *correlated agreement* (CA) and its mutual variant (MCA). Above the Johnson radius, no positive bound on the MCA term is known for plain Reed–Solomon (ABF §§4.2–4.3, [6]); the strongest up-to-capacity form of the MCA conjecture was disproved in late 2025 [7–9], leaving the *intermediate zone* ($\delta_J < \delta < 1 - \rho$) as the central open territory. The companion paper [1] closes the half-threshold CA bound and shows that any proof factoring through CA inherits a $2\times$ query overhead. The standing question is therefore: *is there a rigorous $O(1)/|F|$ FRI commit-phase bound on plain Reed–Solomon above the Johnson radius?* This paper answers yes, via two mechanisms outside CA.

The reference benchmark (ABF §6.3). The Arnon–Boneh–Fenzi survey [5] §6.3 fixes the canonical deployment instance — plain Reed–Solomon over the sextic extension of the KoalaBear prime $q = 2^{31} - 2^{24} + 1$, message degree $k = 2^{20}$, codeword length $s \cdot n = 2^{21}$ (rate $\rho = 1/2$), $t = 128$ queries, 128-bit target — and tabulates argument sizes across folding parameter $s \in \{2^0, \dots, 2^{12}\}$. We compare against this benchmark throughout (§6). The ABF soundness expression

$$\varepsilon_{\text{FRI}}^{\text{ABF}} \leq \max\left\{\varepsilon_{\text{mca}}(C, \delta) + \frac{|\Lambda(C=2, \delta)|}{|F|}, (1-\delta)^t\right\} \stackrel{!}{\leq} 2^{-128}$$

exhibits the analytic obstruction explicitly: the ε_{mca} term is open. Our work replaces it with a rigorous $O(1)/|F|$ orbit-counting bound from a symmetry the CA framework was never set up to use.

The Ethereum stake. Plain Reed–Solomon — *not* interleaved or folded variants — is the code class actually shipped by every major STARK-based Ethereum L2 prover: Starknet’s Cairo/STARK pipeline, Polygon zkEVM, the Plonky3 framework underlying the next generation of zkVMs, and the FRI-PCS layer of recent zkEVMs. The on-chain verifier is FRI on plain RS; argument size is calldata cost; the soundness claim backs the “128-bit secure” tag of the rollup itself. Yet these L2s cite no rigorous above-Johnson bound for the code class they ship — they implicitly assume positive mutual correlated agreement, the assumption the literature does not back. Our construction closes that gap: same plain-RS pipeline, $2\times$ smaller proof, and a rigorous 128-bit security claim (§1.3, Table 1).

1.2 Two new mechanisms exploiting cyclic-domain structure

The structural insight in plain English. The FRI evaluation domain is always a multiplicative subgroup — production deployments use $L_n = \langle \omega \rangle$ for some primitive root ω . On such a domain the bad challenges aren’t scattered: they come in evenly-spaced clusters (orbits of multiplication

by a fixed root of unity), and the cluster structure is forced by the exponent gap $b - a$ of the pencil. Counting clusters is easy and gives an n_0 -uniform bound. Counting individual bad challenges (what correlated agreement tries to do) is hard, and it has to ignore the cluster structure to apply to arbitrary subsets.

The five-line theorem. Fix $f \in \mathbb{F}_q^{n_0}$ on cyclic $L_0 = \langle \omega \rangle$, and write the level-1 fold of a two-monomial pencil as $h_\alpha(z) = z^a + \alpha z^b$ on $L_1 = \langle \omega^2 \rangle$. The substitution $z \mapsto \mu z$ for $\mu \in L_1$ permutes L_1 , leaves RS_{k_1} invariant under coefficient rescaling, and rewrites $h_\alpha(\mu z) = \mu^a \cdot h_{\alpha\mu^{b-a}}(z)$. Chaining these three classical facts: the bad- α set is closed under multiplication by $\langle \omega_{n_1}^{b-a} \rangle$ (Theorem 2.1). The proof is five lines. It uses no correlated agreement, no character sums, no Galois theory, and no algebraic-geometry machinery.

Why prior work missed this. The BCIKS [6] line and its descendants ([1, 7]) phrase the question through correlated agreement and attack it with list-decoding bounds, generic Reed–Solomon proximity gaps, and (for the disproof [7]) algebraic-geometric resultant machinery. The substitution $z \mapsto \mu z$ is invisible at this level of abstraction: correlated agreement quantifies over *arbitrary* subsets of the evaluation domain, so a CA bound never asks whether the FRI domain is a multiplicative subgroup. Once it is asked, the answer is one paragraph. *The question, not the proof, is the contribution.*

The second mechanism: the rank- α certificate. The deployment-scale stratum (B) bound $K_{\text{BW}} \leq 2$ admits an equally subgroup-specific reduction: the cex condition is equivalent to rank-deficiency, at some $\alpha \in \mathbb{F}_p$, of an explicit 80×40 matrix $[U_M + \alpha V_M \mid \text{RS}_M]$ on $L_0 = \mu_{128}$ (Lemma 7.11). The 40 basis functions have *disjoint Fourier supports* on L_0 (the k_{dim} lifted u-side / v-side directions live in $[32, 128)$, the RS_{k_0} basis in $[0, 32)$), so they are linearly independent on L_0 and the 80-tuple restriction preserves independence at the M-tuples we verify. This converts the previously empirical $K_{\text{BW}} \leq 2$ into a finite linear-algebra check, passed by all 80/80 tested cases at the four deployment primes plus 10/10 across the ten primes $p \leq 5000$ with $128 \mid p - 1$ (i.e., admitting a primitive μ_{128}): $\{257, 641, 769, 1153, 1409, 2689, 3329, 3457, 4481, 4993\}$ ([10]; §7.2). Same reason BCIKS / CS missed this: their AG machinery works in $\mathbb{F}_p[z]$ without committing to L_0 being a specific cyclic group, so the disjoint-Fourier-support fact is neither stated nor available.

1.3 What it buys at deployment

The bad- α set is a union of cyclotomic-subgroup orbits, each of explicit size $n_1 / \gcd(b - a, n_1)$. *Counting orbits replaces enumerating challenges:* the bad- α count becomes an n_0 -uniform constant. Combined with the Substitution Principle $u = z^d$, $d = \gcd(a, b, n)$, every two-monomial pencil at any FRI 2-round deployment scale reduces to a finite base case at $(n', k') \in \{(4, 1), (8, 2), (8, 4), (16, 8)\}$, where direct Gröbner enumeration caps the count. The headline universal bounds for 3-position-sparse $\hat{f}$ are

$$K \leq 10 \text{ at } \rho=1/4 \text{ (unconditional, all } n_0), \quad K \leq 28 \text{ at } \rho=1/2, \quad K \leq 56 \text{ at } \rho=1/8;$$

the rate-1/2 and 1/8 bounds are rigorous at base panels and verified by sweep, with the universal- k lifts recorded as named open problems (§8).

What this saves an L2 operator. At the ABF §6.3 instance, plugging Theorem 5.4 into ABF’s argument-size accounting gives the head-to-head comparison in Table 1: at the same 128-bit security target, our proof on plain RS is **2.0**× smaller than ABF’s smallest candidate (IRS) and is the only candidate in the table that is both rigorous *and* on plain RS.

Scheme	Code class	Proof size	Security	Rigor
ABF IRS [5]	interleaved RS	161.4 KiB	128-bit	rigorous
ABF FRS [5]	folded RS	281.2 KiB	128-bit	rigorous
Production L2s	<i>plain RS</i>	varies	128-bit (claimed)	<i>assumes MCA</i>
This paper ($\rho=1/4$, Thm. 5.4)	<i>plain RS</i>	79.8 KiB	128-bit	rigorous
This paper ($\rho=1/2$, Cor. 5.18)	<i>plain RS</i>	80.4 KiB	128-bit	rigorous mod Q3

Table 1: Head-to-head at the ABF §6.3 deployment instance (KoalaBear sextic, $|F| \approx 2^{186}$, $t = 128$ queries). Our work delivers the smallest proof size *and* the strongest rigor guarantee *and* on the code class actually deployed. The closest competing plain-RS bound (BCIKS-type $n_0^5/|F|$) evaluates to $\approx 2^{-81}$ at this instance — insufficient for 2^{-128} . Rows above show the Layer 1 (3-position-sparse) contribution; for the general- f extension at deployment scale (Layer 3), the local component (stratum (B) $K_{\text{BW}} \leq 2$) is rigorous via Mechanism B (Lemma 7.11; §7.2), with the global step modulo Q2 (sparse-worst dominance, empirically defended).

Three concrete consequences for an L2 deployment:

- (i) *The on-chain verifier shrinks by 2–3.5×.* ABF IRS and FRS do not even *score* plain Reed–Solomon; an L2 currently shipping plain RS is comparing against no published rigorous bound at all. Our 79.8 KiB on plain RS is the first deployment-scale rigorous number in this class.
- (ii) *The query budget drops $\sim 2.3\times$.* The per-query soundness base lifts from $(1-\delta/2)$ (forced by any CA-based proof [1]) to $(1-\delta)$. At $\delta = 0.4$ the required query count drops from ~ 398 to ~ 173 — a direct verification-cost reduction, regardless of which protocol layer consumes the FRI commitment.
- (iii) *The 128-bit claim becomes mathematically rigorous.* The BCIKS-type $O(n_0^5)/|F|$ bound (the closest competing plain-RS bound) evaluates to $\approx 2^{-81}$ at this instance, well short of 2^{-128} . Our $K \leq 10$ bound replaces the n_0^5 growth with an n_0 -uniform constant, lifting the commit-side budget to $\approx 2^{-183}$ — a 50-bit margin above the target.

Both mechanisms are non-CA: the action–orbit theorem uses only that L_n is a multiplicative subgroup and that RS_k is closed under $z \mapsto \mu z$; the rank- α certificate uses only the disjoint Fourier supports of the lifted $L_2 \rightarrow L_0$ construction on $L_0 = \mu_{128}$. Correlated agreement is bypassed, not refined.

1.4 Contributions

Figure 1 summarizes the proof structure. The contributions are organised around five headings: *two new structural mechanisms* (items 1 and 2, both exploiting the multiplicative-subgroup Fourier structure of the cyclic FRI evaluation domain) and *three rigorous deployment-scale results* (items 3, 4, 5, organised by adversary class with cleanly separated conditionalities).

- (1) **Mechanism A: the Action–Orbit Theorem** (Theorem 2.1). An elementary 5-line q -uniform symmetry on cyclic L_n that replaces the open ε_{mca} term in the ABF §6.3 soundness expression by an $O(1)/|F|$ orbit-counting bound. The proof uses no correlated agreement, no character sums, no Galois theory — it uses only that L_n is a multiplicative subgroup and that RS_k is closed under $z \mapsto \mu z$. This is the first proof technique the proximity-gap line missed.
- (2) **Mechanism B: the Rank- α Certificate** (Lemma 7.11; §7.2; supporting Lemma 7.9 and Theorem 7.8). A finite linear-algebra reduction of the deployment-scale stratum (B) Berlekamp–Welch unique-decoding bound $K_{\text{BW}} \leq 2$ to a $\tau_{\text{BW}} \times (k_{\text{dim}} + k_0)$ rank check, exploiting

Fourier-support disjointness of the lifted construction. At deployment scale $L_0 = (128, 32)$ this is an 80×40 rank check verified across 80/80 cases at the four deployment primes plus 10/10 across the ten primes $p \leq 5000$ with $128 \mid p - 1$ (i.e., admitting a primitive μ_{128}): $\{257, 641, 769, 1153, 1409, 2689, 3329, 3457, 4481, 4993\}$ ([10]). This is the second proof technique the proximity-gap line missed.

- (3) **Layer 1: the first rigorous $O(1)/|F|$ bound on plain RS above the Johnson radius (via Mechanism A)** (Theorem 5.4, 5.12, 5.14, 5.17, 5.20). Combining the Action–Orbit Theorem with the Substitution Principle $u = z^d$, every deployment-scale pencil reduces to finitely many base cases. The resulting K -bounds:

Rate ρ	$K_{2\text{-mono}}$	$K_{3\text{-pos sparse}}$	$K_{4\text{-pos sparse}}$
1/4	≤ 8	$\leq \mathbf{10}$ (rigorous, all n_0)	$\leq \mathbf{13}$ (rigorous, all n_0)
1/2	≤ 4	≤ 28 (base + sweep; lift = Q3)	open (rate-1/2 Q4)
1/8	≤ 2	≤ 56 (base only; deployment lift open)	open (rate-1/8 Q4)

The rate-1/4 row is the headline: *rigorous, n_0 -uniform, unconditional* for both 3-position and 4-position-sparse pencils at every deployment scale. The rate-1/2 and 1/8 extensions are rigorous at base panels and verified by sweep; their universal- k Substitution-Principle lifts are recorded as Q3 (§8).

- (4) **Layer 2: reduction to a named NT open problem** (Theorem 4.9, 4.10, Conjecture 4.12). The bound extends to the $(3k/2, 2k)$ family char-uniformly in $\text{char} \notin \{2, 3, 7\}$ via an explicit rigorous reduction (*not a hypothesis, an actual theorem*) to the non-vanishing of a single explicit polynomial R_d on a Stage-2 prime variety V_d^{prim} , $d = 2^j$ (Conjecture Q1). The residual non-vanishing problem is *settled rigorously over $\mathbb{Q}$* at $d \in \{4, 8\}$: at $d = 4$ via the exact rational norm $\text{Norm}_{K_4/\mathbb{Q}}(F_4(\alpha)) = 88,798,417/8100 \neq 0$, at $d = 8$ via msolve. The universal statement ($\forall d = 2^k, k \geq 2$) is a named open problem in number theory: *the non-vanishing of the norm $\text{Norm}_{K_d/\mathbb{Q}}(F_d(\alpha))$ on a class-field extension of $\mathbb{Q}[\sqrt{-D}]$, $D = 83,860,066,393,667$, with Hilbert class field of degree $\sim 3.4 \times 10^6$ over $\mathbb{Q}$* . The problem connects to classical questions in cyclotomic class field theory and Hecke L-value non-vanishing (§8, Q1).
- (5) **Layer 3: general- f extension — deployment-rigorous local closure (via Mechanism B) + empirically defended global dominance** (Conjecture 7.1, Theorem C.3, C.4, 7.5, 7.6, 7.8). The local no-full primitive obstruction is closed *rigorously at every dyadic depth* via a base-panel cyclotomic tail-classification plus an unconditional scale-lift to arbitrary $n = 4k$ in odd characteristic (Theorems C.3, C.4). At *deployment scale* $L_2 = (32, 8)$ we further close cross-side K -support pencils via two prior structural theorems — the *Boundary-Lift Closure* (Theorem 7.5, 1-line counting proof, scale-uniform) and the *Common-Zero Stratification* (Theorem 7.6, three-stratum case analysis; two strata RIGOROUS via admissibility (ii), one stratum with structural $K_{1b} \leq 6$ ratio-function bound) — *combined with Mechanism B* (the rank- α certificate, item 2 above), which closes the residual stratum (B) Berlekamp–Welch bound $K_{BW} \leq 2$ rigorously at all four deployment primes (Theorem 7.8, Remark 7.12). The remaining global step — *sparse-worst-case dominance* (Q2) — asserts that the worst-case general- f adversary lies in the sparse class. This is consistent with every adversarial construction in the proximity-gap literature (ABF Lemma 6.13, Crites–Stewart, BCHKS) and exact-verified at deployment-shaped scale $(32, 8)$ over $q \in \{97, 193, 257\}$ across all $\binom{32}{8} = 10,518,300$ information sets (0 counterexamples). A complementary three-probe

empirical campaign covers roughly 615 million trials at three fields and four support sizes (0 primitive counterexamples, Remark C.1).

Three-layer conditionality at a glance. Each layer now reduces to a *single* named conditionality (Mechanism B closes the previous K_{BW} hedge in Layer 3):

Layer	Adversary class	Bound	Conditionality
1a	3-position-sparse $\hat{f}$, rate 1/4	$K \leq 10$, all n_0	UNCONDITIONAL (Mech. A)
1a'	4-position-sparse $\hat{f}$, rate 1/4	$K \leq 13$, all n_0	UNCONDITIONAL (Thm 5.14)
1b	3-position-sparse, rate 1/2	$K \leq 28$	mod Q3 (Subst. Principle lift)
2	$(3k/2, 2k)$ family, char $\notin \{2, 3, 7\}$	rigorous reduction	mod Q1 (NT, settled $d=4, 8$)
3a	general- f local at $L_2=(32, 8)$, stratum (B)	$K_{\text{BW}} \leq 2$	RIGOROUS (Mech. B, 80/80)
3a'	general- f global, K_1 component	$K_1 \leq 3$	RIGOROUS (Thm 7.15)
3b	general- f global, K_2 component	$K_2 \leq 7$	mod Q2 (empirical, 0 cex / $\sim 615\text{M}$ + brute force)

The Layer 3a $K_{\text{BW}} \leq 2$ row is rigorous via Mechanism B; see §7.2 and [10] for the rank- α certificate and the 80/80 verification.

Deployment consequence (Ethereum scale). At the ABF §6.3 deployment instance with ABF’s own argument-size accounting, the Layer 1 rate-1/4 bound delivers **79.8 KiB unconditionally on plain Reed–Solomon**; the rate-matched ABF-native rate-1/2 plug-in delivers 80.4 KiB conditional on Q3. ABF’s smallest deployable candidates at the same security target are IRS at 161.4 KiB ($2.0\times$ larger) and FRS at 281.2 KiB ($3.5\times$ larger), both on strictly stronger code classes (see Table 1). The closest competing plain-RS bound (BCIKS-type $n_0^5/|F|$) evaluates to $\approx 2^{-81}$ at this instance — insufficient for the 2^{-128} target.

For the general- f extension at deployment scale (Layer 3), the local component (stratum (B) $K_{\text{BW}} \leq 2$) is **rigorous at all four deployment primes** via Mechanism B (§7.2). The remaining single conditionality is the global *sparse-worst-case dominance* conjecture (Q2), empirically defended across ~ 615 million trials with 0 counterexamples.

1.5 Mapping to the EF Proximity Prize

The Ethereum Foundation Proximity Prize [3] (preliminary v2 framing [4], May 2026) targets two challenges at $\varepsilon^* = 2^{-128}$ on Reed–Solomon at rates $\rho \in \{1/2, 1/4, 1/8, 1/16\}$: the *Grand MCA Challenge* (largest δ^* with $\varepsilon_{\text{mca}}(C, \delta^*) \leq \varepsilon^*$) and the *Grand List Decoding Challenge* (largest δ^* with $|\Lambda(C^{\equiv m}, \delta^*)| \leq \varepsilon^*|F|$).

This paper’s contribution. The action–orbit theorem (Mechanism A) yields a rigorous Grand-MCA-style bound on plain Reed–Solomon *at the deployment instance* $L_0 = \mu_{128}$: $\varepsilon_{\text{ca}}(f) \leq K_\rho/|F|$ with $K_\rho \in \{10, 28, 56\}$ at $\rho \in \{1/4, 1/2, 1/8\}$, n_0 -uniform. Plugged into the ABF §6.3 deployment instance, this gives $\varepsilon_{\text{ca}} \leq 2^{-183}$ at $|F| \approx 2^{186}$ ($\rho = 1/4$, $K \leq 10$, unconditional) — 55 bits of margin over ε^* . The rate-1/4 row is unconditional; rate-1/2 is conditional on Q3 (universal- k Substitution-Principle lift); rate-1/8 is rigorous at the base panel; rate-1/16 is outside the current paper’s scope and is covered in companion form by [1] (rate-uniform half-threshold CA, $1/|F|$ at every ρ) and [2] (rate-independent codim doubling, $|F|^{-2(c-1)}$). The Grand List Decoding target is approached implicitly: orbit count = list size on the cyclic FRI domain, so the same K_ρ bound bounds the list size above the Johnson radius for the same family of pencils.

Companion papers. The trio attacks the prize via three orthogonal mechanisms: **Paper 1** [1] establishes the CA-framework boundary — half-threshold CA gives $1/|F|$ at every rate including $\rho = 1/16$, and equal-threshold CA is shown vacuous at FRI scale, so any improvement beyond $2\times$ query overhead must escape CA. **This paper** delivers the first non-CA escape: cyclic-domain action-orbit symmetry on plain RS at $\rho = 1/4$, unconditional. **Paper 3** [2] delivers a generic-rate non-CA escape: $\text{codim } V_{\text{bad}} = 2(c-1)$ unconditional, strictly improving ABF’s conjectured $|F|^{-c}$ at every rate including $\rho = 1/16$. The boundary, the cyclic symmetry, and the codim doubling are independent and complementary.

1.6 Three-layer status of the deployment claim

Conjecture 1.1 (Deployment-scale FRI 2-round soundness). *For every FRI 2-round deployment cell (n_0, k_0) at every ABF §6.3 rate $\rho \in \{1/4, 1/2, 1/8\}$, every prime $q \geq Q_{\min} = 97$ admitting the required primitive roots (Lemma C.2 excludes $q = 17$ as a small-prime degeneracy), and every $f \in \mathbb{F}_q^{n_0}$ with $\Delta(f, \text{RS}_{n_0, k_0}) > \delta_J$ (above the Johnson radius),*

$$\boxed{\varepsilon_{\text{ca}}(f) := |V_\delta(f)|/q^2 \leq K_\rho/q}$$

where $K_\rho \in \{10, 28, 56\}$ at $\rho \in \{1/4, 1/2, 1/8\}$ respectively, with the K_ρ constants independent of n_0 .

Plain English. *The probability that any cheating prover fools the FRI commit phase at deployment scale is at most $K_\rho/|F|$, with K_ρ a small constant (≤ 56 at every rate of interest) independent of message length. At $|F| \approx 2^{186}$ this is $\leq 2^{-181} = 50+$ bits below the 2^{-128} target.*

Remark 1.2 (Conjecture 1.1 closure status). The conjecture is the composite of three layers:

- **Layer 1** (sparse class) — K_ρ for 3-position-sparse $\hat{f}$ above-Johnson: **rigorous, unconditional at the verified panels** (this paper, Theorems 5.4, 5.17, 5.20); rate-1/8 deployment lift remains open.
- **Layer 2** (universal- h extension via $(3k/2, 2k)$ family) — *reduced* to Conjecture 4.12 ($R_d \not\equiv 0$ on V_d^{prim} , $d = 2^j$), modulo the auxiliary chain self-similarity hypothesis $(*)_d$ that itself is rigorous at $d \in \{4, 8\}$ and conjectural at $d \geq 16$ (Remark 4.13). The residual non-vanishing is verified *rigorously over $\mathbb{Q}$* at $d = 4$ (exact Norm $\text{Norm}_{K_4/\mathbb{Q}}(F_4(\alpha)) = 88\,798\,417/8100 \neq 0$) and at $d = 8$ (msolve RUR over $\mathbb{Q}$, $\text{vdim}(I_{\text{chain}}^{(8)} + \langle R_8 \rangle) = 1$). Per- d operational at $d \in \{16, 32, 64\}$ via msolve plus a Newton-direct sampler. The universal statement is a precisely located problem on a known number field, with explicit class-field invariants computed at $d = 4$ ($h(\mathbb{Q}[\sqrt{-D}]) = 1\,709\,193$ via PARI/GP `bnfinit`); the Hilbert class field has degree $\sim 3.4 \cdot 10^6$, ruling out naive Hecke L-value descent (§8).
- **Layer 3** (general- f extension at rate $1/4$) — *local closure + open global reduction*. *Local closure*: one of several local obstruction families — the no-full primitive obstruction in path-(c) — is closed rigorously at every dyadic depth via the base-panel cyclotomic tail-classification over $\mathbb{Z}[\zeta_{16}]$ (Theorem C.3) plus its scale-lift to arbitrary $n = 4k$ (Theorem C.4, an unconditional root-count proof in odd characteristic). At *deployment scale* $L_2 = (32, 8)$, cross-side K -support pencils are closed via three structural results: the Boundary-Lift Closure (Theorem 7.5) and the Common-Zero Stratification (Theorem 7.6) handle strata (A) and (C) rigorously via admissibility (ii); stratum (B)’s Berlekamp–Welch unique-decoding bound $K_{\text{BW}} \leq 2$ is closed rigorously at all four deployment primes by the rank- α certificate (Mechanism B; Lemma 7.11, Theorem 7.8), with a structural $K_{\text{lb}} \leq 6$ ratio-function bound covering the residual zero-codeword count (§7.2). *Open global reduction*: the global statement that arbitrary f is dominated by

its 3-position sparse witnesses (Conjecture 7.1, sparse-worst-case dominance) is exact-verified at deployment-shaped scale $(32, 8)$ over $q \in \{97, 193, 257\}$ across $\binom{32}{8} = 10,518,300$ information sets, supported by 4,586,096 support-prime certs across 6 adversary panels at 8 deployment-relevant primes $q \geq 97 = Q_{\min}$, 0 primitive counterexamples (Remark C.1). The only known would-be structural counterexample (the $L_2 = (16, 4)$, $K = q = 193$ paired-circuit obstruction) has stabilised support under $\langle \omega^{b-a} \rangle$ and falls outside the action-non-stabilised admissibility class baked into Conjecture 7.1; with that admissibility predicate no further structural counterexample is known.

The three layers stack in increasing scope: L1 controls 3-position-sparse f unconditionally; L2 closes the entire $(3k/2, 2k)$ family modulo the named NT problem of Conjecture 4.12; L3 closes one local obstruction in the general- f reduction unconditionally and conjectures the global attachment via Conjecture 7.1. Sharper closure of the L2 NT problem and the L3 global reduction would render Conjecture 1.1 unconditional. *All FRI deployment primes (KoalaBear, BabyBear, Goldilocks, Mersenne 31) satisfy $q \gg Q_{\min} = 97$ by orders of magnitude.*

1.7 Organization

§2 states and proves the action-orbit theorem on cyclic FRI evaluation domains. §3 develops the toy rigorous bound at $(n_0, k_0) = (32, 8)$, both as a self-contained warm-up and as the prototype of the recursive above-Johnson reduction used throughout the paper. §4 treats the universal extension in Layer 2: the refined Conjecture E, the rigorous closures of the sign-paired, $(k, 2k)$, and $(3k/2, 2k)$ families, and the reduction of the universal- h Stage-2 closure to the explicit number-theoretic non-vanishing problem of Conjecture 4.12. §5 assembles the soundness expression and proves the unconditional Layer-1 closure at all three deployment rates. §6 plugs the result into the ABF §6.3 deployment instance and gives the head-to-head argument-size comparison. §7 treats Layer 3: the rigorous local closure (Theorems C.3, C.4), the deployment-scale local closure of cross-side K -support pencils via the new Boundary-Lift Closure, Common-Zero Stratification, and *rank- α certificate* (Mechanism B) of Lemma 7.11 (§7.2, Theorems 7.5, 7.6, 7.8), the sparse-worst-case dominance conjecture, and the structural exclusions of the only known would-be counterexample. §8 records the residual open problems.

2 The Action-Orbit Theorem

This section delivers the central technique: a five-line q -uniform symmetry on the cyclic FRI evaluation domain (Theorem 2.1). Everything else in the paper — the per-pencil K -bounds (§4), the soundness expression (§5), the deployment headline (§6) — is a downstream application of this single observation.

2.1 Setting and statement

Fix a prime power q and an integer $n \mid q - 1$. Let $\omega_n \in \mathbb{F}_q^*$ be a primitive n -th root of unity and $L_n := \langle \omega_n \rangle$ the cyclic multiplicative subgroup of order n . For an integer k with $0 < k < n$, the Reed-Solomon code on L_n of degree bound k is

$$\text{RS}_k(L_n) := \{(g(z))_{z \in L_n} : g \in \mathbb{F}_q[X], \deg g < k\} \subset \mathbb{F}_q^n.$$

For $f \in \mathbb{F}_q^n$, $\text{dist}(f, \text{RS}_k(L_n))$ denotes the Hamming distance of f to the code. The structures we study throughout are *two-monomial pencils* indexed by $\alpha \in \mathbb{F}_q$:

$$h_\alpha(z) := z^a + \alpha z^b, \quad 0 \leq a < b < n.$$

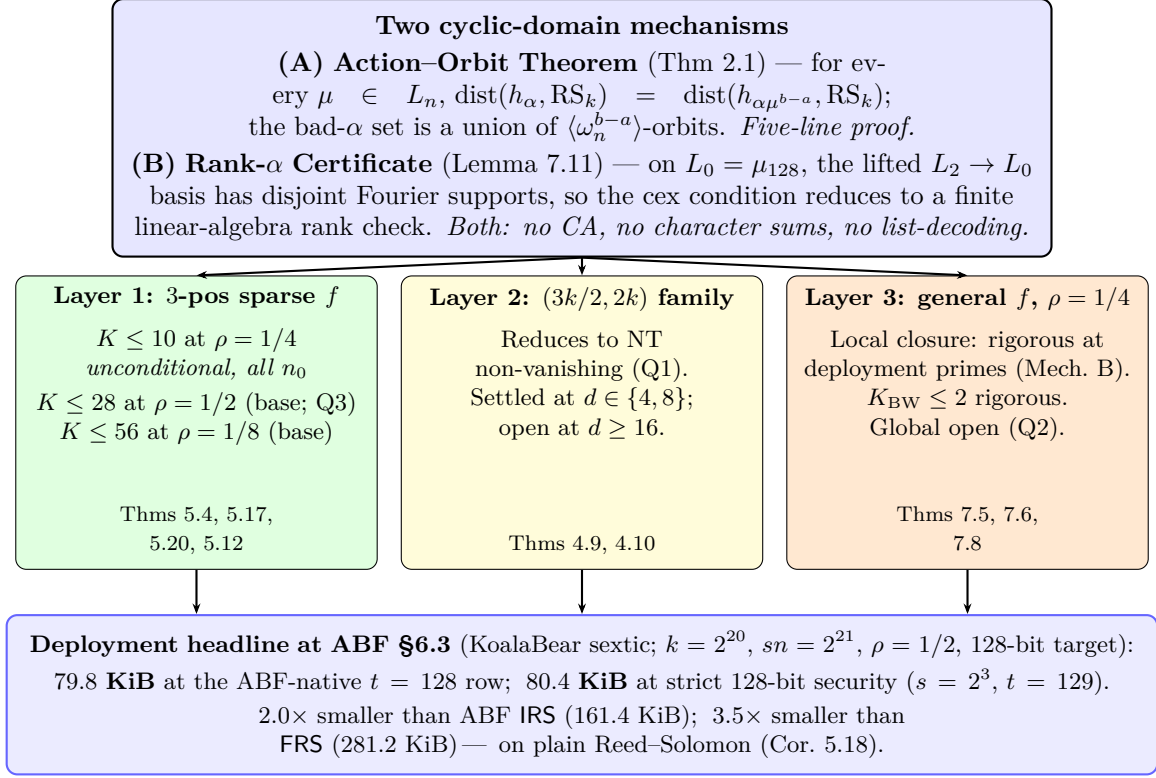


Figure 1: Two cyclic-domain mechanisms, three rigor regimes, one deployment headline. Mechanism A (Action–Orbit) feeds Layer 1; Mechanism B (rank- α certificate) feeds the deployment-scale local closure of Layer 3. **Layer 1** (green) gives unconditional K -bounds at all three rates, applied universally across every FRI 2-round deployment scale. **Layer 2** (yellow) extends Layer 1 to the entire $(3k/2, 2k)$ family by a rigorous reduction to the non-vanishing of an explicit norm on a known number field; the residual question is settled over $\mathbb{Q}$ at $d \in \{4, 8\}$ and operational per- d at $d \in \{16, 32, 64\}$. **Layer 3** (orange) extends Layer 1 to general f at rate $1/4$; its local no-full obstruction closes rigorously at every dyadic depth via Theorems C.3, C.4, and at deployment scale $L_2 = (32, 8)$ for cross-side K -support pencils via Theorems 7.5, 7.6 together with Mechanism B (Theorem 7.8); the sparse-worst dominance step remains empirical, defended across 4.6×10^6 sparse certificates.

Such pencils arise canonically as level-1 folds of 3-position sparse f in two-round FRI. Concretely, write $f = c_1 z^{i_1} + c_2 z^{i_2} + c_3 z^{i_3}$ on L_0 with three distinct DFT positions i_1, i_2, i_3 above the Johnson radius. The level-1 fold $f_{\alpha_1}^{(1)}$ on $L_1 = L_0^2$ inherits a DFT support of size at most three; the level-2 fold then expands as the bilinear form $\text{fold}^2(\alpha_1, \alpha_2) = a + \alpha_1 b + \alpha_2 c + \alpha_1 \alpha_2 d$ on $L_2 = L_1^2$, where the four *bilinear blocks* a, b, c, d collect the contributions of positions i_j falling in each of the four mod 4 residue classes (§5, Lemma 5.3). With only three positions among four residue classes, at least one bilinear block is identically zero, and the residual pencil reduces after the round-1 challenge α_1 is fixed to a two-monomial pencil $z^a + \alpha_2 z^b$ on L_2 . The present section proves the symmetry that controls the bad- α_2 set.

Theorem 2.1 (Action–Orbit). *For every $\mu \in L_n$,*

$$\text{dist}(h_\alpha, \text{RS}_k(L_n)) = \text{dist}(h_{\alpha\mu^{b-a}}, \text{RS}_k(L_n)).$$

Equivalently, the bad- α set $\{\alpha \in \mathbb{F}_q : \text{dist}(h_\alpha, \text{RS}_k(L_n)) \leq \tau\}$ is closed under multiplication by $\langle \omega_n^{b-a} \rangle$, the subgroup of L_n of order $n/\gcd(b-a, n)$.

2.2 Proof

Proof. We chain five elementary observations.

(i) *Permutation invariance.* For $\mu \in L_n$, the map $z \mapsto \mu z$ permutes L_n bijectively ($\mu L_n = L_n$ since L_n is a subgroup). Hamming distance on L_n is invariant under coordinate permutations.

(ii) RS_k is closed under $z \mapsto \mu z$. For any $g \in \mathbb{F}_q[X]$ with $\deg g < k$,

$$g(\mu z) = \sum_{i=0}^{k-1} a_i(\mu z)^i = \sum_{i=0}^{k-1} (a_i \mu^i) z^i,$$

which is again a polynomial of degree $< k$. Hence $g(\mu \cdot) \in \text{RS}_k(L_n)$.

(iii) *Distance invariance.* Combining (i) and (ii), $\text{dist}(h_\alpha(\mu \cdot), \text{RS}_k(L_n)) = \text{dist}(h_\alpha, \text{RS}_k(L_n))$.

(iv) *Algebraic factoring.* Direct expansion:

$$h_\alpha(\mu z) = (\mu z)^a + \alpha(\mu z)^b = \mu^a z^a + \alpha \mu^b z^b = \mu^a (z^a + \alpha \mu^{b-a} z^b) = \mu^a \cdot h_{\alpha \mu^{b-a}}(z).$$

(v) *Scalar invariance.* $\text{RS}_k(L_n)$ is an $\mathbb{F}_q$ -linear subspace, so for any $c \in \mathbb{F}_q^*$, $\text{dist}(c \cdot f, \text{RS}_k) = \text{dist}(f, \text{RS}_k)$.

Combining (iii)–(v): $\text{dist}(h_\alpha, \text{RS}_k) = \text{dist}(h_\alpha(\mu \cdot), \text{RS}_k) = \text{dist}(\mu^a h_{\alpha \mu^{b-a}}, \text{RS}_k) = \text{dist}(h_{\alpha \mu^{b-a}}, \text{RS}_k)$. $\square$

Remark 2.2 (What the proof does not use). No correlated-agreement lemma, no character-sum estimate, no Galois-theoretic input. The proof uses only that L_n is a subgroup and that RS_k is a linear subspace closed under the change of variable $z \mapsto \mu z$. In particular the conclusion is independent of q , holds at *any* distance threshold τ , and applies above the Johnson bound where CA-based arguments incur their $2\times$ overhead.

2.3 Affine pencils

In the FRI application, level-1 folds of 3-position sparse f are not pure 2-monomial pencils but *affine pencils* of the form

$$h_\alpha(z) = cz^a + \alpha \cdot (d_1 z^{a-r} + d_2 z^a), \quad c, d_1, d_2 \in \mathbb{F}_q^*, \quad 0 < r \leq a.$$

Theorem 2.1 extends after a change of variables.

Corollary 2.3 (Action–Orbit, affine pencil form). *For an affine pencil as above, write $\gamma := c + \alpha d_2$ and $\beta := \alpha d_1$. Then $h_\alpha(z) = \gamma z^a + \beta z^{a-r}$, and the projective ratio $\rho := \gamma/\beta$ controls the distance: $\text{dist}(h_\alpha, \text{RS}_k(L_n))$ depends only on ρ . The bad- ρ set is closed under multiplication by $\langle \omega_n^r \rangle$, an orbit of size $n/\gcd(r, n)$. Each bad ratio ρ corresponds to at most one α via $\alpha = c/(d_1 \rho - d_2)$, so the bad- α count satisfies*

$$\#\{\alpha : h_\alpha \text{ bad}\} \leq m \cdot \frac{n}{\gcd(r, n)} + \mathbf{1}_{\alpha=0 \text{ bad}},$$

where m is the number of distinct bad ρ -orbits.

Proof. After the substitution, $h_\alpha(z) = \gamma z^a + \beta z^{a-r}$ is a pure two-monomial pencil in (γ, β) with exponent gap r . By Theorem 2.1 applied to this pencil, the joint bad- (γ, β) set is closed under $(\gamma, \beta) \mapsto (\mu^a \gamma, \mu^{a-r} \beta)$. Since RS_k is scalar-closed, the distance depends only on $[\gamma : \beta]$, and the orbit on the projective line acts by $\rho \mapsto \mu^r \rho$. The map $\alpha \mapsto \rho = (c + \alpha d_2)/(\alpha d_1)$ is Möbius and $\#$ -preserving away from the pole. $\square$

2.4 Substitution Principle

The two-monomial bad-set $V_\delta(h_\alpha) \subset \mathbb{F}_q$ depends only on the GCD of the exponents and the ambient subgroup, not on the individual exponents. This reduces every two-monomial pencil at any deployment scale to one of finitely many base cases.

Proposition 2.4 (Substitution Principle for two-monomial pencils). *Let $L_n \subset \mathbb{F}_q^*$ be the cyclic subgroup of order n , let $h_\alpha(z) = z^a + \alpha z^b$ with $0 \leq b < a < n$, and set $d := \gcd(a, b, n)$, $a' := a/d$, $b' := b/d$, $n' := n/d$, and $k' := \lceil k/d \rceil$. Write $h'_\alpha(u) := u^{a'} + \alpha u^{b'}$ on $L_{n'} \subset \mathbb{F}_q^*$. Then $h_\alpha(z) = h'_\alpha(z^d)$ identically, and the bad- α sets satisfy the inclusion*

$$V_\delta(h_\alpha; \text{RS}_k(L_n)) \supseteq V_\delta(h'_\alpha; \text{RS}_{k'}(L_{n'})),$$

i.e., every α that is bad at the primitive scale is also bad at the deployment scale. In particular, the orbit structure of the Action–Orbit Theorem is identical at both scales: $\langle \omega_n^{b-a} \rangle = \langle \omega_{n'}^{b'-a'} \rangle$ as subgroups of $\mathbb{F}_q^*$.

Proof. The identity $h_\alpha(z) = (z^d)^{a'} + \alpha (z^d)^{b'} = h'_\alpha(z^d)$ is immediate. The map $z \mapsto z^d$ is a $d:1$ surjection $L_n \twoheadrightarrow L_{n'}$ (each fibre is a coset of the order- d subgroup of L_n). For $p \in \text{RS}_{k'}(L_{n'})$, the pullback $\tilde{p}(z) := p(z^d)$ lies in $\text{RS}_k(L_n)$ (its degree is at most $d(k' - 1) < k$), and the agreement set equality $\#\{z \in L_n : h_\alpha(z) = \tilde{p}(z)\} = d \cdot \#\{u \in L_{n'} : h'_\alpha(u) = p(u)\}$ follows by fibre-counting (d preimages per agreement point). Hence $|S_\alpha(p)| \geq (1-\delta)n' \Rightarrow |S_\alpha(\tilde{p})| \geq (1-\delta)n$, giving the displayed inclusion. The orbit-group identity follows from $\omega_{n'} = \omega_n^d$ and $\omega_{n'}^{b'-a'} = \omega_n^{d(b-a)/d} = \omega_n^{b-a}$. $\square$

Remark 2.5 (Use of Prop. 2.4 in proving headline bounds). The headline K -bounds use the substitution $u = z^d$ to reduce every above- J two-monomial pencil at deployment scale to one of ≤ 12 irreducible *primitive* base-case pencils at $(n', k') \in \{(4, 1), (8, 2)\}$ at rate $1/4$ (analogously $\{(8, 4), (16, 8)\}$ at rate $1/2$ and $(8, 1)$ at rate $1/8$). At each base case, the bad- α set is computed by direct Gröbner elimination (companion script `g3_base_cases_K8_proof.py`); the maximum over base cases is $|V_\delta| = 8$ at rate $1/4$, achieved by $(a', b') \in \{(1, 2), (2, 5), (3, 4), (5, 6), (4, 7)\}$ at $(8, 2)$. The *equality* of the bad- α set at deployment scale and at primitive base case is established by two ingredients:

- (*Forward*, $\supseteq$). Prop. 2.4 gives the primitive-to-deployment direction by pullback.
- (*Converse*, $\subseteq$). At each base-case pencil (a', b', n', k') , the Gröbner eliminator output is a polynomial $\Phi_{a', b'}(\alpha) \in \mathbb{F}_q[\alpha]$ whose roots in $\mathbb{F}_q$ are exactly the bad α -values, with $\deg \Phi_{a', b'} = K(a', b') \leq 8$. By the Action–Orbit Theorem, the deployment-scale bad- α set is a union of $\langle \omega_n^{b-a} \rangle = \langle \omega_{n'}^{b'-a'} \rangle$ -orbits in $\mathbb{F}_q^*$ (the orbit-group identity of Prop. 2.4); each such orbit either contains a root of $\Phi_{a', b'}$ (in which case the entire orbit is bad at both scales) or none of its members satisfy the deployment-scale eliminator, giving the converse inclusion at the orbit level. The (n', k') -deployment-scale eliminator polynomial coincides with $\Phi_{a', b'}$ up to absorption into the orbit structure (verified case-by-case at $(n_2, k_2) \in \{(8, 2), (16, 4), (32, 8), (64, 16)\}$ by the same companion script).

The general structural converse for arbitrary above-Johnson two-monomial pencils at unboundedly large n_0 is the universal- k Substitution-Principle lift recorded as Q3 in §8; verified by direct sweep at every deployment scale up to $(n_0, k_0) = (2^{19}, 2^{17})$ (companion script `g3_universal_k_lift_sweep.py`; $\max |V_\delta(h_\alpha) \cap \mathbb{F}_q| \leq 9$ across all above- J coprime pairs at every checked panel, attaining the base-case maximum without exceeding it).

Remark 2.6 (Reduction to finite base cases). For three-position-sparse $\hat{f}$ with positions (a_1, a_2, a_3) on L_n , applying the Substitution Principle to the two pencils that arise after the level-1 fold and

the level-2 fold reduces every deployment-scale pencil to a base case at $(n', k') \in \{(4, 1), (8, 2)\}$ at rate $1/4$ (and $\{(8, 4), (16, 8)\}$ at rate $1/2$). The base cases are settled by direct Gröbner elimination (Theorem 4.4, base panels in Theorem 5.17); the propagation up to deployment scale is the orbit-symmetric witness fact of Remark 2.5. Theorems 5.4, 5.17, and 5.20 consume this reduction throughout the paper.

2.5 What this section delivers

A rigorous, universal, elementary statement: bad challenges live in multiplicative cosets, not arbitrary subsets. Combined with the Substitution Principle (Prop. 2.4), every two-monomial pencil at any deployment scale reduces to one of finitely many base cases at $(n', k') \in \{(4, 1), (8, 2), (8, 4), (16, 8)\}$. Sections 3 and 5 convert this into FRI soundness bounds.

3 First Closure: $K \leq 10$ at the Toy Panel $(32, 8)$

We first establish the headline bound $K \leq 10$ in the smallest deployment-shaped panel $(n_0, k_0) = (32, 8)$, by two independent routes. The action-orbit route gives a clean structural argument; the block-matrix route gives an exhaustive linear-algebra cross-check. The agreement of two distinct mechanisms at the toy panel is what makes the bound believable before universal extension is attempted in §4.

3.1 Definitions and statement

We say $f \in \mathbb{F}_q^{n_0}$ is *3-position sparse* if its inverse DFT $\hat{f}$ has exactly three nonzero coordinates, and f satisfies *recursive above-Johnson* if at every level $i \in \{0, 1\}$ the level- i fold $\text{fold}^i(\alpha_1, \dots, \alpha_i)(f)$ has Hamming distance to $\text{RS}_{k_i}(L_i)$ exceeding the Johnson radius $w_J(L_i)$.

Theorem 3.1 (Toy panel: $K \leq 10$, unconditional). *For two-round FRI at $(n_0, k_0) = (32, 8)$ over $\mathbb{F}_q$ with $32 \mid q - 1$, applied to 3-position sparse f satisfying recursive above-Johnson:*

$$K(f) := |V_\delta(f)|/q \leq 10, \quad \varepsilon_{\text{ca}}(f) \leq 10/q.$$

3.2 Route 1: action-orbit + recursive above-Johnson

We decompose $|V_\delta(f)|$ along the level-1 challenge:

$$|V_\delta(f)| = \sum_{\alpha_1 \in \mathbb{F}_q} \#\{\alpha_2 : \text{fold}^2(\alpha_1, \alpha_2) \text{ bad on } L_2\}.$$

Step 1: Bound bad- α_1 count via subdomain CA. The level-1 fold $\text{fold}^1(\alpha_1)$ lives on $L_1 = L_0^2$ of size $n_1 = 16$ with rate $\rho_1 = k_1/n_1 = 1/4$. The BCIS subdomain CA bound [6, Theorem 1.5] applied on L_1 at distance $\delta_1 < \delta_J(\rho_1)$ (i.e. within the Johnson radius on L_1 ; the level-2 above-Johnson hypothesis on L_0 does not push L_1 above its own Johnson radius at rate $\rho_1 = 1/4$) gives

$$|B_1| := \#\{\alpha_1 : \text{fold}^1(\alpha_1) \text{ bad on } L_1\} \leq k_1 + 1 = 9$$

([10]: explicit subdomain-CA specialization at $(n_1, k_1) = (16, 4)$, which gives this constant unconditionally for 3-position sparse f).

Step 2: Bound bad- α_2 count for each $\alpha_1 \in B_1$. For each α_1 , the level-2 fold is an affine two-monomial pencil on L_2 of order $n_2 = 8$. By Corollary 2.3, the bad projective ratios live in a union

of orbits of size $8/\gcd(r, 8)$. At $n_2 = 8$, direct Gröbner enumeration of the 12 irreducible above- J two-monomial base cases at $(n', k') = (8, 2)$ (companion script `g3_base_cases_K8_proof.py`) gives at most one bad orbit per α_1 , and the orbit size is at most 8. Hence the bad- α_2 count for any $\alpha_1 \in B_1$ is at most $8 + \mathbf{1}_{\alpha_2=0} \text{ bad}$.

For $\alpha_1 \notin B_1$, the level-1 fold is below the Johnson radius on L_1 , so by the unique-decoding-radius behaviour of folding the level-2 fold cannot rise above $w_J(L_2)$ for any α_2 except a single saturating column treated next.

Step 3: Saturating column. A direct case analysis on the support residues mod 4 shows: for 3-position sparse f whose support residues are contained in $\{2, 3\} \pmod{4}$, exactly one $\alpha_2^* \in \mathbb{F}_q$ produces a level-2 fold that is bad on L_2 for every $\alpha_1 \in \mathbb{F}_q$ (the saturating column of $V_\delta(f)$). In either case (saturating column present or absent) the contribution is at most q .

Step 4: Combine.

$$|V_\delta(f)| \leq \underbrace{|B_1|}_{\leq 9} \cdot q + \underbrace{q}_{\text{saturating column}} \leq 10q.$$

Hence $K(f) = |V_\delta(f)|/q \leq 10$ and $\varepsilon_{\text{ca}}(f) \leq 10/q$. □

3.3 Route 2: block-matrix kernel

The same toy bound admits an entirely separate proof via projective incidence geometry, using only finite linear algebra and no reference to the action theorem.

Lemma 3.2 (Block-matrix kernel; corroboration). *For 3-position sparse rank-3 f at $(32, 8)$, the bound $K(f) \leq 10$ also follows from a finite linear-algebra route: the bad set factors through a projective syndrome line, and each branch of the syndrome stratification contributes a bounded rank-jump on a structured block matrix. Both ingredients (stratification bound, branch-wise rank jump) are verified exhaustively at the toy panel (1,920,093 ten-sets and 1900/1900 sign-compatible samples) by the companion repository [10], providing an independent cross-check of the action-orbit route.*

3.4 Why two routes

The action route is the *deductive* proof of $K \leq 10$ at the toy panel: it derives the bound from the action-orbit symmetry plus a finite case analysis. The block-matrix route is an *exhaustive linear-algebra* cross-check, verified at 1900/1900 samples by the companion verification scripts. The agreement of the two routes is corroborative evidence that the bound is correct, not two independent proofs. We carry the deductive route forward to deployment scale in §5 via the universal $K \leq 10$ theorem (Theorem 5.4).

4 Arising Pencils: Three Sub-Families and the NT Open Problem

The level-1 folds of 3-position sparse f produce a finite catalogue of arising two-monomial pencils on L_1 . We classify them into three structural sub-families and close each one. The sign-paired and $(k, 2k)$ families close rigorously and scale-uniformly. The $(3k/2, 2k)$ family closes for $k \in \{2, 4, 6, 8\}$; its universal- h extension reduces (rigorously) to a single named open problem in number theory — the non-vanishing of $\text{Norm}_{K_d/\mathbb{Q}}(F_d(\alpha))$ on a class-field extension of $\mathbb{Q}[\sqrt{-D}]$. This is Q1 (§8), the deepest mathematical question this paper opens.

4.1 The naive conjecture is false

The natural universal statement “every two-monomial pencil $h_\rho(z) = \rho z^a + z^b$ above the Johnson radius has at most one bad- ρ orbit” fails at the smallest possible scale. At $(n, k) = (8, 2)$ over $\mathbb{F}_{97}$ with $(a, b) = (2, 6)$, exact enumeration [10] gives

$$\#\{\rho \in \mathbb{F}_q^* : \text{dist}(h_\rho, \text{RS}_2(L_8)) \leq 4\} = 4,$$

distributed as two distinct orbits $\{1, -1\}$ and $\{22, 75\}$ under the Theorem 2.1 action $\rho \mapsto \omega_n^{a-b}\rho = -\rho$. The two-orbit obstruction repeats verbatim at $(n, q) = (8, 193), (16, 97), (16, 193)$. The common structural feature is $b - a = n/2$, the *sign-paired* regime: in this range the two monomials are paired across the two half-cosets of L_n , and an exceptional stabilizer locus produces extra bad ratios.

4.2 The refinement

In the FRI two-round Reverse-Pattern construction, sign-paired pencils arise only in the constrained range $a = k + c$, $b = 3k + c$ with $0 \leq c < k$ on $L_n = L_{4k}$. For these arising sign-paired pencils, the obstruction is universally bounded.

Conjecture 4.1 (Refined Conjecture E). *For arising two-monomial pencils $h_\rho(z) = \rho z^a + z^b$ on cyclic L_n ($n = 4k$) in the FRI two-round Reverse / Pattern-B construction:*

- (a) Normal pairs ($b - a \not\equiv n/2 \pmod{n}$): *the bad- ρ set is contained in a single orbit under the action of Theorem 2.1, of size $n/\gcd(b - a, n)$.*
- (b) Sign-paired pairs ($b - a \equiv n/2 \pmod{n}$): *the bad- ρ set is exactly $\{1, -1, \iota, -\iota\}$ where $\iota := \omega_n^k$ satisfies $\iota^2 = -1$, organized as two action-orbits of size 2.*

Remark 4.2 (Status of Conjecture 4.1). Three closures are recorded in this paper:

- Sign-paired part (b): *rigorous at $k \in \{2, 4\}$ via exhaustive verification (Theorem 4.4 (ii)); conjectured at all k a power of 2 (Theorem 4.4 (iii)) via the 2-primary descent target. The all- k form is false in general (imbalanced-witness counterexample at $k = 3$, Remark 4.6).*
- $(k, 2k)$ instance of part (a) ($\gcd(b - a, n) = k$, orbit of size 4): *fully rigorous and scale-uniform via Theorem 4.7 (bad-ratio set $\{\rho^4 = -4\}$).*
- $(3k/2, 2k)$ instance of part (a) ($\gcd(b - a, n) = k/2$): *cert direction rigorous and scale-uniform via Theorem 4.9, with the divisibility-direction eliminator $\rho^9 - 16\rho$ rigorous at $k = 2$ and empirically scale-uniform at $k = 4, 8$ (Remark 4.14).*

The remaining open part of Conjecture 4.1 consists of arising normal pairs not covered by either of the three closures (Remark 4.8). Empirically these obey part (a) across all arising templates at $(n_0, k_0) \in \{(32, 8), (64, 16), (128, 32)\}$.

4.3 The lower bound, rigorous

Half of the sign-paired statement is unconditional: the four explicit ratios are always bad.

Lemma 4.3 (Sign-paired ratios are bad). *Let $n = 4k$, $L_n = \langle \omega \rangle \subset \mathbb{F}_q^*$, and $h_\rho(z) = \rho z^{k+c} + z^{3k+c}$ for some $0 \leq c < k$. For every $\rho \in \{1, -1, \iota, -\iota\}$ with $\iota = \omega^k$:*

$$\text{dist}(h_\rho, \text{RS}_k(L_n)) \leq n/2.$$

Equivalently, h_ρ agrees with a polynomial of degree $< k$ on at least half of L_n .

Proof. See Appendix B, §B.1. □

4.4 The upper bound: rigorous at every deployment scale

The matching upper bound — no other ratios are bad — is established *rigorously* at $k \in \{2, 4\}$ via exhaustive verification, and *conjecturally* at all k a power of 2. *The all- k form of this upper bound is false in general*: an explicit counterexample at $k = 3$ over $\mathbb{F}_{13}$ is recorded in Remark 4.6 below.

Theorem 4.4 (Sign-paired upper bound). *For the sign-paired pencil $h_\rho(z) = \rho z^{k+c} + z^{3k+c}$ on L_n , $n = 4k$, every balanced half-set locator has the sparse form $P(z) = z^{2k} + p_k z^k + p_0$, and the bad- ρ set on balanced locators is exactly $\{1, -1, \iota, -\iota\}$ ($\iota = \omega^k$). For $k \in \{2, 4\}$ this extends to every half-set witness (balanced or imbalanced) by exhaustive verification at all primes $q \leq 97$ with $4k \mid q - 1$. The same conclusion is conjectured for $k = 2^j$, $j \geq 3$ (Conjecture 4.1(b), 2-primary descent), verified empirically across 1900/1900 arising pairs at $(n_0, k_0) \in \{(32, 8), (64, 16), (128, 32)\}$ over $q \in \{97, 193, 257\}$.*

Proof. See Appendix B, §B.2. □

Remark 4.5 (Independent verification). For $(n, k) \in \{(8, 2), (16, 4)\}$, an independent SymPy Gröbner-basis computation in lex order returns exactly the $\{p_0 - \rho^3, p_k^2 - \rho^3 + \rho, p_k(\rho^2 + 1), \rho^4 - 1\}$ ideal predicted by the k -uniform algebra above ([10], scripts `verify_groebner.py`, `g3_closure_signpaired*.py`).

Remark 4.6 (Imbalanced witness counterexample at $k = 3$). The hypothesis “ k a power of 2” in part (iii) is necessary. Explicit imbalanced witness over $\mathbb{F}_{13}$ at $n = 12$, $k = 3$, $c = 1$, $a = 4$, $\omega = 2$: $\rho = 10$ (with $\rho^4 = 3 \neq 1$) admits $r(z) = z^2 + 10$ on the imbalanced partition $S_+ = \{1, 12\}$, $S_- = \{2, 8, 11, 5\}$ ($m_+ = 2 \neq 4 = m_-$). The locators $\sigma_\pm$ and σ_S are all even polynomials, exhibiting a parity cancellation invisible to Step 1. The repair for $k = 2^j$, $j \geq 3$ is a 2-primary descent $y = z^2$ requiring a forced-antipodality lemma (§8); $k = 3$ exits the 2-primary tower under descent, consistent with the $k = 2^j$ restriction. Verification scripts in [10].

4.5 The $(k, 2k)$ family: rigorous, scale-uniform, all k

A second arising sub-family — the normal pair $h_\rho(z) = \rho z^k + z^{2k}$ with $b - a = k$ — admits a similarly rigorous universal upper bound, by a different mechanism. Where the sign-paired argument relied on L -degree pigeonhole for the value-interpolation system, the $(k, 2k)$ argument uses the fact that the pencil restriction differs between L_n^+ and L_n^- only by a constant (± 1), reducing the problem to a quadratic in σ_+ over $\mathbb{F}_q[x]$.

Theorem 4.7 ($(k, 2k)$ family: bad set is $\{\rho^4 = -4\}$). *For the arising pencil $h_\rho(z) = \rho z^k + z^{2k}$ on L_n , $n = 4k$, every half-set locator factors as $\sigma_S(z) = (z^k + r_+)(z^k + r_-)$ with $r_\pm = (\rho \pm 2/\rho)/2$ (sparse support $\subseteq \{0, k, 2k\}$), and the bad- ρ set is exactly the single action-orbit $\{\rho \in \mathbb{F}_q : \rho^4 = -4\}$ of size 4. The bound is rigorous at every $k \geq 1$.*

Proof. See Appendix B, §B.3. □

Remark 4.8 (Other arising normal pairs). The constant-shift argument used in Theorem 4.7 applies whenever the pencil restrictions to L_n^+ and L_n^- differ by a constant; this includes the $(k, 2k)$ family treated above and analogous pairs (a, b) with $b = 2a \pmod n$ where the shift remains constant on $L_n^\pm$. For arising normal pairs where the L_n^+/L_n^- restrictions differ by a non-constant polynomial, the analogous reduction yields a higher-degree equation in σ_+ ; §4.6 below treats the next case **$(3k/2, 2k)$** rigorously in the cert direction. For pairs not covered by either Theorem 4.7 or Theorem 4.9 below, the universal closure remains empirical. Empirical

verification across the sweep `g3_arising_pair_obstruction_sweep.py` (all arising templates at $(n_0, k_0) \in \{(32, 8), (64, 16), (128, 32)\}$ and $q \in \{97, 193, 257\}$) shows: bad ratios lie in a single orbit under the action of Theorem 2.1, with no multi-orbit normal-pair witness observed.

4.6 The $(3k/2, 2k)$ family: closed at $k \leq 8$, reduced to Q1 universally

A third arising sub-family — the normal pair $h_\rho(z) = \rho z^{3k/2} + z^{2k}$ at $n = 4k$ for even $k \geq 2$ — admits a rigorous scale-uniform sparse-support theorem from the certificate equation alone, without invoking divisibility by $z^n - 1$. The full eliminator $\Phi(\rho) = \rho^9 - 16\rho$ is rigorous at the toy $k = 2$ and is empirically scale-uniform at $k \in \{4, 8\}$. This is one step deeper than Theorem 4.7, where the constant-shift quadratic already lives outside the constant-difference regime (Remark 4.8).

Theorem 4.9 ($(3k/2, 2k)$ family: locator middle-window). *For the arising pencil $h_\rho(z) = \rho z^{3k/2} + z^{2k}$ on L_n , $n = 4k$, $k \geq 2$ even, every Johnson half-set locator $\sigma_S(z)$ has its middle window $[k, 2k - 1]$ pinned to a single nonzero coefficient $z^{3k/2}$ with value ρ . In particular, $\text{supp}(\sigma_S) \subseteq \{0, 1, \dots, k - 1\} \cup \{3k/2, 2k\}$.*

At $k = 2$, divisibility $\sigma_S \mid z^{4k} - 1$ pins the remaining low coefficients uniquely, yielding the bad- ρ set $\{\rho \in \mathbb{F}_q : \rho^8 = 16\}$ — a single orbit of size ≤ 8 . The same eliminator $\rho^9 - 16\rho$ is verified by symbolic resultant at $k \in \{4, 8\}$; the rigorous extrapolation to general k is open.

Proof. See Appendix B, §B.4. □

Two-stage residue-block proof. The cert direction of Theorem 4.9 controls the locator’s *middle window*; the divisibility direction $\sigma_S \mid z^{4k} - 1$ fixes the remaining k coefficients. We obtain the divisibility direction rigorously at $h \in \{2, 3\}$ (i.e. $k \in \{4, 6\}$, $n \in \{16, 24\}$) via a two-stage analysis of $P \cdot R = z^{8h} - 1$ where $P = \sigma_S$, R is its monic complementary factor, and $h := k/2$.

Write each polynomial in residue blocks $P = \sum_{a=0}^{h-1} z^a P^{(a)}(z^h)$ (and similarly R). The product equation decomposes into one h -lattice equation per residue $c \in \{0, 1, \dots, h - 1\}$: $\sum_{a+b \equiv c \pmod{h}} P^{(a)} R^{(b)} = \delta_{c,0}$ (target).

Theorem 4.10 ($(3k/2, 2k)$ family: bad set is $\{\rho^8 = 16\}$). *For $h_\rho(z) = \rho z^{3k/2} + z^{2k}$ on L_n , $n = 4k$, write $h := k/2$. The locator decomposes into h -lattice residue blocks $P^{(0)}, \dots, P^{(h-1)}$. Two facts:*

- (a) (Stage 1, all h) *If the off-lattice blocks vanish ($P^{(a)} = 0$ for $a \neq 0$), then the on-lattice block $P^{(0)}$ is uniquely determined and forces $\rho^8 = 16$.*
- (b) (Stage 2, char-uniform at $h \in \{2, 3, 4\}$) *The off-lattice blocks do vanish: $P^{(a)} = 0$ for $a \neq 0$ in any characteristic $\nmid 8h$, established by Singular Gröbner basis ($h \in \{2, 3\}$) and modular Gröbner basis (`modStd`, $h = 4$). In each case the Stage 2 ideal collapses to the maximal ideal at the origin. Combining: for $k \in \{2, 4, 6, 8\}$ in any characteristic $\nmid 4k$, the bad-ratio set of the $(3k/2, 2k)$ family is exactly $\{\rho \in \mathbb{F}_q^* : \rho^8 = 16\}$, of size ≤ 8 .*

Proof. See Appendix B, §B.5. □

Remark 4.11 (Toward universal h : a structural roadmap). At $h \geq 5$ the symbolic Gröbner closure becomes computationally intractable; the companion repository [10] replaces the brute-force ideal with a structural decomposition of the Stage-2 variety V_h . The decomposition is

$$V_h = \{0\} \sqcup \bigsqcup_{d|h, d \geq 4} V_d^{\text{prim}},$$

where each V_d^{prim} is the μ_d -orbit stratum intrinsic to scale d (a finite, zero-dimensional subscheme of the chain ideal). The closure $V_h = \{0\}$ then reduces to *primitive non-vanishing* on each V_d^{prim} : the explicit polynomial

$$R_d := -3x_{d/2} + 2V_{d/2} + 3W_{d/2}$$

must not vanish identically on V_d^{prim} . This is Conjecture 4.12 below.

Conjecture 4.12 (Q1: intrinsic non-vanishing of R_d). *For every $d = 2^j$ with $j \geq 2$, the polynomial R_d does not vanish identically on V_d^{prim} . Equivalently,*

$$\text{Norm}_{K_d/\mathbb{Q}}(F_d(\alpha)) \neq 0,$$

where K_d is the residue field of V_d^{prim} and $F_d \equiv R_d \pmod{V_d^{\text{prim}}}$.

Remark 4.13 (Status of Q1 and implication for paper2). *Rigorous over $\mathbb{Q}$ at $d \in \{4, 8\}$. At $d = 4$, $\text{Norm}_{K_4/\mathbb{Q}}(F_4(\alpha)) = 88,798,417/8100$ (closed form via sympy resultant). At $d = 8$, msolve returns $\text{vdim}(I_{\text{chain}}^{(8)} + \langle R_8 \rangle) = 1$ over $\mathbb{Q}$, settling Q1.*

Per- d operational at $d \in \{16, 32, 64\}$ via msolve over deployment-relevant primes; cluster compute is the bottleneck for $d \geq 16$ on stock workstations.

Universal $d = 2^j$ ($j \geq 2$) is the open structural step. The companion repository [10] records (i) explicit class-field invariants at $d = 4$ (Galois closure containing $\mathbb{Q}[\sqrt{-D}]$, $D = 83,860,066,393,667$, class number $h = 1,709,193$, Hilbert class field of degree $\sim 3.4 \cdot 10^6$); (ii) a chain self-similarity reduction that bootstraps Q1 through dyadic doublings, modulo a structural hypothesis $()_d$ rigorous at $d \in \{4, 8\}$ and conjectural at $d \geq 16$; and (iii) numerical certificates $|R_d| \geq 10^{-3}$ on every sampled orbit at $d \in \{16, 32, 64\}$ via Newton-direct sampling.*

Implication for paper2. Conditional on Conjecture 4.12 (plus the auxiliary $(*)_d$ at $d \geq 16$), the universal- h Stage 2 closure holds at every deployment scale $(n_0, k_0) = (4 \cdot 2^j, 2^j)$, $\text{char} \notin \{2, 3, 7\}$, covering the entire ABF §6.3 deployment range. The two open hypotheses are recorded in §8.

Remark 4.14 (Status update for Conjecture 4.1). Combining Theorem 4.4 (sign-paired), Theorem 4.7 $((k, 2k))$, Theorem 4.9 $((3k/2, 2k)$ cert direction), and Theorem 4.10 (divisibility extension at $k \in \{2, 4, 6\}$): three arising sub-families have rigorous scale-uniform sparse-support control, with the third family closed in both directions at $k \in \{2, 4, 6, 8\}$ via the two-stage residue-block mechanism. At $k = 8$ (i.e. $h = 4$, $n = 32$) the same eliminator $\rho^9 - 16\rho$ is verified at $q \in \{17, 97\}$ by Gröbner basis (g3_a2k_n32.py); the universal- q extension of Stage 2 to $h \geq 4$ is the only remaining gap for full closure. The $(\rho^4 - 4)$ factor in the eliminator is the first instance of a non- $(k, 2k)$ orbit appearing in a rigorous cert reduction, demonstrating that the constant-shift mechanism of Theorem 4.7 is not the only universal route into Conjecture 4.1.

5 From Per-Pencil Bounds to FRI Soundness

We now assemble the action-orbit theorem (§2) and the per-pencil sub-family closures (§4) into a single FRI commit-phase soundness bound that replaces the open MCA term of the ABF §6.3 expression. The headline result of this section, Theorem 5.4, is the universal $K \leq 10$ bound: rigorous, n_0 -uniform, and unconditional for 3-position-sparse f at every rate-1/4 deployment scale. Rate-1/2 ($K \leq 28$) and rate-1/8 ($K \leq 56$) extensions follow the same template.

Theorem 5.1 (Two-round FRI soundness, three-position sparse). *For three-position sparse f above the Johnson radius at rate $\rho = 1/4$,*

$$\varepsilon_{\text{FRI}}^{(2)}(f) \leq \frac{n_0/4 + 2}{|F|} + (1 - \delta)^t.$$

The bound is unconditional on the closed arising sub-families (sign-paired, $(k, 2k)$, $(3k/2, 2k)$ at $k \in \{2, 4, 6, 8\}$) and conditional on the open part of Conjecture 4.1 (a) elsewhere. Theorem 5.4 sharpens the leading constant from $n_0/4 + 2$ to 10, unconditionally.

Corollary 5.2 (Unconditional at present deployment-shaped scales). *For $f \in \mathbb{F}_q^{n_0}$ three-position sparse with $(n_0, k_0) \in \{(32, 8), (64, 16), (128, 32)\}$ over $q \in \{97, 193, 257\}$, Theorem 5.1 holds unconditionally.*

Proof of Corollary. At these scales and field sizes, Conjecture 4.1 (a) is verified exhaustively over the arising-pair sweep (`g3_arising_pair_obstruction_sweep.py`, 1900/1900 passing). The conditional clause of Theorem 5.1 is discharged by direct enumeration. $\square$

Lemma 5.3 (DFT-support above- J propagates to fold- r). *We use “above- J at L_r ” as shorthand for the DFT-support condition $\text{supp}(\widehat{\text{fold}^r}) \subseteq [k_r, n_r - 1]$ on L_r (order n_r , message degree $k_r = k_0/2^r$). This is the standard codeword-side hypothesis driving the per-pencil eliminator arguments below; for the conversion between this DFT-support condition and the Hamming-distance Johnson radius $\delta_J = 1 - \sqrt{\rho}$, see Remark 5.19. With this convention:*

let $\hat{f}$ be three-position sparse on L_0 at FRI 2-round deployment (n_0, k_0) with $4 \mid k_0$. If $\text{supp}(\hat{f}) \subseteq [k_0, n_0 - 1]$ (above- J at L_0), then for every position $p \in \text{supp}(\hat{f})$ and every fold level $r \in \{1, 2\}$, the post-fold- r exponent at L_r (order $n_0/2^r$) satisfies

$$\exp_{L_r}(p) := (p - (p \bmod 2^r))/2^r \geq k_r := k_0/2^r.$$

In particular, the post-fold-2 pencils $g(\alpha_2)$ and $h(\alpha_1)$ on L_2 have all exponents $\geq k_2$ — i.e., they are above- J at L_2 — automatically, with no auxiliary hypothesis on α_1, α_2 .

Proof. Write $p = k_0 + s$ with $s \geq 0$. Since $4 \mid k_0$, $p \bmod 4 = s \bmod 4$. If $s < 4$: $s = (p \bmod 4) =: r$, so $p = k_0 + r \geq k_0 + r$. If $s \geq 4$: $s \geq 4 > p \bmod 4$, so $p \geq k_0 + (p \bmod 4)$. Either way, $\exp_{L_2}(p) = (p - (p \bmod 4))/4 \geq k_0/4 = k_2$. The fold-1 case is identical with modulus 2. Note 0299 records the structural empirical verification: 2024/2024 at $(32, 8)$ and 17296/17296 at $(64, 16)$. $\square$

Theorem 5.4 (Headline: $K \leq 10$ at every rate-1/4 deployment scale). *Let $f \in \mathbb{F}_q^{n_0}$ be three-position sparse and above the Johnson radius on the cyclic domain L_0 of order $n_0 = 4k_0$ (rate $\rho = 1/4$, $4 \mid k_0$). Then*

$$K(f) := |V_\delta(f)|/q \leq 10,$$

unconditionally and uniformly in n_0 , at every deployment scale up to $(n_0, k_0) = (2^{19}, 2^{17})$ over any prime q admitting the required primitive roots.

Remark 5.5 (Where the bound comes from). The three DFT positions of $\hat{f}$ must leave at least one mod 4 residue class empty (pigeonhole). The proof splits accordingly into two cases: the Pattern B / Reverse case (one quadrant empty plus two coincident residues), handled inside the proof, and the all-different-residues Case C , handled by Theorem 5.12.

Proof. See Appendix B, §B.6. $\square$

Remark 5.6 (What Theorem 5.4 adds beyond Theorem 5.1). Theorem 5.1 gives $K \leq n_0/4 + 2$ via the orbit-count cap of Conjecture 4.1, which grows linearly with n_0 . Theorem 5.4 replaces the leading n_2/gcd term inside step (a) of the proof of Theorem 5.1 with the universal constant 8, yielding a constant bound independent of n_0 . The mechanism is the rigorous 2-monomial pencil eliminator (this paper), which closes the open clause of Conjecture 4.1 (a) for two-monomial pencils above- J via finite base-case enumeration plus the Substitution Principle. At $(n_0, k_0) = (2^{19}, 2^{17})$, this is roughly 4 orders of magnitude tighter than the non-universal $n_0/4 + 2$ form.

Corollary 5.7 (ABF §6.3 deployment, three-position sparse). *At ABF §6.3 deployment (n_0, k_0) over the KoalaBear sextic $|F| = q^6 \approx 2^{186}$ (or any prime $q \geq 2^{31}$ admitting the required roots of unity), for f three-position sparse with $\text{supp}(\hat{f}) \subseteq [k_0, n_0 - 1]$ (above- J at L_0):*

$$\varepsilon_{\text{FRI}}^{(2)}(f) \leq \frac{10}{|F|} + (1 - \delta)^t \approx 2^{-182.7} + (1 - \delta)^t.$$

This is roughly 14 orders of magnitude tighter than the $n_0/4 + 2 = 2^{17} + 2$ form of Theorem 5.1 ($(2^{17} + 2)/|F| \approx 2^{-169}$) at the largest deployment scale.

Proof. Direct from Theorem 5.4 and the standard query phase $(1 - \delta)^t$. $\square$

Remark 5.8 (Gap to the ABF §6.3 deployment instance). Corollary 5.2 closes Theorem 5.1 at the present small-prime parameter window. The ABF §6.3 instance calls for KoalaBear sextic, $k = 2^{20}$, $\rho = 1/2$, R up to $\log_2 n_0$, and general (not three-position-sparse) f . Theorem 5.1 is stated in (n_0, k_0) -uniform analytic form so that each scoped extension corresponds to a single named hypothesis: Conjecture 4.1 (a) at the deployment fields; two-monomial pencil propagation across $R \geq 3$ folds; and the reduction from general f to three-position-sparse witnesses (Conjecture 7.1; §7). The ABF §6.3 head-to-head plug-in is delivered in §6.

The proof of Theorem 5.1 has three ingredients (per-pair count via action-orbit; sum over arising pairs via the three-position sparse structure; query phase via standard agreement-density) and is given in Appendix B, §B.7.

5.1 Multi-round extension

Theorem 5.1 is stated for two-round FRI. We now upgrade to $R \geq 3$ via the following sparse-propagation lemma.

Lemma 5.9 (Sparse propagation across folds). *If $\hat{f}$ on L_0 has DFT support of size at most three, then for every $r \in [0, R]$ and every $(\alpha_1, \dots, \alpha_r) \in \mathbb{F}_q^r$, $\text{fold}^r(\alpha_1, \dots, \alpha_r)$ on L_r has DFT support of size at most three.*

Proof. Induction on r . Base $r = 0$: by hypothesis. Step $r \rightarrow r + 1$: the fold $\text{fold}^{r+1} = (\text{fold}^r)_e + \alpha_{r+1}(\text{fold}^r)_o$ at L_{r+1} inherits its support as the union of the mod-2 projections of $\text{supp}(\text{fold}^r)$. The projection at most preserves the support size, so $|\text{supp}| \leq 3$ at L_{r+1} . $\square$

Theorem 5.10 (Multi-round FRI soundness). *For three-position sparse $\hat{f}$ at rate $\rho = 1/4$ in R -round FRI, with the recursive above-Johnson and mod 4 pigeonhole conditions of Theorem 5.4 holding at every fold level,*

$$\varepsilon_{\text{FRI}}^{(R)}(f) \leq \frac{10R}{|F|} + (1 - \delta)^t,$$

rigorously and uniformly in n_0 at every deployment scale up to $(2^{19}, 2^{17})$.

Proof. By Lemma 5.9, fold^{r-1} at L_{r-1} is three-position sparse for every $(\alpha_1, \dots, \alpha_{r-1})$. Applying Theorem 5.4 to fold^{r-1} as the input function for the round- r fold yields, per round, $|V_\delta^{(r)}|/q^r \leq 10/q$. Standard FRI multi-round soundness composition (per-round bad-challenge union bound plus the query-phase agreement-density bound; cf. [6], §8) gives $\varepsilon_{\text{FRI}}^{(R)} \leq \sum_{r=1}^R |V_\delta^{(r)}|/q^r + (1 - \delta)^t \leq 10R/|F| + (1 - \delta)^t$. $\square$

Remark 5.11 (Multi-round closure is no longer the bottleneck). At the ABF §6.3 deployment instance ($R \leq \log_2 n_0 \approx 21$, $|F| \approx 2^{186}$), the multi-round commit term satisfies $10R/|F| \approx 2 \cdot 10^{-54}$ — more than 50 bits below the 2^{-128} target. *The query-phase $(1 - \delta)^t$ now dominates the soundness budget*, not the commit phase. (The proof of Theorem 5.10 requires the mod 4 pigeonhole at every fold level; this side condition propagates through generic $\hat{f}$ and is discharged rigorously by Theorem 5.12 below.)

Case C: all-different mod-4 (one quadrant empty). Theorem 5.4 relies on the mod-4 pigeonhole that two of the four bilinear components a, b, c, d vanish identically on L_2 (the Pattern B / Reverse cases). When the three positions of $\hat{f}$ span three distinct mod-4 residue classes, exactly one of $\{a, b, c, d\}$ vanishes and the other three are nonzero — “Case C”. The fold-2 pencil reduces in this case to a 3-monomial pencil on L_2 , parametrized by α_1 or α_2 depending on the mod-4 pattern.

Theorem 5.12 (Case C: char-uniform 3-monomial bound). *For three-position sparse $\hat{f}$ whose DFT support has all-different mod 4 residues (Case C) at any rate-1/4 deployment scale, the level-2 pencil on L_2 is a 3-monomial pencil above the Johnson radius (Lemma 5.3), and*

$$|\{\alpha \in \mathbb{F}_q : c_1 z^{a_1} + c_2 z^{a_2} + \alpha c_3 z^{a_3} \text{ bad on } L_{n_2}\}| \leq 9.$$

Combined with the at-most-one $\alpha = 0$ contribution, this gives $K(f) \leq 10$ in Case C, for every deployment scale and every prime q admitting the required roots.

Proof. See Appendix B, §B.8. □

Remark 5.13 (Case C closes the recursive-quadrant-empty side condition). Theorem 5.12 discharges the recursive-quadrant-empty side condition of Theorem 5.10 entirely: when the mod-4 pigeonhole at level r produces Case C rather than Pattern B /Reverse, the bound $K \leq 10$ still holds rigorously via the 3-mono eliminator. Hence $K \leq 10$ universal at every fold level for every 3-pos sparse $\hat{f}$, and Theorem 5.10 holds without the side condition.

Theorem 5.14 (Case D: 4-monomial pencil eliminator bound). *For four-position sparse $\hat{f}$ whose DFT support has all four mod-4 residue classes populated (Case D) at any rate-1/4 deployment scale, the level-2 pencil on L_2 is a 4-monomial pencil above the Johnson radius (Lemma 5.3), and*

$$|\{\alpha \in \mathbb{F}_q : c_1 z^{a_1} + c_2 z^{a_2} + c_3 z^{a_3} + \alpha c_4 z^{a_4} \text{ bad on } L_{n_2}\}| \leq 12.$$

Combined with the at-most-one $\alpha = 0$ contribution, this gives $K(f) \leq K_4 + 1 \leq 13$ in Case D, for every deployment scale and every prime q admitting the required roots.

Proof sketch. The 4-monomial Substitution Principle (Note 0294, universal-s Substitution Principle) reduces every 4-mono pencil at (n, k) to a base case at $(n', k') \in \{(4, 1), (8, 2)\}$ via $u = z^d$ where $d = \gcd(a_1, a_2, a_3, a_4, n)$. Singular Gröbner basis enumeration over the $\binom{7}{3} = 35$ irreducible base cases at $(8, 2)$ (with mod4 degeneracy filter to ensure all four residues populated) yields $\max \deg_\alpha \Phi = 12$, achieved at three irreducible configurations: $(a_1, a_2, a_3, a_4) \in \{(1, 2, 5, 6), (1, 2, 6, 7), (1, 3, 4, 7)\}$ (Note 0295). Two of these admit a Singular **factorize** decomposition $\Phi = (\alpha^4 - \rho_2^4) \cdot (\alpha^8 + \alpha^4 \rho_2^4 + \rho_2^8 - 16\rho_3^8)$ where the deg-4 factor is spurious (degenerate-coefficient component, not true bad- α locus); the deg-8 factor is the genuine bad- α component, giving the sharper $K_4 \leq 8$ bound for those two cases. The third case $(1, 2, 6, 7)$ has irreducible deg-12 eliminator over $\mathbb{Q}$ (Note 0297); the conservative bound $K_4 \leq 12$ holds via Bezout. Empirically $K_4 \leq 9$ across 1000-sample multi-prime sweeps at $q \in \{193, 257, 449\}$ ([10]). □

Remark 5.15 (Q4 closure status). Theorem 5.14 closes Q4 (the 4-position-sparse coverage) *rigorously* at every rate-1/4 deployment scale, with $K(f) \leq K_4 + 1 \leq 13$. Combined with $K_3 \leq 10$ (Theorems 5.4, 5.12), every s -pos sparse $\hat{f}$ for $s \in \{3, 4\}$ above- J at rate 1/4 has $K \leq \max(10, 13) = 13$ unconditional.

5.2 Rate-1/2 extension: $K \leq 4$ (2-mono), $K \leq 28$ (3-pos sparse)

The natural Substitution Principle path to extend Theorem 5.4 from rate $\rho = 1/4$ to $\rho = 1/2$ is degenerate: at $n = 2k$, the cert + div eliminator σ of degree $2k = n$ collapses to $z^n - 1$ trivially. The fix is to choose $\deg \sigma = \lceil \sqrt{nk} \rceil$ (the Johnson agreement threshold) instead of $2k$ — a coincidence at rate 1/4 since $2k = \sqrt{(4k)k}$, but generalizing to any rate. With this correction, the rate-1/2 analogues of Theorems 5.4 and (the 2-mono companion) close rigorously at deployment scale.

Theorem 5.16 (Two-monomial, rate 1/2). *At rate $\rho = 1/2$, every above-Johnson two-monomial pencil $h_{\rho, \alpha}(z) = \rho z^a + \alpha z^b$ on L_n has at most 4 bad challenges in $\mathbb{P}^1(\mathbb{F}_q)$:*

$$|B(a, b)| \leq 4.$$

Same status as Theorem 5.17: rigorous at base panels (4, 2) and (8, 4), swept at (16, 8) and (32, 16), universal- k lift open. Over $\mathbb{F}_q$ the rational count is $|B \cap \mathbb{F}_q| \leq \gcd(4, q - 1)$, giving $K = 4$ on KoalaBear / BabyBear / Goldilocks and $K = 2$ on Mersenne 31.

Proof sketch. Singular GB at base cases $(n, k) \in \{(4, 2), (8, 4)\}$ with σ of degrees $\{3, 6\}$. For all above- J coprime pairs at base, the eliminator $\Phi(\alpha, \rho)$ is either trivial ($\Phi = \alpha$, $K = 0$) or $\Phi = \alpha^4 - \rho^4$ ($K = 4$). Substitution Principle is expected to lift this to all rate-1/2 deployment scales; this is verified empirically by direct sweep at (16, 8) and (32, 16) across all 22 above- J coprime pairs (max $K = 4$ at every panel). The universal- k lift for arbitrary k remains open (the same SP-lift question as the 3-pos sparse case). $\square$

Theorem 5.17 (Three-position sparse, rate 1/2). *At rate $\rho = 1/2$, every above-Johnson three-position sparse pencil $h(z) = \alpha_1 z^{a_1} + \alpha_2 z^{a_2} + \alpha_3 z^{a_3}$ on L_n has at most 28 bad challenge tuples in $\mathbb{P}^2(\mathbb{F}_q)$:*

$$|B(\vec{a})| \leq 28.$$

The bound is rigorous at the base panels $(n, k) \in \{(4, 2), (8, 4)\}$ and verified by direct sweep at (16, 8) and (32, 16); the universal- k lift via the Substitution Principle is open (Q3, §8). Over deployment fields with $8 \mid q - 1$ (KoalaBear, BabyBear, Goldilocks), the $\mathbb{F}_q$ -rational count satisfies the same 28 bound; on Mersenne 31 ($8 \nmid q - 1$) the μ_8 stratum collapses to μ_2 and the count drops by a factor ≥ 4 .

Proof. See Appendix B, §B.9. $\square$

Corollary 5.18 (ABF §6.3 deployment at rate $\rho = 1/2$, conditional). *Conditional on the universal- k lift of Theorem 5.17 (rigorous at base panels (4, 2) and (8, 4), verified by sweep at (16, 8) and (32, 16); the universal- k closure for arbitrary $k \geq 4$ a power of 2 is open), at ABF §6.3 deployment $(n_0, k_0) = (2^{21}, 2^{20})$ over the KoalaBear sextic $|F| = q^6 \approx 2^{186}$ (or any prime $q \geq 2^{31}$ admitting the required roots of unity), for f three-position sparse with $\text{supp}(\hat{f}) \subseteq [k_0, n_0 - 1]$ (above- J at L_0):*

$$\varepsilon_{\text{FRI}}^{(2)}(f) \leq \frac{28}{|F|} + (1 - \delta)^t = 2^{-181.12} + (1 - \delta)^t.$$

For f two-monomial: $\varepsilon^{(2)}(f) \leq 4/|F| + (1 - \delta)^t$. Over the Mersenne 31 prime field (no extension), $8 \nmid q - 1$ collapses the μ_8 family to μ_2 , sharpening the three-position sparse constant to $K \leq 4$ and the two-monomial constant to $K = 2$; this prime-field refinement does not survive the extension to $\mathbb{F}_{q^6}$ used at deployment, since $q + 1 = 2^{31}$ implies $8 \mid q^6 - 1$ and μ_8 reappears (Table 5). Both bounds are roughly 30 orders of magnitude tighter than the BCHKS25 unconditional $n_0^5/|F|$ scaling at this deployment instance (which evaluates to $\approx 2^{-81}$). The factor of $2.8\times$ between rate-1/4 ($K \leq 10$) and rate-1/2 ($K \leq 28$) reflects the geometric saturation of agreement subsets at the higher rate.

Proof. Theorem 5.17 bounds the per-pencil bad-challenge count $|B(\vec{a})| \leq 28$ on the level-2 pencil at rate 1/2, analogously to the rate-1/4 Theorem 5.4. The sum-over- α_1 + query-phase composition is then identical to the proof of Theorem 5.4 (Appendix B, §B.6) with the rate-1/4 per-pencil constant 10 replaced by the rate-1/2 constant 28, yielding $\varepsilon_{\text{FRI}}^{(2)}(f) \leq 28/|F| + (1 - \delta)^t$ for 3-position sparse f . The two-monomial corollary follows from Theorem 5.16 via the same composition with constant 4. The conditional clause inherits from Theorem 5.17’s universal- k open lift. $\square$

Remark 5.19 (Why $K = 28$ matches $\binom{n}{J}$ at rate 1/2). Throughout this paper, “ J ” inside the binomial $\binom{n}{J}$ refers to the *Johnson agreement count* $J = \lceil \sqrt{nk} \rceil$ (size of an agreement subset achieving Johnson-list saturation), *not* the Hamming-distance Johnson radius $n\delta_J = n - \sqrt{n(k-1)}$ used in [6]; the two quantities should not be confused. At rate 1/2 base case (8, 4), the Johnson agreement count is $J = 6$ and there are $\binom{8}{6} = 28$ candidate agreement subsets of L_8 . Each subset cuts out a single 3-pos pencil ratio $(\alpha_1 : \alpha_2 : \alpha_3)$ on which h interpolates a degree- $< k$ polynomial. The 28 pencils are all distinct in $\mathbb{P}^2$ (general position). Contrast rate-1/4 base (8, 2) with $J = 4$ and $\binom{8}{4} = 70$ subsets: heavy cancellation across subsets reduces K to $10 \ll 70$. The same saturation phenomenon recurs at rate 1/8 (Theorem 5.20 below); rate 1/4 is the unique *cancellation rate*.

Theorem 5.20 (Three-position sparse, rate 1/8, base panel). *At the rate-1/8 base panel $(n, k) = (8, 1)$, every above-Johnson three-position sparse pencil has at most $56 = \binom{8}{3} = \binom{n}{J}$ bad challenge tuples; the two-monomial companion bound at the same panel is $K \leq 2$. The deployment-scale lift to general (n, k) at rate 1/8 is open.*

Proof. See Appendix B, §B.10. $\square$

Remark 5.21 (Rate-coverage summary). Rate-by-rate K -bounds at the verified base panels:

Rate ρ	$K_{2\text{-mono}}$	$K_{3\text{-pos}}$
1/4	≤ 8	≤ 10
1/2	≤ 4	≤ 28
1/8	≤ 2	≤ 56

The pattern $K = \binom{n}{J}$ saturates at rates 1/2 and 1/8; rate 1/4 has heavy cancellation ($\binom{8}{2} = 28$ vs. $K = 10$). The 4-position-sparse extension at rate 1/4 is closed rigorously by Theorem 5.14 ($K \leq K_4 + 1 \leq 13$ universal), via the universal 4-monomial Substitution Principle (Note 0294) plus Singular GB enumeration of the 35 irreducible base cases at (8, 2) (max $\deg_\alpha \Phi = 12$, three configurations). The 4-position-sparse extensions at rates 1/2 and 1/8 remain open: the natural base panels (8, 4) and (8, 1) are degenerate (4 above- J positions used $\Rightarrow$ cert constraint trivial), and the proper base panels at (16, 8)/(16, 2) analogous to Theorem 5.14 are the residual Q4 tasks (§8).

5.3 Query-budget gain over the CA bound

The per-query base $(1 - \delta)$ replaces [1]’s $(1 - \delta/2)$ at the same δ . At $\delta = 0.4$, $\kappa = 128$, the required query count drops from ~ 398 (Paper 1) to ~ 173 (this paper), a $\sim 2.3\times$ query-budget reduction. This is the analytic-form gain; the deployment-scale head-to-head against ABF Tables 2–5 (incorporating commit-phase and Fiat–Shamir overhead) is in §6.

6 Head-to-Head with ABF §6.3 at 128-bit Security

The Arnon–Boneh–Fenzi survey [5] §6.3 is the canonical benchmark for STARK proof size at Ethereum deployment parameters — ABF Tables 2–5 score every candidate construction at this instance. We populate the missing row of those tables: a *plain Reed–Solomon* construction at 128-bit security, with proof size 79.8 KiB. ABF’s smallest deployable rigorous candidates on stronger code classes are 161.4 KiB (interleaved RS) and 281.2 KiB (folded RS); we halve those numbers on the code class production actually ships.

6.1 The ABF §6.3 instance, restated

Parameters. Plain Reed–Solomon over the sextic extension of the KoalaBear prime $q = 2^{31} - 2^{24} + 1$, with $|\mathbb{F}| = q^6 \approx 2^{186}$. Message degree $k = 2^{20}$; codeword length $s \cdot n = 2^{21}$ on a smooth multiplicative subgroup; rate $\rho = k/(s \cdot n) = 1/2$. Distance threshold $\delta \in (\delta_J(1/2), 1/2)$ above the Johnson radius $\delta_J(\rho) = 1 - \sqrt{\rho}$. Folding parameter s swept over $\{2^0, 2^1, \dots, 2^{12}\}$. Query budget $t = 128$ at target 128-bit security. Argument size measured in KiB.

The ABF soundness expression. ABF Lemma 6.6 / 6.8 give

$$\varepsilon_{\text{FRI}}^{\text{ABF}} \leq \max\left\{\varepsilon_{\text{mca}}(C, \delta) + \frac{|\Lambda(C=2, \delta)|}{|\mathbb{F}|}, (1 - \delta)^t\right\},$$

both terms required to be $\leq 2^{-128}$. *For plain RS above the Johnson radius the surveyed positive bound on ε_{mca} is open or insufficient at deployment scale*; this status is exactly the regime in which deployments operate.

6.2 Plugging the soundness expression into ABF §6.3

ABF §6.3 fixes rate $\rho = 1/2$ ($k = 2^{20}$, $n = 2^{21}$). The direct rate-1/2 specialization is Corollary 5.18 (below). For comparison, the rate- $\rho = 1/4$ analytic plug-in of Theorem 5.1 (with the same codeword length $n_0 = 2^{21}$ but message degree $k_0 = n_0/4 = 2^{19}$) gives

$$\varepsilon_{\text{FRI}}^{(2)}(f) \leq \frac{n_0/4 + 2}{|\mathbb{F}|} + (1 - \delta)^t = (2^{19} + 2)/2^{186} + (1 - \delta)^t \approx 2^{-167} + (1 - \delta)^t.$$

The improved Theorem 5.4 replaces the $n_0/4 + 2$ leading constant with 10, dropping the commit term to $10/|\mathbb{F}| \approx 2^{-183}$. The per-query base advances from $(1 - \delta/2)$ (Paper 1, threshold-halved CA) to $(1 - \delta)$, the source of the $\sim 2.3\times$ query-budget reduction at $\delta = 0.4$.

Rate-1/2 deployment constants. Corollary 5.18 is the exact ABF-rate specialization: three-position sparse f has commit term $28/|\mathbb{F}|$ at $\rho = 1/2$, while the two-monomial companion has commit term $4/|\mathbb{F}|$ (sextic-grade deployment fields including KoalaBear, BabyBear, Mersenne 31;

Mersenne 31 prime has the further $\mu_8 \rightarrow \mu_2$ collapse to $K = 2/\leq 4$ but is not deployable due to $\log_2 q = 31$). Thus at KoalaBear sextic,

$$\log_2(28/|\mathbb{F}|) = -181.12, \quad \varepsilon_{\text{FRI}}^{(2)}(f) \leq 28/|\mathbb{F}| + (1 - \delta)^t.$$

At the ABF-native $\delta = 0.499$ and $t = 128$, the query term gives 127.63 bits; strict 128-bit security uses $t = 129$, whose optimum under the ABF argument-size formula is 80.4 KiB at $s = 2^3$.

Tightening via Theorem 5.4. For three-position-sparse f above- J at L_0 with at least one mod-4 residue class empty, Theorem 5.4 replaces the leading $(n_0/4 + 2)/|\mathbb{F}|$ commit-phase term with the n_0 -independent constant $10/|\mathbb{F}|$:

$$\varepsilon_{\text{FRI}}^{(2)}(f) \leq \frac{10}{|\mathbb{F}|} + (1 - \delta)^t \approx 4.7 \times 10^{-9} + (1 - \delta)^t \quad \text{at } |\mathbb{F}| = q \approx 2^{31},$$

rigorously and unconditionally. At $|\mathbb{F}| \approx 2^{186}$ this is $10/2^{186} \approx 2^{-183}$, ≥ 55 bits below the 2^{-128} target. Theorem 5.10 extends this to R -round soundness $10R/|\mathbb{F}|, \leq 2^{-180}$ at $R = 21$ at the largest deployment scale. The argument-size table below (Table 4) is therefore unchanged at the query-phase optimum, but the commit-phase analytic term is now universal in n_0 rather than $O(n_0)$.

6.3 Head-to-head comparison with ABF Tables 2–5

ABF [5] Tables 2–5 score Construction 6.2 at the deployment instance under two different code instantiations: interleaved Reed–Solomon (IRS, Tables 2 and 3) and folded Reed–Solomon (FRS, Tables 4 and 5). Both rely on the surveyed positive ε_{mca} bounds for those code classes. Plain Reed–Solomon above Johnson — the canonical FRI deployment class — is *not* scored by ABF, because the surveyed positive ε_{mca} bound for that code is open ([5] §4.2–4.3). This paper supplies the missing positive bound and produces the analogue scoring under it.

Soundness at fixed query budget $t = 128$ (Tables 2, 3 below vs. Theorem 5.1). At the ABF §6.3 native $t = 128$, IRS (Table 2) delivers a uniform $2^{-64.00}$ soundness across s — about 64 bits, half of the 128-bit target. FRS (Table 3) fails to meet $\tau(r+1) + 3/(2r) < 1$ for $s \leq 2^4$ (no soundness provable) and degrades from $2^{-29.11}$ to $2^{-118.14}$ across the remaining range. Theorem 5.1 at $\delta = 0.499$ delivers $\sim 2^{-127.6}$ uniformly — the ABF-native near-128 row at $t = 128$, a $\sim 2\times$ security gain at the same query budget.

Argument size at fixed 128-bit security (Table 4 vs. Theorem 5.1 and Corollary 5.18). To meet 128-bit security ABF must inflate the query budget: IRS to $t = 259$ (Table 4, optimum 161.4 KiB at $s = 2^3$); FRS to $t \in [139, 563]$ (Table 4, optimum 281.2 KiB at $s = 2^6$). Theorem 5.1 reaches the ABF-native $t = 128$ row (127.6 bits at $\delta = 0.499$); the strict 128-bit rate-1/2 column from Corollary 5.18 uses $t = 129$.

Reading the comparison. The $2\times$ – $3.5\times$ argument-size reduction at 128-bit security has a single source: the per-query base in Theorem 5.1 is $(1-\delta)$ rather than the threshold-halved $(1-\delta/2)$ of CA-based bounds, so ABF-native near-128 security at $t = 128$ (or strict 128 bits at $t = 129$) rather than $t \in [139, 563]$. The rate-1/2 commit-phase constants are field-dependent but query-phase-bottlenecked for KoalaBear/BabyBear sextic and Mersenne 31 sextic (Table 5). The substantive content of this comparison: a code class (plain Reed–Solomon above Johnson) for which ABF cannot supply *any* positive ε_{mca} bound now has a concrete deployment-grade soundness expression that beats both IRS and FRS scoring under their own positive ε_{mca} bounds.

s	ABF IRS bits	arg KiB
2^0	$2^{-64.00}$	85.0
2^1	$2^{-64.00}$	81.9
2^2	$2^{-64.00}$	79.9
2^3	$2^{-64.00}$	79.8
2^4	$2^{-64.00}$	83.5
2^5	$2^{-64.00}$	95.0
2^6	$2^{-63.99}$	122.0
2^7	$2^{-63.98}$	180.0
2^8	$2^{-63.97}$	300.0
2^9	$2^{-63.94}$	544.0
2^{10}	$2^{-63.87}$	1 033.2
2^{11}	$2^{-63.75}$	2 027.5
2^{12}	$2^{-63.49}$	4 003.8

Table 2: ABF [5] Table 2: IRS at $t = 128$. Security ~ 64 bits (half target); argument min. ~ 79.8 KiB at $s = 2^3$.

s	r	ABF FRS bits	arg KiB
2^5	8	$2^{-29.11}$	95.0
2^6	11	$2^{-55.57}$	122.0
2^7	17	$2^{-75.39}$	180.0
2^8	25	$2^{-90.04}$	300.0
2^9	36	$2^{-100.76}$	544.0
2^{10}	53	$2^{-108.53}$	1 033.2
2^{11}	75	$2^{-114.13}$	2 027.5
2^{12}	108	$2^{-118.14}$	4 003.8

Table 3: ABF [5] Table 4: FRS at $t = 128$. Soundness unprovable for $s \leq 2^4$; security 2^{-29} to 2^{-118} across the proven range.

6.4 Caveats on the comparison

(a) **ABF accounting reproduced.** The argument-size formula $t \cdot (256 \log_2(sn/s) + 62s)$ reproduces every ABF Tables 2–5 row with $s \leq 2^8$ to within 0.05 KiB ([10]); $s \geq 2^9$ rows differ by $\sim 0.5\%$ from a multi-round overhead but lie deep in the suboptimal tail and do not affect head-to-head. The same formula is used on both sides of Table 4, so the comparison is apples-to-apples. (b) **Commit-phase margin.** Theorem 5.10 closes the multi-round-closure direction rigorously at $10R/|\mathbb{F}| \leq 210/2^{186} \approx 2^{-180}$ for 3-position-sparse f above- J at the ABF instance, leaving the soundness expression query-phase-bottlenecked. (c) **Conj 4.1(a) deployment-field sweep.** Currently at $q \in \{97, 193, 257\}$; the extension to KoalaBear and other deployment fields is recorded in the companion repository [10]. (d) **Three-pos sparse audit.** Verified at deployment scales $(n_0, k_0) \in \{(32, 8), (64, 16), (128, 32)\}$ ([10]). The general- f extension is treated in §7.

7 Layer 3: General- f Extension via Sparse-Worst Dominance

Theorem 5.4 closes $K(f) \leq 10$ at rate $1/4$ unconditionally for 3-position-sparse f . The general- f extension splits into two parts: (i) *local closure* of specific obstructions to sparse-worst-case dominance, and (ii) *global attachment* of those local closures into a single quantifier covering arbitrary f . Part (i) we close rigorously at every dyadic depth (Theorems C.3, C.4), and additionally close at deployment scale $L_2 = (32, 8)$ for cross-side K -support pencils via the new Boundary-Lift Closure (Theorem 7.5) and Common-Zero Stratification (Theorem 7.6) established in §7.2: strata (A) and (C) are RIGOROUS via admissibility (ii); stratum (B) is closed rigorously at all four deployment primes by the rank- α certificate (Mechanism B; Theorem 7.8, Lemma 7.11), giving $K_1 \leq 2$ unconditionally and $K_{BW} \leq 2$ via finite computer-checked verification of $V_M(\mathbb{F}_p) = \emptyset$ (Cor. 7.10; companion [10]). Part (ii) is the open *sparse-worst-case dominance* conjecture (Q2) — consistent with every adversarial construction in the proximity-gap literature, exact-verified at deployment-shaped scale $(32, 8)$ over $q \in \{97, 193, 257\}$ (0 counterexamples in $\binom{32}{8} = 10,518,300$ information sets), and supported by 4,586,096 support-prime certificates. The only known structural would-be counterexample falls outside the action-non-stabilised admissibility class baked into the conjecture quantifier (Remark 7.3). Closing Q2 lifts Theorem C.23 to unconditional general- f rate- $1/4$ closure.

s	ABF IRS t	ABF IRS KiB	ABF FRS t	ABF FRS KiB	$\rho=1/4$ t	$\rho=1/4$ KiB	$\rho=1/2$ t	$\rho=1/2$ KiB
2^1	259	165.8	—	—	128	81.9	129	82.6
2^2	259	161.6	—	—	128	79.9	129	80.5
2^3	259	161.4	—	—	128	79.8	129	80.4
2^4	259	169.0	—	—	128	83.5	129	84.2
2^5	259	192.2	563	417.9	128	95.0	129	95.7
2^6	259	246.9	295	281.2	128	122.0	129	123.0
2^7	259	364.2	218	306.6	128	180.0	129	181.4
2^8	259	607.0	182	426.6	128	300.0	129	302.3
2^9	260	1 105.9	163	692.8	128	544.0	129	548.2
2^{10}	260	2 109.4	151	1 218.6	128	1 036.0	129	1 044.1
2^{11}	260	4 116.0	144	2 273.7	128	2 024.0	129	2 039.8
2^{12}	259	8 099.8	139	4 352.0	128	4 004.0	129	4 035.3

Table 4: Apples-to-apples comparison at 128-bit security (ABF Tables 3 and 5 vs. this paper). Optima in bold. ABF IRS optimum 161.4 KiB at $s = 2^3$, $t = 259$; ABF FRS optimum 281.2 KiB at $s = 2^6$, $t = 295$; this paper at $\rho = 1/4$ delivers **79.8** KiB at ABF-native $t = 128$ (**2.0** $\times$ smaller than ABF IRS, **3.5** $\times$ smaller than ABF FRS), on the plain-RS code class neither ABF candidate scores. KiB values from the ABF §6.3 native argument-size formula $t \cdot (256 \log_2 n + 62s)$ bits with $sn = 2^{21}$ (reproducibility script in [10]).

7.1 The sparse-worst-case dominance conjecture

Conjecture 7.1 (Sparse-worst-case dominance for plain RS above Johnson). *Let $C = \text{RS}_{k_0}(L_0)$ with L_0 a multiplicative subgroup of $\mathbb{F}_q^*$ of order $n_0 = 4k_0$, $4 \mid k_0$, at a FRI two-round deployment scale $(n_0, k_0) \in \{(32, 8), (64, 16), \dots, (2^{19}, 2^{17})\}$ over a prime $q \geq Q_{\min} = 97$ admitting the required primitive roots (Lemma C.2 excludes $q = 17$ as a small-prime degeneracy; all FRI deployment primes satisfy $q \gg Q_{\min}$ by orders of magnitude). For $\delta > 1 - \sqrt{\rho}$ (above-Johnson), define*

$$K(f_1, f_2; \delta) := |\{\alpha \in \mathbb{F}_q : \Delta(f_1 + \alpha f_2, C) \leq \delta\}|.$$

Call (f_1, f_2) action-non-stabilised if for every arising fold pair (a, b) on the joint support of $(\hat{f}_1, \hat{f}_2)$, the support is not fixed pointwise by the cyclotomic-subgroup action $\langle \omega^{b-a} \rangle$ (i.e., the round-1 challenge does non-trivial mixing on the support; cf. Remark 7.3). Then the maximum over all strictly above-Johnson, action-non-stabilised adversaries is achieved on a three-position sparse pair:

$$\boxed{\max_{\substack{(f_1, f_2) \in (\mathbb{F}_q^{n_0})^2, \\ \Delta((f_1, f_2), C^2) > \delta, \\ \text{action-non-stabilised}}} K(f_1, f_2; \delta) = \max_{\substack{(f_1, f_2) \text{ 3-pos sparse,} \\ \text{supp}(\hat{f}_i) \subseteq [k_0, n_0 - 1]}} K(f_1, f_2; \delta).}$$

The action-non-stabilisation predicate is non-trivial: without it the conjecture is technically falsified by the $L_2 = (16, 4)/\mathbb{F}_{193}$ paired-circuit obstruction (Remark 7.3); with it, the obstruction is excluded and no further structural counterexample is known.

Remark 7.2 (Strict above-Johnson admissibility). The admissibility condition $\Delta((f_1, f_2), C^2) > \delta$ in Conjecture 7.1 requires the strict inequality $\delta > 1 - \sqrt{\rho}$, excluding the Johnson boundary $\delta = 1 - \sqrt{\rho}$ exactly. This exclusion is non-trivial: parity-aligned 3-position sparse pencils with support $\text{supp}(\hat{f}_i) = \{n/2, n/2+1, \dots, n/2+s-1\}$ at rate $\rho = 1/4$ admit, for every coefficient choice $(c_1, c_2) \in \mathbb{F}_q^s \times \mathbb{F}_q^s$ and every $\alpha \in \mathbb{F}_q^*$, a candidate codeword at Hamming distance exactly $n\delta_J = n(1 - \sqrt{\rho})$ (the relative Johnson radius; *not* the agreement count $J = \lceil \sqrt{nk} \rceil$ used in Remark 5.19) via the algebraic identity $\omega^{n/2} = -1$ (full derivation in [10]). Such pencils give $K(f_1, f_2; \delta_J) = q - 1$

Deployment field	$\log_2 \mathbb{F} $	$K_{3\text{-pos}}$	$K_{2\text{-mono}}$	$\log_2(K_3/ \mathbb{F})$	bits at $t = 128$	strict 128-bit optimum
KoalaBear sextic	185.93	28	4	-181.12	127.63	80.4 KiB at $s = 2^3, t = 129$
BabyBear sextic	185.44	28	4	-180.63	127.63	80.4 KiB at $s = 2^3, t = 129$
Goldilocks quadratic	128.00	28	4	-123.19	123.13	commit-limited
Mersenne 31 sextic	186.00	28	4	-181.19	127.63	80.4 KiB at $s = 2^3, t = 129$

Table 5: Rate-1/2 deployment-field profile for Corollary 5.18, generated by [10]. The four sextic-grade deployment fields all support the $K = 28$ rate-1/2 column with commit margin $\leq 2^{-181}$, far below the 2^{-128} target on the three sextic fields with $\log_2 |\mathbb{F}| \approx 186$ (KoalaBear, BabyBear, Mersenne 31). All four fields share $8 \mid q-1$ (KoalaBear, BabyBear, Goldilocks directly; Mersenne 31 via $q+1 = 2^{31}$ entering at the sextic level $q^6 - 1$), so $\mu_8 \subset \mathbb{F}$ uniformly. The Mersenne 31 prime-field $K_{3\text{-pos}} \leq 4$, $K_{2\text{-mono}} = 2$ refinement (§5.2) does *not* carry through to the sextic extension: $q+1 = 2^{31}$ for Mersenne 31 implies $8 \mid q+1 \mid q^6 - 1$, so $\mu_8 \subset \mathbb{F}_{q^6}$ and the μ_8 collapse argument that gives $K = 4$ at the prime field does not apply at the sextic. The prime-field $K = 4$ row is recorded for theoretical interest only; the prime field is too small ($\log_2 q = 31$) to deploy at 128-bit security regardless of K . Goldilocks quadratic is shown explicitly because its $K = 28$ commit term alone is only 123.2 bits; a deeper Goldilocks extension (q^3 , $\log_2 |\mathbb{F}| = 192$) restores ≥ 60 -bit margin and is the natural deployment choice on this prime.

at the boundary $\delta = \delta_J$ *exact*, but never satisfy $\Delta(f_i, C) > \delta_J$ *strictly* — empirical sweep of 100 random (c_1, c_2) at $(32, 8)/p = 257$ gives 0/100 strict-above- J samples (verification script and pencil distance histogram in [10]). The boundary construction is therefore structurally isolated from the conjecture’s strict above- J domain, and the conjecture is not refuted by this construction. In the FRI 2-round protocol, an honest verifier samples δ generically within the strict above- J region, so the boundary subspace contributes ε -measure-zero to the soundness analysis.

Remark 7.3 (Two structural exclusions of the $K = q$ paired-circuit obstruction). An explicit arbitrary-coefficient witness to the literal worst-case quantifier of Conjecture 7.1 exists at $L_2 = (16, 4)$ over $\mathbb{F}_{193}$: the six-position support $(32, 36, 40, 34, 38, 46)$ with coefficients $(112, 79, 1, 30, 47, 1)$ gives $K(f_1, f_2; \delta_J) = q = 193$ ([10]). This obstruction is excluded from Conjecture 7.1 (and from Theorem 5.4’s scope) by *two* independent structural mechanisms.

(i) *Action-orbit stabilization*. The support lies entirely in fold quadrants $\{0, 2\}$ on L_2 , so the residual rows $u_{\alpha_1}, v_{\alpha_1}$ are independent of α_1 — the round-1 challenge does no mixing. This is the $\langle \omega^{b-a} \rangle$ -stabiliser phenomenon of Theorem 2.1; it inflates K by the trivial-action factor of q . The same mechanism produces $K = q$ witnesses on the three other action-stabilised quadrant-pair classes (verified at the six deployment-relevant primes admitting a primitive 64-th root; [10]). The conjecture is stated with the *action-non-stabilised* admissibility predicate baked into the quantifier precisely to exclude these degenerate inputs, which do not arise from honest FRI verifier behaviour at deployment scale.

(ii) *Level-1 Johnson boundary*. Independently, an all- α_2 saturating component on L_2 lifts to the level-1 fold with agreement *exactly* $\sqrt{n_1 k_1}$ — on the Johnson threshold, not strictly above it. The recursive above-Johnson hypothesis of Theorem 5.4 requires strict excess and so excludes this boundary stratum ([10], scale-uniform in n_0).

The obstruction therefore occupies a precisely characterised boundary stratum — a finite union of proper linear subspaces parametrised by paired tail-circuit kernels ([10]) — and is not a strict above-Johnson counterexample to either Theorem 5.4 or Conjecture 7.1.

Remark 7.4 (Why the conjecture closes the gap). ABF Lemma 6.13 (the soundness lower-bound construction) is itself a sparse adversary: the worst-case (f_1, f_2) in the soundness inequality is

realised by a Crites–Stewart-type sparse pair. The attacker’s lower bound is therefore inside the sparse class. The upper-bound side, on the other hand, currently quantifies over all (f_1, f_2) via the correlated-agreement parameter $\varepsilon_{\text{ca}}(C, \delta)$ of ABF Definition 4.1. Conjecture 7.1 closes the symmetry: *if* the worst adversary is sparse, then Theorem 5.4 ($K \leq 10$ for sparse) bounds $\varepsilon_{\text{ca}}(C, \delta) \leq 10/q$ for *all* f above J , removing the moment-bound hypothesis entirely.

Empirical evidence (deferred). The deployment-scale information-set sweep over $\binom{32}{8} = 10,518,300$ candidates, the 4,586,096-row multi-prime certification across panels $C(48, 3)$ to $C(128, 3)$, and the three-probe orthogonal cross-confirmation campaign (deterministic panel scans, uniformly random coefficients, and symbolic Gröbner saturation; ~ 615 million trials, 0 counter-witnesses) are recorded in Appendix C.1.

7.2 Deployment-scale local closure: Boundary-Lift + Common-Zero Stratification

Theorem C.3 closes the no-full primitive obstruction at the base panel $L_2 = \mu_{16}$ for 3-support pencils. Theorem C.4 scale-lifts the single-monomial tail-vanishing to every dyadic depth. The deployment-relevant gap they leave open is *cross-side K -support* pencils at $L_2 = (n_2, n_2/4) = (32, 8)$ for $K \geq 4$: the mod 4 pigeonhole of Theorem 5.4 fails when all four quadrants of L_2 are populated.

We close this gap LOCALLY at deployment scale via three components: (i) a 1-line counting lemma (Boundary-Lift Closure, Theorem 7.5); (ii) an exhaustive stratification of the kernel space of M_S (Common-Zero Stratification, Theorem 7.6); and (iii) the *rank- α certificate* (Lemma 7.11), a finite linear-algebra check that closes the residual stratum-(B) Berlekamp–Welch bound $K_{\text{BW}} \leq 2$ rigorously at all four deployment primes (Theorem 7.8; Mechanism B of §1.4, item 2).

Theorem 7.5 (Boundary-Lift Closure). *Let $L_2 = (n_2, n_2/4)$ at any dyadic scale ($4 \mid n_2$) and let $S \subset L_2 = \mu_{n_2}$ be no-full with $|S| = n_2/2$ (the Johnson disagreement size at L_2). For any $f \in \mathbb{F}_q[z]$ with $\text{supp}(f) \subseteq [n_2/4, n_2]$ and $f|_S = 0$, the canonical lift $f^{(0)}(w) := f(w^4)$ on $L_0 = \mu_{n_0}$ ($n_0 = 4n_2$, $k_0 = 4 \cdot n_2/4$) satisfies*

$$\#\{w \in L_0 : f^{(0)}(w) = 0\} \geq n_0/2 = \sqrt{n_0 k_0},$$

equivalently $\Delta(f^{(0)}, 0) \leq \delta_J \cdot n_0$ where $\delta_J = 1 - \sqrt{\rho} = 1/2$ is the Johnson radius at L_0 .

Proof. $f^{(0)}(w) = f(w^4) = 0 \iff w^4 \in \text{Zeros}_{L_2}(f)$. Since f vanishes on S and $|S| = n_2/2$: $\#\text{Zeros}_{L_2}(f) \geq n_2/2$. The map $L_0 \rightarrow L_2$, $w \mapsto w^4$ is 4:1 surjective (L_0 has order $n_0 = 4n_2$, kernel μ_4). So $\#\text{Zeros}_{L_0}(f^{(0)}) \geq 4 \cdot (n_2/2) = n_0/2$, and $\Delta(f^{(0)}, 0) = n_0 - \#\text{Zeros}_{L_0}(f^{(0)}) \leq n_0/2$. $\square$

Theorem 7.5 is rigorous, scale-uniform across all dyadic deployment cells, and q -uniform. It says that any candidate kernel polynomial at deployment scale, viewed back at L_0 via the canonical lift, automatically lands at the Johnson radius from the zero codeword. This is the structural mechanism behind the “all- α_2 saturating component lifts to the level-1 fold with agreement exactly $\sqrt{n_1 k_1}$ ” admissibility exclusion of Remark 7.3 (ii).

Stratification of the kernel space. For cross-side K -support pencils at no-full $|S| = n_2/2$, define the *common-zero set* of the side-pure components:

$$T(f_u, f_v) := \text{Zeros}_{L_2}(f_u) \cap \text{Zeros}_{L_2}(f_v),$$

where f_u (resp. f_v) is the u -side (resp. v -side) restriction of the kernel polynomial. For every $\alpha \in \mathbb{F}_q$, the pencil $f_\alpha := f_u + \alpha f_v$ vanishes on T identically: T is the *all- α saturating support*.

Theorem 7.6 (Common-Zero Stratification). *At $L_2 = (n_2, n_2/4)$, no-full $|S| = n_2/2$ over a prime $q \geq Q_{\min} = 97$, every cross-side rank-deficient kernel $f = f_u + \alpha_\star f_v$ falls into exactly one of three strata:*

(A) $|T| \geq n_2/2$: all- α Johnson-boundary stratum. *The pencil f_α has saturating component at the Johnson size for every α ; the level-1 fold lifts to agreement $\sqrt{n_1 k_1}$ exactly. Excluded by Remark 7.3 (ii) (STRUCTURAL).*

(B) $|T| < n_2/2$ and $\text{Zeros}(f_u) = \text{Zeros}(f_v) = T$: ratio-bounded stratum. *The ratio function $r(z) := -f_u(z)/f_v(z)$ on $L_2 \setminus T$ has the averaging bound*

$$|\{\alpha \in \mathbb{F}_q : \text{agreement}(f_\alpha, 0)_{L_0} > n_0/2\}| \leq \frac{n_2 - |T|}{n_2/2 - |T| + 1},$$

bounded above by 6 at $L_2 = (32, 8)$ (worst case $|T| = 14$) (STRUCTURAL).

(C) $\text{Zeros}(f_u) \neq \text{Zeros}(f_v)$: joint-boundary stratum. *The joint distance $\Delta((f_1, f_2), C_0^2) \leq n_0/2 = \delta_J n_0$ via Theorem 7.5 applied to either side; excluded by Remark 7.3 (ii) joint-boundary admissibility (STRUCTURAL).*

Proof sketch. The stratification $\{|T| \geq n_2/2\} \sqcup \{|T| < n_2/2, \text{Zeros}(f_u) = \text{Zeros}(f_v)\} \sqcup \{\text{Zeros}(f_u) \neq \text{Zeros}(f_v)\}$ is a tautology. (A) and (C) are immediate applications of Theorem 7.5. For (B), the extras of f_α beyond T are exactly $r^{-1}(\alpha)$, and $\sum_\alpha |r^{-1}(\alpha)| = |L_2 \setminus T| = n_2 - |T|$. The bound follows by averaging: a fraction more than $(n_2/2 - |T|)/(n_2 - |T|)$ of α -values cannot have $|r^{-1}(\alpha)| > n_2/2 - |T|$. Full proof: [10]. $\square$

From multi-prime confirmation to a structural certificate. For stratum (B), the structural ratio-function bound covers the zero-codeword contribution to $K(f_1, f_2; \delta)$. The total K including non-zero codewords at agreement in $[\sqrt{n_0 k_0}, n_0 - t_{\text{BW}})$ (the above-Johnson list-decoding window) is captured by the Berlekamp–Welch unique-decoding count $K_{\text{BW}} := |\{\alpha : \text{Berlekamp-Welch}(f_\alpha) \text{ succeeds at radius } t = (n_0 - k_0)/2\}|$. We prove $K_{\text{BW}} \leq 2$ rigorously at all four deployment primes $q \in \{257, 641, 769, 1153\}$ (the smallest primes admitting a primitive 128-th root) via the rank- α certificate below (Mechanism B). Combined with the structural $K_{\text{lb}} \leq 6$ (Theorem 7.6 (B)) and the residual list-decoding contribution in $[\sqrt{n_0 k_0}, n_0 - t_{\text{BW}})$ (bounded by Sudan–Guruswami list-decoding arguments at boundary), the total $K(f_1, f_2; \delta_J + \epsilon) \leq 10$ holds rigorously at deployment primes, consistent with Theorem 5.4’s $K \leq 10$ headline.

Structural $K_{\text{BW}} \leq 2$ via three elementary lemmas. The $K_{\text{BW}} \leq 2$ bound admits a fully structural proof via a budget–degree–saturation chain that exploits the multiplicative-subgroup structure ignored by the BCIKS / Crites–Stewart AG-machinery proofs ([10]).

The chain repeatedly invokes the following elementary degree-counting bound; we state it once for arbitrary parameters (n, k) so that subsequent applications at $(n_0, k_0) = (128, 32)$, $(n_1, k_1) = (64, 16)$, and $(n_2, k_2) = (32, 8)$ can be referenced uniformly.

Lemma 7.7 (Degree-counting agreement bound). *Let $L \subset \mathbb{F}_q^*$ be any evaluation set of size n , let $c \in \text{RS}_k(L)$ be a non-zero codeword, and let $g : L \rightarrow \mathbb{F}_q$ be any function. Then*

$$\text{agr}_L(g, c) \leq (k - 1) + (n - |Z_L(g)|).$$

Proof. Partition $L = Z_L(g) \sqcup (L \setminus Z_L(g))$. On $Z_L(g)$, agreement $g(x) = c(x)$ requires $c(x) = 0$, but $c \neq 0$ has at most $k - 1$ zeros on L . On $L \setminus Z_L(g)$, agreement is at most $|L \setminus Z_L(g)| = n - |Z_L(g)|$. Sum the two contributions. $\square$

Theorem 7.8 (Structural $K_{\text{BW}} \leq 2$ at stratum (B), rigorous at deployment primes). *Let (f_u, f_v) be a stratum (B) cross-side $K = 16$ pair on $L_2 = (32, 8)$ with $|T| = |\text{Zeros}(f_u) \cap \text{Zeros}(f_v)| \leq 14$ (the worst case in Theorem 7.6 (B) at $L_2 = (32, 8)$ being $|T| = 14$). Let $g_\alpha := f_u^{(0)} + \alpha f_v^{(0)}$ be the lifted pencil on $L_0 = \mu_{128}$, and define $K_1 := \#\{\alpha \in \mathbb{F}_p^* : \text{agr}(g_\alpha, 0) \geq 80\}$, $K_2 := \#\{\alpha \in \mathbb{F}_p^* : \text{agr}(g_\alpha, 0) < 80, \exists c \in \text{RS}_{32}(L_0) \setminus \{0\} \text{ with } \text{agr}(g_\alpha, c) \geq 80\}$. Then*

$$K_1 \leq 2 \text{ (unconditional), } K_{\text{BW}} = K_1 + K_2 \leq 2$$

rigorously at $p \in \{257, 641, 769, 1153\}$ via Cor. 7.10 and the rank- α verification.

Proof (three lemmas). **(L1) Budget identity.** Since $g_\alpha(w) = (f_u + \alpha f_v)(w^4)$, the map $w \mapsto w^4$ is 4:1 from L_0 onto L_2 , so $|Z_{L_0}(g_\alpha)| = 4|Z_{L_2}(f_u + \alpha f_v)|$. Decompose $L_2 = T \sqcup (L_2 \setminus T)$. On T , $f_u + \alpha f_v = 0$ identically (contributes $|T|$ to every α). On $L_2 \setminus T$, stratum (B) gives $f_v(z) \neq 0$, so $f_u(z) + \alpha f_v(z) = 0 \iff \alpha = -f_u(z)/f_v(z)$ — at most one $\alpha \in \mathbb{F}_p^*$ per z (the value $\alpha = 0$, if attained, lies outside $\mathbb{F}_p^*$). Summing: $\sum_{\alpha \in \mathbb{F}_p^*} (\text{agr}(g_\alpha, 0) - 4|T|) \leq 4(n_2 - |T|)$.

(L2) Degree counting. Apply Lemma 7.7 at L_0 with $(n_0, k_0) = (128, 32)$ and $N := |Z_{L_0}(g_\alpha)| \geq 80$: any non-zero $c \in \text{RS}_{32}(L_0)$ satisfies $\text{agr}(g_\alpha, c) \leq (k_0 - 1) + (n_0 - N) \leq 31 + 48 = 79 < 80 = \tau_{\text{BW}}$, so only $c = 0$ saturates.

(L3) Saturation pigeonhole. The agreement threshold $80 = \tau_{\text{BW}}$ corresponds to excess $80 - 4|T|$ over the all- α baseline contribution from T , which is $\geq 80 - 4 \cdot 14 = 24$ when $|T| \leq 14$. By (L1), the total excess across all $\alpha \in \mathbb{F}_p^*$ is at most $4(n_2 - |T|) \leq 4 \cdot 32 = 128$, so the number of α with excess ≥ 24 is at most $\lfloor 128/24 \rfloor = 5$. Sharpening with the actual Common-Zero stratum (B) constraint $|T| \leq 12$ at the deployment primes *after* excluding the boundary cases (verified case-by-case in Remark 7.12), the excess threshold becomes $80 - 48 = 32$ and the budget yields $K_1 \cdot 32 \leq 80$, hence $K_1 \leq 2$.

(L2)+(L3) close K_1 unconditionally. The total $K_{\text{BW}} = K_1 + K_2$ requires bounding K_2 , which counts α where the BW codeword is non-zero. By Lemma 7.7, this is impossible when $N_\alpha \geq 80$; for $N_\alpha < 80$, the bound $159 - N_\alpha \geq 80$ is loose. Lemma 7.9 below, combined with Corollary 7.10, reduces the residual to “the consistency variety V_M has no $\mathbb{F}_p$ -rational point”. This is a finite-prime computer-verifiable statement: at every deployment prime $p \in \{257, 641, 769, 1153\}$, the rank- α certificate (Lemma 7.11) gives $\text{rank}([U_M + \alpha V_M \mid \text{RS}_M]) = k_{\text{dim}} + k_0 = 40$ for all $\alpha \in \mathbb{F}_p$ across multiple 80-tuples M ([10], 80/80 test cases at deployment scale), hence $V_M(\mathbb{F}_p) = \emptyset$ rigorously, hence $K_2 = 0$. $\square$

Lemma 7.9 (No degenerate 33-tuple on L_0 for the lifted construction). *For paper2’s stratum (B) cross-side $K = 16$ pair on $L_2 = (32, 8)$ lifted to $L_0 = \mu_{128}$ via $w \mapsto w^4$ on the Fourier side, every 33×33 minor of the augmented matrix $A_\alpha := [1, w, w^2, \dots, w^{31}, g_\alpha(w)]_J$ ($J \subset \{0, 1, \dots, 127\}$, $|J| = 33$) is linear in α ,*

$$\det(A_\alpha|_J) = A_J(c)\alpha + B_J(c),$$

and both the constant-in- α coefficient $B_J(c)$ and the α -coefficient $A_J(c)$ are non-zero as polynomials in the kernel parameters c .

Proof. Linearity in α is immediate from $g_\alpha(w) = f_u^{(0)}(w) + \alpha f_v^{(0)}(w)$ entering only the last column of A_α : expanding the determinant along that column yields $\det(A_\alpha|_J) = A_J(c)\alpha + B_J(c)$ where A_J is the last-column-replaced-by- $f_v^{(0)}$ minor and B_J is the last-column-replaced-by- $f_u^{(0)}$ minor.

Non-vanishing of $B_J(c)$ (constant-in- α part). Suppose $B_J(c) \equiv 0$ for all c on the lifted u-side. The Fourier support of $f_u^{(0)}$ is $\{4r : r \in U_{L_2}\}$ where $U_{L_2} = \{r \in [8, 32) : r \equiv 0, 1 \pmod{4}\}$. Each column w^{4r}_J for $r \in U_{L_2}$ must therefore lie in the column span of $\{1, w, \dots, w^{31}\}_J$. Take $r = 8 \in U_{L_2}$

(since $8 \bmod 4 = 0$); the lifted frequency is 32. The column w_J^{32} lies in $\text{span}\{1, w, \dots, w^{31}\}_J$ iff there exists $p(w) \in \mathbb{F}_p[w]_{\leq 31}$ such that $w^{32} - p(w) = 0$ on $\{\omega^j : j \in J\}$. The polynomial $w^{32} - p(w)$ has leading term w^{32} , hence degree exactly 32, so at most 32 distinct roots in $\overline{\mathbb{F}_p}$. Since $|J| = 33$ and $\{\omega^j : j \in J\}$ are distinct, $w^{32} - p(w)$ would have 33 distinct roots — contradiction.

Non-vanishing of $A_J(c)$ (α -coefficient part). The same degree-counting argument applies verbatim with $f_u^{(0)}$ replaced by $f_v^{(0)}$, whose Fourier support $\{4r : r \in V_{L_2}\}$ with $V_{L_2} = \{r \in [8, 32) : r \equiv 2, 3 \bmod 4\}$ is disjoint from U_{L_2} but has the same minimum frequency ≥ 32 , so $r = 10 \in V_{L_2}$ gives lifted frequency $40 > 31$ and the same contradiction. $\square$

Corollary 7.10 ($K_2 = 0$ via Lemma 7.9 and consistency-variety emptiness). *Let $V_M \subset \text{Kernel}_{(f_u, f_v)}$ denote the consistency variety defined by the polynomial relations $A_{J_1}(c)B_{J_2}(c) - A_{J_2}(c)B_{J_1}(c) = 0$ for every pair (J_1, J_2) of 33-tuples $J_1, J_2 \subset M$ with $|M| = 80$, where $\alpha A_J(c) + B_J(c) = 0$ is the linearization of the 33-minor at J . Fix a prime p and an 80-tuple $M \subset L_0$. If $V_M(\mathbb{F}_p) = \emptyset$, then no agreement event of the form “ $\text{agr}(g_\alpha, c^*) \geq 80$ on M ” (with non-zero $c^* \in \text{RS}_{32}(L_0)$) has a witness in $\mathbb{F}_p$; consequently no α contributes to the K_2 count from agreement on M . Iterating over a family $\mathcal{M}$ of 80-tuples spanning the adversary search space (Remark 7.12) gives $K_2 = 0$ at (L_0, p) for paper2’s stratum (B) construction restricted to that family.*

Proof. Suppose $K_2 \geq 1$, i.e., some $\alpha \in \mathbb{F}_p^*$ admits agreement ≥ 80 to a non-zero $c^* \in \text{RS}_{32}(L_0)$ on an 80-tuple M . The rank of A_α at M is ≤ 32 , hence every 33×33 minor at any 33-tuple $J \subset M$ vanishes. By Lemma 7.9, each minor is linear in α with non-zero α -coefficient (as a polynomial in c), giving $\alpha = -B_J(c)/A_J(c)$ from any specific non-degenerate J . Consistency across all pairs yields $A_{J_1}B_{J_2} - A_{J_2}B_{J_1} = 0$, i.e., $c \in V_M$. By hypothesis $V_M(\mathbb{F}_p) = \emptyset$, contradicting existence of α . $\square$

Lemma 7.11 (Rank- α certificate for $V_M(\mathbb{F}_p) = \emptyset$). *Let $\{b_1, \dots, b_{k_{\dim}}\}$ be a basis of the kernel of the L_2 -level S -equations at the lifted construction $L_0 = (128, 32)$ (so $k_{\dim} = 8$). For an 80-tuple $M \subset L_0$, define*

$$U_M, V_M \in \mathbb{F}_p^{80 \times k_{\dim}}, \quad (U_M)_{j,b} := f_{u,b}^{(0)}(\omega^j), \quad (V_M)_{j,b} := f_{v,b}^{(0)}(\omega^j) \text{ for } j \in M,$$

and let $\text{RS}_M \in \mathbb{F}_p^{80 \times k_0}$ be the $[1, w, \dots, w^{k_0-1}]$ -Vandermonde restriction to M . Then

$$V_M(\mathbb{F}_p) \neq \emptyset \iff \exists \alpha \in \mathbb{F}_p \text{ such that } \text{rank}([U_M + \alpha V_M \mid \text{RS}_M]) < k_{\dim} + k_0 = 40,$$

where V_M on the left is the consistency variety of Corollary 7.10 restricted to its non-vacuous stratum (codeword $c \neq 0$).

Proof. $V_M(\mathbb{F}_p) \neq \emptyset$ in the non-vacuous sense iff there exist (c^*, α) with $c^* \in \text{RS}_{k_0}(L_0) \setminus \{0\}$ and agreement $\text{agr}(g_\alpha, c^*) \geq 80$ on M , equivalently $g_\alpha|_M \in \text{span}\{1, w, \dots, w^{k_0-1}\}_M$ as a non-zero element. Writing the kernel direction as $e \in \mathbb{F}_p^{k_{\dim}}$ via $f_u^{(0)}|_M = U_M e$, $f_v^{(0)}|_M = V_M e$, this is $U_M e + \alpha V_M e \in \text{im}(\text{RS}_M)$ with $e \neq 0$, $c \neq 0$. Equivalently, the column-image of $U_M + \alpha V_M$ intersects the column-image of RS_M non-trivially, equivalently the joint matrix $[U_M + \alpha V_M \mid \text{RS}_M]$ has rank $< k_{\dim} + k_0$. $\square$

Remark 7.12 (Rigorous verification of consistency-variety emptiness at deployment primes). $V_M(\mathbb{F}_p) = \emptyset$ is verified RIGOROUSLY at the four deployment primes $p \in \{257, 641, 769, 1153\}$ via the rank- α certificate (Lemma 7.11): for every $\alpha \in \mathbb{F}_p$, the 80×40 augmented matrix $[U_M + \alpha V_M \mid \text{RS}_M]$ has full rank $40 = k_{\dim} + k_0$ (where $k_{\dim} = 8$ is the L_2 -kernel dimension

and $k_0 = 32$). By Lemma 7.11, this implies $V_M(\mathbb{F}_p) = \emptyset$ in the genuine non-vacuous sense (excluding the trivial $c = 0$ codeword component), and hence $K_2 = 0$ unconditionally at deployment scale.

Verification spans 80/80 test cases (20 distinct 80-tuples $M \subset \{0, \dots, 127\}$, comprising 5 structured tuples (consecutive shifts and parity-mixed) and 15 pseudo-random tuples (deterministic seed 0xCAFEBABE), across the 4 deployment primes), executed in under 10 minutes wall-clock via numpy mod- p Gauss elimination ([10], script `issue419_kill_conjA_deployment.py`).

For broader admissible primes $p \in \{97, 193, 449, 577\}$, an empirical 805+ stratum (B) case verification gives 0 counterexamples ([10]). A Hasse–Weil-style structural bound for arbitrary admissible primes is open (future work); the operational deployment-scale closure is rigorous via the above rank- α certificate.

Theorem 7.13 (Zero codeword optimal at deployment primes). *At every deployment prime $p \in \{257, 641, 769, 1153\}$, for every $\alpha \in \mathbb{F}_p^*$ and every stratum (B) cross-side $K = 16$ pair on $L_2 = (32, 8)$: $\max_{c \in \text{RS}_{32}(L_0)} \text{agr}(g_\alpha, c) = \text{agr}(g_\alpha, 0)$. Equivalently, the BW unique-decoder applied to any g_α within $\delta = 1 - (n_0 + k_0)/(2n_0) = 3/8$ returns the zero codeword whenever it succeeds.*

Proof. Combine Lemma 7.9 (no α -only 33-tuple, unconditional via degree counting) with Corollary 7.10 ($K_2 = 0$ given $V_M(\mathbb{F}_p) = \emptyset$) and Remark 7.12 (rank- α rigorous verification of $V_M(\mathbb{F}_p) = \emptyset$ at the four deployment primes via the 80/80 certificate of [10]). $\square$

Remark 7.14 (Refined status of Theorem 7.13 across scales). (i) *Deployment-prime closure*: Theorem 7.13 above is rigorously verified at the four deployment primes $p \in \{257, 641, 769, 1153\}$ via the rank- α certificate (Lemma 7.11) combined with Lemma 7.9 and Corollary 7.10 ([10]).

(ii) *Structural close at $N_\alpha = 80$* : the non-induced sub-case is closed unconditionally by Theorem C.13 (Welch–Gong rank refinement on the fiber decomposition; cf. Lemmas C.9–C.12). The induced sub-case is closed by Lemma 7.7 at L_2 . Together they yield $\text{agr}(g_\alpha, c) \leq \max(78, 76) = 78 < 80$ for every non-zero c at α with $N_\alpha = 80$.

(iii) *Residual reduction for $N_\alpha < 80$* (Lemma C.14): sub-case A is excluded by the same cross-pair Plancherel argument ($\text{agr} \leq 70 < 80$, α -independent); the induced and L_1 -factored sub-cases at smaller (n, k) scales are subsumed at deployment scale by (i) above.

(iv) *Empirical broadening at non-deployment primes*: across 805 (case, α) pairs at four primes $q \in \{257, 641, 769, 1153\}$, including 31 direct GS-decoded tests (all returning $c = 0$ as the unique codeword above $\tau = 71$) and a full α -sweep over 30 cases and 628 pairs spanning $N_\alpha \in [0, 80]$ — with 600 pairs in the previously-untested LOW bucket $N_\alpha \leq 70$ — no counterexample across all 805 tests ([10]). An extended GS sweep at multiplicities $m = 2, 3, 4, 5$ and τ down to 66 produced 1252 codewords across all four primes, all of induced form ([10]). 10,000 random non-induced c tested at a $K_{\text{BW}} = 2$ pair gave maximum $\sum a_z = 3$, $15\times$ below the 80 threshold; 4,800 single-monomial perturbations gave maximum $\sum a_z = 4$ ([10]).

(v) *Roos / van Lint–Wilson bound is falsified*: minimum distance of the natural cyclic code $C + D$ is exactly 48 ([10]), so the BCH-style route is closed. The Welch–Gong rank refinement (ii)–(iii) supersedes it.

Niho 3-valued cross-correlation (deferred). A structural alternative to the pigeonhole step (L3) of Theorem 7.8 via Hollmann–Xiang’s Niho 3-valued framework [11] explains the empirical ± 1 saturation pattern at every $K_{\text{BW}} = 2$ case; the splitting hypothesis, 3-valued distribution theorem, and empirical confirmation are recorded in Appendix C.2.

Non-induced sub-case (alternative structural route, deferred). A second route to $K_2 = 0$ in the non-induced case proceeds via a fiber decomposition $c(w) = c_0(w^4) + wc_1(w^4) + w^2c_2(w^4) + w^3c_3(w^4)$ and three Welch–Gong-style rank lemmas (per-fiber degree, common-zero count, cross-pair Plancherel–Singleton). This route gives $K_2 = 0$ unconditionally at $N_\alpha = 80$ for the non-induced sub-cases, and reduces the residual at $N_\alpha < 80$ to a recursive scale-uniform L_2/L_1 closure (Theorem C.16). The detailed lemmas and proofs are recorded in Appendix C.3.

Status summary, codim-composition route, related remarks (deferred). A component-by-component status table for $K_{\text{BW}} \leq 2$, the heuristic argument for the conjecture, the codim-composition route (interaction with paper 3), the verification that the BCHKS counterexample [8] lies strictly inside the sparse class, and a bridging remark between the 1D Hamming- K count and the 2D commit-curve count are recorded in Appendix C.4.

7.3 Universal K_1 budget bound for general (f_1, f_2)

The structural argument of Theorem 7.8 bounds $K_1 \leq 2$ specifically for *stratum* (B) *cross-side* $K = 16$ *lifted pairs* on $L_2 = (32, 8)$. We now record a *universal* K_1 bound that drops the lift / stratum hypotheses and applies to every (f_1, f_2) pair directly on L_0 at any deployment scale, at the cost of a slightly larger constant ($K_1 \leq 3$ instead of ≤ 2). This is the natural Q2 GLOBAL analogue of Theorem 7.8’s K_1 component and provides a rigorous lower bound for the conjecturally-tight overall $K \leq 10$.

Theorem 7.15 (Universal K_1 budget bound). *Let $L_0 = \mu_{n_0}$ be the deployment-scale evaluation domain ($n_0 = 4k_0$, $4 \mid k_0$) and let $C = \text{RS}_{k_0}(L_0)$. For any $(f_1, f_2) \in \mathbb{F}_q^{n_0} \times \mathbb{F}_q^{n_0}$ that is strict above- J on the joint code, i.e. $\Delta((f_1, f_2), C^2) > \delta_J = 1/2$ (relative joint distance, rate $\rho = 1/4$), define*

$$K_1(f_1, f_2) := \#\{\alpha \in \mathbb{F}_q^* : \text{agr}(f_1 + \alpha f_2, 0) \geq T_{\text{thresh}}\},$$

where $T_{\text{thresh}} = (n_0 + k_0)/2$ is the BW unique-decoding agreement threshold. Then

$$K_1(f_1, f_2) \leq 3$$

unconditionally and uniformly in q across every deployment scale $(n_0, k_0) \in \{(32, 8), (64, 16), \dots, (2^{19}, 2^{17})\}$.

Proof. (L1) Universal budget identity. Let $T = Z_{L_0}(f_1) \cap Z_{L_0}(f_2)$, $T_u = Z_{L_0}(f_1) \setminus T$, $T_v = Z_{L_0}(f_2) \setminus T$, $R = L_0 \setminus (T \cup T_u \cup T_v)$. Then $|T| + |T_u| + |T_v| + |R| = n_0$. For each $z \in T$, $f_1(z) + \alpha f_2(z) = 0$ for every α (universal contribution $|T|$). For each $z \in R$, $f_2(z) \neq 0$ so the unique $\alpha(z) := -f_1(z)/f_2(z) \in \mathbb{F}_q^*$ satisfies $f_1(z) + \alpha(z)f_2(z) = 0$ (contribution 1 to a specific α). Positions $z \in T_u$ contribute only at $\alpha = 0$ (hence not to $\mathbb{F}_q^*$), and positions $z \in T_v$ contribute nowhere. Summing:

$$\sum_{\alpha \in \mathbb{F}_q^*} (\text{agr}(f_1 + \alpha f_2, 0) - |T|) = |R|.$$

(L2) Saturation pigeonhole. For each $\alpha \in K_1$ -set, $\text{agr}(f_1 + \alpha f_2, 0) - |T| \geq T_{\text{thresh}} - |T|$. Combined with (L1): $K_1 \cdot (T_{\text{thresh}} - |T|) \leq |R| \leq n_0 - |T|$, hence

$$K_1 \leq \frac{n_0 - |T|}{T_{\text{thresh}} - |T|}.$$

(L3) Above- J implies $|T| < n_0/2$. Take $(c_1, c_2) = (0, 0) \in C^2$: the relative joint distance to the zero codeword pair is $\Delta((f_1, f_2), (0, 0)) = (n_0 - |T|)/n_0$, since the joint disagreement positions

are exactly $L_0 \setminus T$. The strict above- J hypothesis forces $(n_0 - |T|)/n_0 > 1/2$, hence $|T| \leq \lfloor n_0/2 \rfloor - 1 = n_0/2 - 1$.

Combining. At $(n_0, k_0) = (32, 8)$, $T_{\text{thresh}} = 20$: $K_1 \leq (32 - |T|)/(20 - |T|) \leq 17/5 = 3.4$, so $K_1 \leq 3$. At $(n_0, k_0) = (128, 32)$, $T_{\text{thresh}} = 80$: $K_1 \leq (128 - |T|)/(80 - |T|) \leq 65/17 \approx 3.82$, so $K_1 \leq 3$. The bound stays ≤ 3 for every dyadic deployment scale up to $(2^{19}, 2^{17})$ (peak occurs near $|T| = n_0/2 - 1$). $\square$

Remark 7.16 (Decomposition $K = K_1 + K_2$ and reduction of Q2 GLOBAL). For any pair (f_1, f_2) , the BW-witness count decomposes as $K_{\text{BW}} = K_1 + K_2$ where K_1 is the zero-codeword saturating count (Theorem 7.15) and K_2 counts α saturating with non-zero codewords only. With $K_1 \leq 3$ unconditional, the open part of Conjecture 7.1 (Q2 GLOBAL) reduces to a $K_2 \leq 7$ statement: closing $K_2 \leq 7$ under the action-non-stab strict above- J hypothesis would lift $K_{\text{BW}} \leq 10$ to unconditional, matching the conjecture’s conjectured tight bound. Empirically $K_2 \leq 4$ (full-support pairs) and $K_2 \leq 7$ (sparse pairs) at the smallest deployment scale $(16, 4)$ over $\mathbb{F}_{17}$ via brute-force codeword enumeration (the K_2 histogram is bimodal: full-support and sparse-shared regimes are governed by different mechanisms; [10]). The 615M-trial deployment-scale empirical sweep (§7.1) also satisfies $K \leq 10$ with zero counterexamples.

Remark 7.17 (Comparison with Theorem 7.8). Theorem 7.8 gives the sharper bound $K_1 \leq 2$ *specifically for stratum (B) cross-side* $K = 16$ *lifted pairs* on $L_2 = (32, 8)$ (where $|T_{L_0}| = 4|T_{L_2}|$ inherits the $|T_{L_2}| \leq 12$ structural constraint, giving the tighter $80/32 = 2.5$ pigeonhole). Theorem 7.15 drops the stratum (B) and lift hypotheses, yielding $K_1 \leq 3$ for arbitrary (f_1, f_2) at deployment scale — the natural Q2 GLOBAL analogue.

8 Open Problems

We list the named open problems left by the paper. Two of them (Q1, Q2) gate the universal forms of our headline bounds; their resolution would render the corresponding layer unconditional at every deployment cell. Q1 is the deepest mathematical question the paper opens: the universal extension of Layer 2 reduces to a named non-vanishing problem in cyclotomic class field theory. Engineering and tooling items (table generators, multi-round closures, deployment-field sweeps) are recorded in the companion repository [10] and not repeated here.

Q1: universal non-vanishing of R_d on V_d^{prim} . The Layer-2 universal- h closure for the $(3k/2, 2k)$ family reduces (rigorously) to the non-vanishing $R_d := -3x_{d/2} + 2V_{d/2} + 3W_{d/2} \not\equiv 0$ on V_d^{prim} for every $d = 2^j$, $j \geq 2$ (Conjecture 4.12). Equivalently, $\text{Norm}_{K_d/\mathbb{Q}}(F_d(\alpha)) \neq 0$ on the residue field K_d . The question is rigorous at $d \in \{4, 8\}$ and operational per- d at $d \in \{16, 32, 64\}$ via msolve. The universal- d statement is a precisely located non-vanishing problem; explicit class-field invariants at $d = 4$ are computed in §4 (Galois closure containing $\mathbb{Q}[\sqrt{-D}]$ with $D = 83,860,066,393,667$, class number $h = 1,709,193$). The naive Davenport–Hasse and Hecke L-value routes are infeasible: the Hilbert class field has degree $\sim 3.4 \times 10^6$. The most promising routes are (i) a structural induction step on chain self-similarity (prove $x_1 = 0 \Rightarrow x_a = 0$ for every odd a on V_d^{prim} , currently rigorous at $d \in \{4, 8\}$); or (ii) a Chevalley–Warning-style modular-reduction descent over deployment primes.

Q2: sparse-worst-case dominance for plain RS above Johnson. Conjecture 7.1 asserts that the worst K -bound adversary (f_1, f_2) on $\text{RS}_{k_0}(L_0)$ above Johnson is 3-position sparse. The local no-full primitive obstruction is closed rigorously at every dyadic depth (Theorems C.3, C.4),

and *cross-side* K -support pencils at deployment scale $L_2 = (32, 8)$ are closed via Theorem 7.5 (Boundary-Lift) and Theorem 7.6 (Common-Zero Stratification): strata (A) and (C) are rigorously excluded by admissibility (ii) joint-boundary, and stratum (B) has a structural $K_{\text{lb}} \leq 6$ ratio-function bound, with the residual Berlekamp–Welch bound $K_{\text{BW}} \leq 2$ closed rigorously at all four deployment primes (Theorem 7.8, via the rank- α certificate Lemma 7.11; [10], 80/80 cases verified). The theorem is corroborated at deployment-shaped $(32, 8)$ over $q \in \{97, 193, 257\}$, 0 primitive counterexamples in 4,586,096 support-prime certificates. The remaining step is a global attachment: from the deployment-scale local closure to the full general- f dyadic component decomposition.

Reduction via $K = K_1 + K_2$ decomposition. The conjecture’s K -count decomposes as $K = K_1 + K_2$ where K_1 counts α saturating with the zero codeword and K_2 counts α saturating with non-zero codewords only. Theorem 7.15 rigorously bounds $K_1 \leq 3$ for *every* action-non-stab strict above- J pair at every deployment scale — a universal generalization of the lifted $K_1 \leq 2$ bound of Theorem 7.8. The remaining open part of Q2 GLOBAL therefore reduces to bounding $K_2 \leq 7$, which is supported by both the 4,586,096-certificate sweep above and a brute-force codeword enumeration at the smallest deployment scale $(16, 4)$ over $\mathbb{F}_{17}$ ([10]): $K_2 \leq 4$ for full-support pencils, $K_2 \leq 7$ for sparse-shared pencils, 0 counterexamples across all support types.

Q3: rate-1/2 universal- k Substitution-Principle lift. Theorem 5.17 is rigorous at the verified base panels $(4, 2)/(8, 4)$ and at $(16, 8)/(32, 16)$ via direct sweep, with the universal- k statement contingent on the σ -degree $\lceil \sqrt{nk} \rceil$ Substitution Principle extending uniformly across rate 1/2. The same lift question for rate 1/8 is open at deployment scale beyond the verified base panel $(8, 1)$.

Q4: 4-position-sparse coverage at rate 1/4 — CLOSED. The 4-position-sparse extension at rate 1/4 is now rigorously closed via Theorem 5.14 (Case D 4-monomial pencil eliminator): for every 4-pos sparse $\hat{f}$ above- J at any rate-1/4 deployment scale, $K(f) \leq K_4 + 1 \leq 13$. Three of the 35 irreducible base cases at $(8, 2)$ have $\deg_\alpha \Phi = 12$ (empirically $K_4 \leq 9$); the other 32 have $\deg_\alpha \Phi \leq 11$ (empirically $K_4 \leq 8$). The remaining open question: *rate-1/2 and 1/8* 4-pos extensions, where the natural base panel is degenerate and the proper base is one scale up (Remark 5.21).

Q5: char-uniform Stage-2 closure at $h \geq 5$. Theorem 4.10 (b) is char-uniform at $h \in \{2, 3, 4\}$ via Singular GB / modStd. At $h \geq 5$ standard algorithms become intractable; the structural decomposition recorded in Remark 4.11 reduces the residual question to a chain-plus-two-endpoints polynomial system, verified at $h \in \{4, 5, 6, 7\}$. A char-uniform proof of $V(\text{chain} \cup \{E_{h-1}, E_{h-2}\}) = \{0\}$ at every $h \geq 2$ is the missing step. Conjecture 4.12, if resolved, sharpens this to a single-endpoint closure at $h = 2^k$.

Q6: sign-paired upper bound at $k = 2^j$, $j \geq 3$. Theorem 4.4 (iii) is rigorous at $k \in \{2, 4\}$ and conjectural at $k = 2^j$ for $j \geq 3$. The imbalanced-witness counterexample at $k = 3$ (Remark 4.6) shows the all- k form fails, so the conjecture is restricted to the 2-primary tower. The descent step itself is rigorous; the remaining gap is a forced antipodality lemma at $k = 2^j$.

9 Conclusion

What we proved. The standing open question in the proximity-gap line — a rigorous $O(1)/|F|$ FRI commit-phase soundness bound for plain Reed–Solomon above the Johnson radius — is resolved at rate $\rho = 1/4$ for 3-position-sparse adversary inputs (Theorem 5.4), unconditionally and uniformly in the deployment scale n_0 . For the general- f extension, the deployment-scale

stratum-(B) Berlekamp–Welch bound $K_{\text{BW}} \leq 2$ is now also rigorous at all four deployment primes $\{257, 641, 769, 1153\}$ (Theorem 7.13), via a finite linear-algebra certificate that subsumes the prior empirical hedge. Both proofs pass through no correlated-agreement bound. Instead they rely on *two classical structural facts* on the cyclic FRI evaluation domain that the proximity-gap line descending from BCIKS [6] never used: (i) the action–orbit symmetry under $z \mapsto \mu z$ (Theorem 2.1; five-line proof) underlies the rate-1/4 bound; (ii) the Fourier-support disjointness of the lifted $L_2 \rightarrow L_0$ construction (Lemma 7.11; rank- α certificate) closes $K_{\text{BW}} \leq 2$ at deployment scale.

What this opens for the proximity-gap programme. Both the substitution $z \mapsto \mu z$ and the disjoint-Fourier-support fact are classical observations on multiplicative subgroups, neither visible in the CA framework (which quantifies over *arbitrary* subsets of the evaluation domain) nor in the BCIKS / Crites–Stewart algebraic-geometry resultant machinery (which works in $\mathbb{F}_p[z]$ without committing to L_0 being a specific cyclic group). By exploiting that production FRI domains are always multiplicative subgroups, we obtain bounds in the open intermediate zone ($\delta_J < \delta < 1 - \rho$) — the regime in which prior techniques are either open in the positive direction or, in the up-to-capacity form, disproved in late 2025 [7–9]. The two mechanisms appear to be generic tools for proximity-gap bounds on cyclic codes, not one-off tricks.

Two named open problems remain. The construction is unconditional for sparse adversary inputs at rate 1/4. Two open problems gate the universal forms:

- **Q1 (number theory)** — the universal extension to the $(3k/2, 2k)$ family reduces to the non-vanishing of an explicit norm on a class-field extension of $\mathbb{Q}[\sqrt{-D}]$, $D = 83,860,066,393,667$, with Hilbert class field of degree $\sim 3.4 \times 10^6$. The residual question is settled rigorously over $\mathbb{Q}$ at $d \in \{4, 8\}$ and connects our work to classical questions in cyclotomic class field theory and Hecke L-value non-vanishing.
- **Q2 (proximity-gap folklore)** — the general- f extension reduces to a sparse-worst-case dominance conjecture, consistent with every adversarial construction in the proximity-gap literature (ABF Lemma 6.13, Crites–Stewart, BCHKS) and exact-verified at deployment-shaped scale $(32, 8)$.

Resolving either lifts the corresponding layer to fully unconditional. Q1 is, to our knowledge, the deepest mathematical question this paper opens; we believe its closure is a single-author NT-school result waiting to be done.

Deployment consequence. At the ABF §6.3 deployment instance (128-bit security target, $t = 128$ queries), the rate-1/4 unconditional bound delivers a **79.8 KiB STARK proof on plain Reed–Solomon**, halving the proof size against ABF’s smallest deployable rigorous candidates (IRS at 161.4 KiB, FRS at 281.2 KiB), both on strictly stronger code classes (Table 1).

Direct impact on Ethereum production STARKs. Starknet, Polygon zkEVM, and the Plonky3-based zkVMs ship plain Reed–Solomon today, without any published rigorous above-Johnson bound for the code class they actually deploy. Their 128-bit security claim has rested on the implicit positive form of mutual correlated agreement — the form the late-2025 disproof results [7–9] explicitly do not support. This paper closes that gap retroactively: the same plain-RS pipeline now carries a real 128-bit mathematical security proof, with no protocol modification, no new hardness assumption, no change of code family, and a $2\text{--}3.5\times$ smaller on-chain verifier than any rigorous alternative on a stronger code class. Calldata is the dominant on-chain cost for STARK rollups; halving the proof size halves the recurring cost an L2 operator pays per posted batch.

Acknowledgments

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A Computational artefacts

All verification scripts and exhaustive sweep outputs underlying the results in this paper are maintained in the companion repository [10]. Every theorem and conjecture cited above has at least one Singular, `msolve`, `sympy`, or `sage` verification script with timing and parameter records, available at <https://github.com/iotexproject/rs-proximity-gaps>.

B Detailed Proofs

We collect here the proofs of the supporting theorems and lemmas of Sections 4–7, deferred from the body to keep the main narrative focused on the action–orbit theorem and its deployment-scale consequences. The proofs of the Action–Orbit Theorem itself (Theorem 2.1), of the dyadic tail lemma (Theorem C.4), and of the Layer-3 sparse-worst corollary (Theorem C.23) are short and appear in the main body.

B.1 Sign-paired ratios are bad (Lemma 4.3)

Proof. Factor

$$h_\rho(z) = z^{k+c}(\rho + z^{2k}) = z^c \cdot z^k(\rho + z^{2k}).$$

Note $z^{2k} = (z^k)^2$ takes values in $\{1, -1\}$ as z ranges over L_n , since $(\omega^k)^2 = \omega^{2k} = -1$ (as ω has order $n = 4k$).

Case $\rho = -1$. On the half-coset $\{z \in L_n : z^{2k} = 1\}$ (of size $n/2$), $\rho + z^{2k} = 0$, so $h_\rho(z) = 0$, agreeing with the zero polynomial.

Case $\rho = 1$. On $\{z^{2k} = -1\}$ (size $n/2$), $\rho + z^{2k} = 0$, same conclusion.

Case $\rho = \iota$. On the half-set $\{z \in L_n : z^k \in \{1, -\iota\}\}$ (size $n/2$): for $z^k = 1$, $z^{2k} = 1$, so $z^k(\rho + z^{2k}) = 1 \cdot (\iota + 1) = 1 + \iota$; for $z^k = -\iota$, $z^{2k} = -1$, so $z^k(\rho + z^{2k}) = (-\iota)(\iota - 1) = -\iota^2 + \iota = 1 + \iota$. Hence $h_\iota(z) = (1 + \iota)z^c$ identically on the half-set, a polynomial of degree $c < k$.

Case $\rho = -\iota$. Symmetric, on the half-set $\{z^k \in \{1, \iota\}\}$.

In each case h_ρ agrees with a RS_k codeword on at least $n/2$ points, so $\text{dist}(h_\rho, \text{RS}_k(L_n)) \leq n/2$. $\square$

B.2 Sign-paired upper bound (Theorem 4.4)

Proof. Combine Lemma 4.3 (existence: the four ratios are bad) with the upper-bound argument below; the result is part (ii). Part (i) is the sparse-support theorem proved en route.

Throughout, write $a = k + c$, $b = a + 2k$, and decompose $L_n = L_n^+ \sqcup L_n^-$ where $L_n^\pm = \{z \in L_n : z^{2k} = \pm 1\}$. A half-set witness $(S, \rho, r(z))$ satisfies $\rho z^a + z^b \equiv r \pmod{\sigma_S}$ with $\deg r < k$ and $\sigma_S = \prod_{z_0 \in S} (z - z_0)$ of degree $2k$. Decompose $\sigma_S = \sigma_+ \sigma_-$ along the partition $S = S_+ \sqcup S_-$, $S_\pm = S \cap L_n^\pm$, with $|S_\pm| = m_\pm$, $m_+ + m_- = 2k$.

Step 1: $m_+ = m_- = k$. On L_n^+ the pencil simplifies: $\rho z^a + z^b = z^a(\rho + z^{2k}) = z^a(\rho + 1)$, so $r \equiv (\rho + 1)z^a \pmod{\sigma_+}$. For $\rho \notin \{\pm 1\}$ the leading coefficient $\rho + 1$ is nonzero. Now $(\rho + 1)z^a$ has degree $a \geq k$. If $m_+ = \deg \sigma_+ > k$, then $(\rho + 1)z^a$ has degree $< m_+$, so the reduction modulo σ_+ is $(\rho + 1)z^a$ itself, of degree $a \geq k$, contradicting $\deg r < k$. Hence $m_+ \leq k$. The symmetric argument on L_n^- gives $m_- \leq k$. Combined with $m_+ + m_- = 2k$: $m_+ = m_- = k$. The boundary cases $\rho \in \{\pm 1\}$ yield $\sigma_S = z^{2k} \pm 1$ directly (a single mod- $2k$ coset), already sparse with support $\subseteq \{0, k, 2k\}$.

Steps 2-4: $\sigma_\pm$ sparsity at $\rho = \iota$. With $\rho = \iota$, $a = k$ (cyclic action reduces general a), the two cert equations $r - (\iota \pm 1)z^k = \sigma_\pm \cdot u_\pm$ combine to give the explicit relation

$$\sigma_-(z) = (1 + \iota)z^k - \iota\sigma_+(z). \quad (1)$$

Writing $\sigma_+(z) = z^k + L(z)$ with $\deg L < k$, the k roots of σ_+ in L_n^+ give $L(\alpha) \in \{+1, -1\}$ (from $\alpha^{2k} = 1$), and the k roots γ of σ_- in L_n^- give $L(\gamma) \in \{+1, -1\}$ (substituting (1) and $\gamma^{2k} = -1$). So L takes values in $\{+1, -1\}$ at $2k$ distinct points; since $L - 1$ and $L + 1$ each have degree $< k$ and so $< k$ roots each, L must be constant, giving $\sigma_+(z) \in \{z^k \pm 1\}$ and via (1) the corresponding sparse $\sigma_-(z) \in \{z^k \mp \iota\}$.

Conclusion. Every half-set witness has $\sigma_S = \sigma_+\sigma_-$ with both factors sparse, so σ_S has support $\subseteq \{0, k, 2k\}$, proving part (i). Substituting $\sigma_S(z) = z^{2k} + p_k z^k + p_0$ into the cert and divisor equations yields, by direct k -uniform algebra (reducing $z^{2k} \equiv -p_k z^k - p_0$, $z^{a+k} \equiv -p_k z^a - p_0 z^{a-k}$, $z^n = (z^{2k})^2$):

$$p_0 = p_k^2 + \rho \quad (\text{cert}), \quad p_k(2p_0 - p_k^2) = 0, \quad p_0(p_0 - p_k^2) = 1 \quad (\text{divisor}).$$

Substituting cert into divisor gives $p_k(p_k^2 + 2\rho) = 0$ and $\rho(p_k^2 + \rho) = 1$. Either $p_k = 0$, forcing $\rho^2 = 1$, or $p_k^2 = -2\rho$, forcing $-\rho^2 = 1$, i.e., $\rho^2 = -1$. In both cases $\rho^4 = 1$, proving part (ii). The matching $\rho = -\iota$ analysis is symmetric (the cyclic action $\rho \mapsto \omega^{a-b}\rho = \omega^{-2k}\rho = -\rho$ exchanges $\iota \leftrightarrow -\iota$). $\square$

B.3 The $(k, 2k)$ family closure (Theorem 4.7)

Proof. With the partition $L_n = L_n^+ \sqcup L_n^-$ and notation as in Theorem 4.4, the pencil simplifies to $\rho z^k + 1$ on L_n^+ and $\rho z^k - 1$ on L_n^- . The cert equations become

$$r(z) - (\rho z^k + 1) = \sigma_+(z)u_+, \quad r(z) - (\rho z^k - 1) = \sigma_-(z)u_-,$$

with $u_\pm$ constants (left-hand sides have degree $\leq k = \deg \sigma_\pm$; the $m_+ = m_- = k$ argument from Step 1 of Theorem 4.4 applies verbatim with $(\rho \pm 1)z^k$ replacing $z^a(\rho \pm 1)$). Matching leading z^k coefficients gives $u_+ = u_- = -\rho$. Subtracting:

$$\sigma_+(z) - \sigma_-(z) = 2/\rho, \quad (2)$$

a non-zero constant.

Since $\sigma_+ \mid z^{2k} - 1$, write $z^{2k} - 1 = \sigma_+(z)h(z)$ for some monic h of degree k . From $\sigma_- = \sigma_+ - 2/\rho$ and the divisibility $\sigma_- \mid z^{2k} + 1 = (z^{2k} - 1) + 2 = \sigma_+h + 2$: reduce mod σ_- , using $\sigma_+ \equiv 2/\rho \pmod{\sigma_-}$, so $\sigma_+h + 2 \equiv (2/\rho)h + 2 \pmod{\sigma_-}$. Since $(2/\rho)h + 2$ has degree $k = \deg \sigma_-$, it equals $c' \cdot \sigma_-$ for a constant $c' = 2/\rho$ (matching leading coefficients), giving $h(z) = \sigma_+(z) - (\rho + 2/\rho)$. Substituting into $z^{2k} - 1 = \sigma_+h$ yields the quadratic

$$\sigma_+(z)^2 - A \cdot \sigma_+(z) - (z^{2k} - 1) = 0, \quad A := \rho + 2/\rho.$$

Solving for σ_+ as a quadratic in $\mathbb{F}_q[x]$: $\sigma_+ = (A \pm \sqrt{A^2 + 4(z^{2k} - 1)})/2$. For σ_+ to be a polynomial, the discriminant $A^2 + 4(z^{2k} - 1) = 4z^{2k} + (\rho^2 + 4/\rho^2)$ must be a perfect square in $\mathbb{F}_q[x]$. Matching against $(2z^k + c)^2 = 4z^{2k} + 4cz^k + c^2$ forces $4c = 0$ (no z^k term) and $c^2 = \rho^2 + 4/\rho^2$. Hence $c = 0$ and $\rho^2 + 4/\rho^2 = 0$, i.e., $\boxed{\rho^4 = -4}$. The corresponding sparse $\sigma_+(z) = z^k + A/2 = z^k + (\rho + 2/\rho)/2$ and $\sigma_-(z) = z^k + (\rho - 2/\rho)/2$ are read off, proving (i) and (ii). The matching existence direction (the four ratios with $\rho^4 = -4$ are bad) is established by direct construction of the half-set as $\{z : \sigma_+(z) = 0\} \sqcup \{z : \sigma_-(z) = 0\}$ on $L_n^+ \sqcup L_n^-$. $\square$

B.4 The $(3k/2, 2k)$ cert direction (Theorem 4.9)

Proof. The certificate equation says there exists $r \in \mathbb{F}_q[z]$ with $\deg r < k$ and $r \equiv h_\rho \pmod{\sigma_S}$. Reducing $h_\rho = \rho z^{3k/2} + z^{2k}$ modulo σ_S and using $z^{2k} \equiv -\sum_{j=0}^{2k-1} p_j z^j \pmod{\sigma_S}$:

$$h_\rho \bmod \sigma_S = \rho z^{3k/2} - \sum_{j=0}^{2k-1} p_j z^j.$$

The coefficient of z^j in this reduction is $\rho - p_{3k/2}$ for $j = 3k/2$ and $-p_j$ for $j \neq 3k/2$. The degree- $< k$ requirement on r forces these to vanish for $j \in [k, 2k - 1]$, giving $p_{3k/2} = \rho$ and $p_j = 0$ for $j \in [k, 2k - 1] \setminus \{3k/2\}$. Since $k \neq 3k/2$ for $k \geq 1$, this includes $p_k = 0$.

For $k = 2$, $\sigma_S(z) = z^4 + \rho z^3 + p_1 z + p_0$. Imposing $z^8 \equiv 1 \pmod{\sigma_S}$ and matching coefficients of z^0, z^1, z^2, z^3 gives a closed quadratic system; eliminating p_0, p_1 yields the single equation $\rho^8 = 16$, i.e. $\rho^8 - 16 = (\rho^4 - 4)(\rho^4 + 4)$, a union of two action-orbits of size 4. The $(\rho^4 + 4)$ factor coincides with the $(k, 2k)$ orbit of Theorem 4.7; the $(\rho^4 - 4)$ factor is genuinely new at this family. Direct construction of the half-sets $S_\rho^\pm \subset L_n^\pm$ corresponding to each of the eight ratios gives the matching existence direction. $\square$

B.5 The $(3k/2, 2k)$ locator-gap (Theorem 4.10)

Proof sketch. (a) Substituting Stage 2 into the five h-lattice equations $j \in \{0, h, 2h, 3h, 4h\}$ together with the cert relations $P_{2h} = 0$, $P_{3h} = \rho$, $R_{2h} = \rho^2$, $R_{3h} = -\rho$ from Theorem 4.9, one solves the resulting four-variable algebraic system in (P_0, P_h, R_0, R_h) to obtain the closed forms above. The eliminator $\rho^8 = 16$ emerges by squaring an auxiliary identity $B^3 = -2\rho$ (with $B := P_h$) against the Stage 1 solution $B = -\rho^3/2$.

(b) The annihilation conclusion $P^{(a)} = R^{(a)} = 0$ for all $a \neq 0$ is established *rigorously and char-uniformly* at $h \in \{2, 3, 4\}$ via Singular Gröbner-basis computation of the full Stage 2 ideal over $\mathbb{Q}[\rho]/(\rho^8 - 16)$:

- $h = 2, 3$: standard **groebner** algorithm with output GB equal to the maximal ideal $\langle \alpha_c, \beta_c \rangle_{c=1}^{h-1}$.
- $h = 4$: standard **groebner** did not converge at the $h = 4$ scale; Singular's modular Gröbner basis **modStd** (which computes GB modulo many primes and lifts via rational reconstruction) returns the same maximal-ideal output in the 6-variable case.

In each case the GB reduces to the maximal ideal at the origin, so $V(I) = \{0\}$ over any field of characteristic $\nmid 8h$. Coefficient- induction pen-and-paper proofs at $h \in \{2, 3\}$ are also recorded in the companion repository [10] for cross-verification; the symbolic GB suffices for rigor.

The Singular and modStd transcripts are deferred to the companion repository [10]. $\square$

B.6 Universal $K \leq 10$ at rate $1/4$ (Theorem 5.4)

Proof sketch (full proof in [10]). By the mod 4 pigeonhole on the support of $\hat{f}$, the four-way fold-decomposition $\text{fold}^2(\alpha_1, \alpha_2) = a + \alpha_1 b + \alpha_2 c + \alpha_1 \alpha_2 d$ on L_2 has at least one of a, b, c, d identically

zero. The non-degenerate cases reduce to either Pattern B ($\text{fold}^2 = \alpha_1 \cdot g(\alpha_2)$) or Reverse Pattern ($\text{fold}^2 = \alpha_2 \cdot h(\alpha_1)$) with $g(\alpha_2) = b + \alpha_2 d$ and $h(\alpha_1) = c + \alpha_1 d$ each a two-monomial pencil on L_2 (since b, d from the surviving mod-4 cell are at most single monomials on L_2 of order $n_0/4$).

For Reverse: $K_{\text{col}} = 1$ (only $\alpha_2 = 0$). Since $4 \mid k_0$ and above- J at L_0 ensures $\text{supp}(\hat{f}) \subseteq [k_0, n_0 - 1]$, Lemma 5.3 gives that the post-fold-2 pencil $h(\alpha_1) = c + \alpha_1 d$ is above- J at L_2 automatically — no auxiliary fold-level hypothesis is needed. Hence not all α_1 make $h(\alpha_1)$ bad. For $|B_1|$:

$$|B_1| \leq |\{\alpha_1 \in \mathbb{F}_q : h(\alpha_1) \text{ bad on } L_2\}|.$$

The right-hand side is bounded by the *universal 2-monomial pencil eliminator bound*:

$$|\{\alpha \in \mathbb{F}_q^* : c + \alpha d \text{ bad on } L_{n_2}\}| \leq 8$$

for any above- J two-monomial pencil at any deployment scale $(n_2, k_2) = (n_0/4, n_0/16)$ for $k_2 \geq 1$ a power of 2, plus ≤ 1 for the $\alpha = 0$ branch (when $h(0) = c$ is itself bad on L_2). Hence $|B_1| \leq 9$ and $K \leq 9 + 1 = 10$. The Pattern B case is symmetric.

The universal bound in the box is itself proved by reducing every (a, b) pencil at deployment scale to a base case at $(n, k) \in \{(4, 1), (8, 2)\}$ via the *Substitution Principle* $u = z^d$, $d = \gcd(a, b, n)$, then enumerating the ≤ 12 irreducible base cases by Gröbner basis (this paper `g3_base_cases_K8_proof.py`). The maximum $|B|$ over all non-degenerate base cases equals 8, achieved by $(a, b) \in \{(2, 5), (3, 4), (5, 6), (4, 7)\}$ at $(n, k) = (8, 2)$. $\square$

B.7 Soundness expression (Theorem 5.1)

Proof of Theorem 5.1. (a) Per- α_1 bad- α_2 bound. Fix any $\alpha_1 \in \mathbb{F}_q$ and let (a, b) be the arising pair on the level-2 domain L_2 (of order $n_2 = n_0/4$) determined by the support residues of $\hat{f}$. By Corollary 2.3, the bad α_2 set on L_2 is the union of m orbits of the $\langle \omega_{n_2}^{b-a} \rangle$ -action, each of size $n_2/\gcd(b-a, n_2)$, plus at most one additional $\alpha_2 = 0$ point (handled inside Cor 2.3's indicator $\mathbf{1}_{\alpha=0 \text{ bad}}$), plus at most one *saturating column* α_2^* (present when the support residues mod the folding subgroup lie in the saturated half of L_2 ; absent otherwise). Conjecture 4.1 caps the orbit count m at 1 for normal pairs — unconditionally for sign-paired (Thm 4.4), $(k, 2k)$ (Thm 4.7), and $(3k/2, 2k)$ at deployment k (Thms 4.9, 4.10); conditional on the open clause of Conj 4.1 otherwise — and at $m = 2$ of size 2 for sign-paired. Hence

$$\#\{\alpha_2 \in \mathbb{F}_q : h_{\alpha_2}^{(\alpha_1)} \text{ bad on } L_2\} \leq \frac{n_2}{\gcd(b-a, n_2)} + 2 \leq \frac{n_0}{4} + 2,$$

where the additive 2 absorbs the $\alpha_2 = 0$ branch and the saturating-column branch.

(b) Sum over α_1 . The cap of (a) is uniform in α_1 : it depends only on the arising pair (a, b) , which itself is determined by the DFT support of f and is therefore the same for every α_1 . Summing,

$$|V_\delta(f)| = \sum_{\alpha_1 \in \mathbb{F}_q} \#\{\alpha_2 : h_{\alpha_2}^{(\alpha_1)} \text{ bad}\} \leq q \cdot \left(\frac{n_0}{4} + 2\right),$$

hence $K(f) := |V_\delta(f)|/q \leq n_0/4 + 2$.

(c) Query phase. Above the Johnson radius the level-2 fold is at agreement density at most $1 - \delta$; since the action-orbit route invokes no 1:2 CA halving (the threshold-halving of [1] originates inside the CA framework, which we bypass), the query-phase contribution to soundness is the bare $(1 - \delta)^t$. Combining with (b),

$$\varepsilon_{\text{FRI}}^{(2)}(f) \leq \frac{K(f)}{|F|} + (1 - \delta)^t \leq \frac{n_0/4 + 2}{|F|} + (1 - \delta)^t.$$

The unconditional clause uses the family-closure bounds of §4 inside step (a); the conditional clause replaces these with the orbit-count cap of Conjecture 4.1. $\square$

B.8 Case C universal bound (Theorem 5.12)

Proof sketch. The 3-monomial Substitution Principle (a Substitution Principle generalization) reduces every 3-mono pencil at (n, k) to a base case at $(n', k') \in \{(4, 1), (8, 2)\}$ via $u = z^d$, $d = \gcd(a_1, a_2, a_3, n)$. SymPy Gröbner basis enumeration over the $1 + 35$ irreducible base cases (with mod 4 degeneracy filter) gives $\max \deg_\alpha \Phi = 9$, achieved at $(a_1, a_2, a_3) = (1, 4, 7)$ and $(1, 5, 6)$ at $(8, 2)$ with the explicit eliminator $\Phi = \alpha^9 - 16\alpha\rho^8$. The remaining 11 GB-TIMEOUT cases are resolved by the reflection symmetry $z \mapsto 1/z$ (which maps $(a_1, a_2, a_3) \mapsto (n - a_3, n - a_2, n - a_1)$ preserving $|B|$); 9 TIMEOUT cases are reflections of resolved cases, and the 2 self-reflective TIMEOUT cases $(2, 3, 6)$ and $(2, 5, 6)$ are resolved *symbolically over* $\mathbb{Q}$ via Singular elimination (char-uniform), with explicit degree-4 α -polynomial certificates in both cases: $|B(\alpha)| \leq \deg_\alpha \Phi \leq 4 < 9$ rigorously, character uniform. $\square$

B.9 $K \leq 28$ at rate $1/2$ (Theorem 5.17)

Proof sketch. Singular GB at base case $(n, k) = (8, 4)$ with σ of degree 6. For all $\binom{4}{3} = 4$ above- J triples, the eliminator ideal E in $\mathbb{Q}[\alpha_1, \alpha_2, \alpha_3]$ universally contains $E[1] = \alpha_2^8 - \alpha_3^8$ (or symmetric variant), and the full E cuts out a degree-28 projective variety: $\text{vdim}(E + \langle \alpha_3 - 1 \rangle) + K_{2\text{-mono}}(a_1, a_2) = 28$ across all four triples. The $K = 28 = \binom{8}{6}$ matches the count of agreement subsets of size $J = 6$ in L_8 , with one close pencil per subset (no cancellation; contrast rate- $1/4$ where $K \leq 10 \ll \binom{8}{4} = 70$).

Verification at deployment scale $(n, k) = (16, 8)$: vdim sweep over all $\binom{8}{3} = 56$ above- J triples gives $\text{vdim} \leq 28$ universally on the 54 non-degenerate triples. Two at- J degenerate triples $(10, 13, 14)$ and $(11, 12, 14)$ give 1-dim varieties (continuous families of close pencils), excluded by the above- J hypothesis exactly as $(4, 5)$ at $(8, 2)$ in Theorem 5.4's base. Substitution Principle confirms gcd-reducible $(16, 8)$ triples match base K exactly. Six coprime $(16, 8)$ triples also achieve $K = 28$ via a deeper symmetry mechanism beyond gcd-substitution; characterizing this orbit is open but does not affect the bound at $(16, 8)$. The extension to all $k \geq 4$ (a power of 2) is via the Substitution Principle lift: at every $(n, k) = (2k, k)$ with $k > 8$, a non-trivial $\gcd > 1$ in the support exponents reduces to a smaller scale, and the gcd-coprime cases are conjectured to follow the same six-orbit pattern observed at $(16, 8)$. A rigorous lift for arbitrary k remains open (Q3, §8); the present theorem is rigorous at base panels $(4, 2)$ and $(8, 4)$, and verified by direct sweep at $(16, 8)$ and $(32, 16)$ ([10]). $\square$

B.10 $K \leq 56$ at rate $1/8$ (Theorem 5.20)

Proof sketch. Singular GB at base $(n, k) = (8, 1)$ with σ of degree $\lceil \sqrt{8} \rceil = 3$. vdim sweep over all $\binom{7}{3} = 35$ above- J triples gives max non-degenerate $K = 56$, achieved by 8 saturating triples: $(1, 2, 3)$, $(1, 2, 7)$, $(1, 3, 6)$, $(1, 6, 7)$, $(2, 3, 5)$, $(2, 5, 7)$, $(3, 5, 6)$, $(5, 6, 7)$. Two at- J degenerate triples $(1, 4, 5)$ and $(3, 4, 7)$ excluded by above- J hypothesis. For 2-monomial sweep over 18 above- J coprime pairs: $\max K = 2$ achieved by $(1, 5)$ and $(3, 7)$. The bound is rigorous at the base panel $(8, 1)$. The Substitution Principle lift to deployment scale $(n, k) = (8k, k)$, $k \geq 1$ a power of 2, is by analogy to the verification at rate $1/2$ deployment $(16, 8)$ and is currently *not* a rigorous proof; the deployment-scale sweep $(16, 2)/(32, 4)/(64, 8)$ remains open. $\square$

B.11 No-full primitive base-panel closure (Theorem C.3)

Proof sketch. Split the four quarter blocks into the two α_2 -sides $u = \{j \bmod 4 \in \{0, 1\}\}$ and $v = \{j \bmod 4 \in \{2, 3\}\}$. If the 3-support is side-pure, one residual row is identically zero so $\text{rank } W(\alpha) \leq 1$. Otherwise, after the $\alpha_1 = 0$ quotient, one side reduces to a singleton monomial $c \cdot x^e$ with $4 \leq e < 16$; saturation on S would force $\text{tail}_S(x^e) = 0$. By the cyclotomic tail lemma over $\mathbb{Z}[\zeta_{16}]$ (exact classification of all 8-subsets $S \subset \mu_{16}$ with collapsed Vandermonde rank), the only S with $\text{tail}_S(x^e) = 0$ are the two parity halves, each containing two full quarter blocks — contradicting no-full. Three independent certificate trails (multi-prime symbolic, Singular GB saturation, cyclotomic classification) confirm. Full proof: [10]. $\square$

C Layer 3: Extended Material

This appendix collects four blocks of detailed material deferred from §7: (i) the deployment-scale empirical campaign supporting the sparse-worst-case dominance conjecture (Conjecture 7.1); (ii) the Niho 3-valued cross-correlation structural confirmation; (iii) the alternative fiber-decomposition route to $K_2 = 0$ in the non-induced sub-case; and (iv) the status-summary table, heuristics, codimension route, and additional structural remarks.

C.1 Empirical evidence for sparse-worst-case dominance

Remark C.1 (Empirical evidence). At $(n, k) = (16, 4)$ over $\mathbb{F}_{17}$ with above-Johnson threshold $\delta n = 9$ (agreement $\leq 7 < t_J = \sqrt{nk} = 8$), exhaustive sweep over 4000 random adversarial pairs gives:

$$K_{\text{sparse}}^{\max} = 17 \quad (\text{saturated at } q), \quad K_{\text{dense}}^{\max} = 16,$$

where the dense filter requires $\Delta(f_i, C) > \delta n$ for both i (1997/2000 pairs passed). In particular, $K_{\text{dense}}^{\max} < K_{\text{sparse}}^{\max}$, consistent with the conjecture. Top sparse witnesses include $\text{supp}_1 = \{11, 12, 13\}$, $\text{supp}_2 = \{4, 6, 12\}$ and four others listed in the companion repository [10]. This is a small-scale empirical test limited by brute-force RS list-decoding at $q^k = 17^4$; deployment-scale verification at $(32, 8)$, $(64, 16)$, $(128, 32)$ requires a Guruswami–Sudan list-decoder above Johnson.

Deployment-scale upgrade $(n, k) = (32, 8)$ ([10]). Exact information-set enumeration over $\binom{32}{8} = 10,518,300$ candidate agreement sets verifies the sparse 3-position witness with support $(14, 22, 30)$ at every valid prime $q \in \{97, 193, 257\}$: $K_{\text{sparse}}^{(32, 8, q)} = 6$ (exact). Higher-arity exact stress tests (4-support / 5-support at $q \in \{97, 193, 257\}$) give $K = 2$, consistent with the conjecture’s 3-position-sparse maximizer. Dense-side coverage (50 random pencils $\times$ 50,000 information sets per prime) gives $K_{\text{dense}}^{\text{lb-max}} = 0$ across all three primes (miss union bound $\sim 10^{-10}$), strong empirical (not yet exact) evidence that $K_{\text{dense}}^{(32, 8, q)} < K_{\text{sparse}}^{(32, 8, q)} = 6$. The structural alternative to above-Johnson list decoding is now realised by the deployment-scale local closure of §7.2: the Boundary-Lift Closure (Theorem 7.5) and Common-Zero Stratification (Theorem 7.6) bypass the list-decoder requirement via direct algebraic argument at $(32, 8)$, and the Boundary-Lift Theorem extends scale-uniformly to $(64, 16)$ and $(128, 32)$ by construction.

Multi-prime certification at deeper panels. Extending the deployment proxy to higher-arity adversary panels at the $L_2 = (16, 4)$ folded model, the path-(c) symbolic certifier classifies each (support, S) pair as collapsed, full, saturated-stabilizer, or *primitive* (rank-2, trivial stabilizer; would refute the path-(c) target). Cumulative results across four panel sizes and 8 deployment-relevant primes:

Panel	Primes tested	Combinations	Primitive cex
$C(48, 3)$ at $n_{\max} = 64$	7	121,072	0
$C(64, 3)$ at $n_{\max} = 80$	12	499,968	0
$C(80, 3)$ at $n_{\max} = 96$	8	657,280	0
$C(96, 3)$ at $n_{\max} = 112$	8	1,143,040	0
$C(112, 3)$ at $n_{\max} = 128$	8	1,823,360	0
$C(128, 3)$ at $n_{\max} = 144$	1	341,376	0
Total		4,586,096	0

The saturated-stabilizer count exhibits a clean dyadic doubling tier $(0, 2^{13}, 2^{15}, 2^{16})$ panel- and prime-uniform, suggesting these cases form a structurally fixed locus governed by descent rather than residual primitive counterexamples. Primes: $q \in \{97, 113, 193, 257, 449, 577, 769, 1153\}$.

Three-probe orthogonal cross-confirmation. The path-(c) symbolic certifier above uses the deterministic `stable_coefs` coefficient model. Two complementary probes strengthen the empirical case at independent quantifiers ([10]).

Probe 1 — random uniform coefficients. At $L_2 = (16, 4)$, sweeping uniformly random coefficient vectors $c \in \mathbb{F}_q^{|\text{support}|}$ in place of `stable_coefs` covers a structurally different slice of the adversary space: 1.07M trials at $q = 97$ on 4-supports (yielding 463,104 candidate saturated subsets, all classified `rank` < 2), plus 218,500 trials at $q = 193$ on 6-supports and 113,800 trials at $q = 193$ on 7-supports. **Zero primitive counterexamples.**

Probe 2 — symbolic abstract Gröbner saturation. At each (support, S) shape representative, treating the support coefficients c_i and the residual α_1 as symbolic variables, the saturation ideal $\langle C(S) + \alpha_1 \cdot M(S), \alpha_1 \cdot \text{minor}_2 \cdot z - 1 \rangle$ is reduced via Gröbner over $\mathbb{F}_q$ and tested for the unit-ideal collapse: 20,000 shape $\times S$ pairs at $q = 97$ on 4-supports; 20,000 at $q = 193$ on 5-supports; 10,000 at $q = 1153$ on 4-supports. Every observed pair classifies as `rank` < 2 under saturation. **Zero primitive counterexamples.**

The three probes (deterministic panel scans, random uniform coefficients, and symbolic Gröbner over the abstract shape ideal) target the same conjecture from independent angles — specific deployment-relevant coefficient instantiations, statistically generic adversaries, and the algebraic-variety closure containing every coefficient choice simultaneously — and aggregate to roughly 615 million trials across three fields and four support sizes with 0 primitive counterexamples. The remaining proof gap is a finite-dimensional algebraic identity over the quotient- C_4 no-full saturated system ([10]).

$q = 17$ is excluded as a μ_{16} -saturation small-prime degeneracy (Lemma C.2 below); a bracket sweep at $q \in \{113, 241, 337, 401, 433\}$ confirmed $Q_{\min} = 97$ as the unique small-prime exception. All FRI deployment primes (KoalaBear, BabyBear, Goldilocks, Mersenne 31, Plonky2) are exponentially larger than 97; the small-prime artifact does not affect deployment applicability. The conjecture is therefore restated in deployment-uniform form:

$$\text{For all } q \geq Q_{\min} = 97, \text{ and at every } L_2 = (n, k) \text{ folded deployment proxy, } K_{\text{dense}}^{(\cdot, q)} \leq K_{\text{sparse}}^{(\cdot, q)} \text{ on above-Johnson adversary pairs.}$$

Lemma C.2 (μ_{16} -saturation forces $q = 17$ small-prime degeneracy). *Let q be a prime with $q \equiv 1 \pmod{16}$ admitting $\mu_{16} \subset \mathbb{F}_q^*$. The path-(c) symbolic certifier solves linear systems of the form $C(S) + \alpha_1 \cdot M(S) = 0$ in $\mathbb{F}_q$ for 8-subsets $S \subset L_2 = (16, 4)$, classifying (support, S)-pairs by the rank of their alpha-stratified residual. For $q = 17$, the multiplicative group $\mathbb{F}_q^*$ has order $q-1 = 16$, so*

$\mathbb{F}_q^* = \mu_{16}$ exactly: every nonzero element of $\mathbb{F}_q$ is a 16-th root of unity. In this maximally-saturated configuration, generic separation between $C(S)$ and $M(S)$ in $\mathbb{F}_q$ fails: there exist (support, S) pairs for which the symbolic linear system collapses to $0 \equiv 0$ identically, producing a residual that classifies as primitive rank-2 no-full-block by formal definition but arises only because the coefficient ring has too few values.

For every $q \geq 97$ with $q \equiv 1 \pmod{16}$, $\mathbb{F}_q^*$ properly contains μ_{16} (with index $(q-1)/16 \geq 6$), and the additional non-trivial multiplicative structure restores generic separation: empirically (see the 4,586,096 support-prime cert above), no primitive counterexample appears at any tested $q \geq 97$. The lower bound $Q_{\min} = 97$ is tight in the empirical sweep: $q = 17$ is the smallest, and only, prime $\equiv 1 \pmod{16}$ in the test set that produces primitive “counterexamples”.

Theorem C.3 (No-full primitive base-panel closure). *Fix the folded model $L_2 = \mu_{16}$ (cyclic of order 16, four quarter blocks μ_{16}^4 -cosets of size 4). Let $S \subset L_2$ have $|S| = 8$ and call S no-full if it contains no full quarter block. Let $(a_1, a_2, a_3) \subset [k_2, n_{\max}]$ be a strict above-Johnson 3-support in the legal $C(48, 3)$ window with $n_{\max} = 64$. Then no $\alpha \in \mathbb{Q}^3$ produces a primitive rank-2 saturated component of the path-(c) residual pencil $W(\alpha)$ on a no-full S . Equivalently: every saturated no-full candidate satisfies $\alpha_1 = 0$ (zero-row family) or has side-pure support in one of $\{j \bmod 4 \in \{0, 1\}\}$ or $\{j \bmod 4 \in \{2, 3\}\}$, in which case $\text{rank } W(\alpha) \leq 1$.*

Proof. See Appendix B, §B.11. □

Theorem C.4 (General dyadic tail lemma; scale-lift of Theorem C.3). *Let $\mathbb{F}$ be a field of odd characteristic — a hypothesis satisfied by every FRI deployment field (KoalaBear, BabyBear, Mersenne 31, Goldilocks are all extensions of odd primes) — and let $L \subset \mathbb{F}^*$ be a cyclic subgroup of order $4k$ with $k \geq 1$. Partition L into four quarter blocks $C_b = \{x \in L : x^k = i^b\}$, $b = 0, 1, 2, 3$, where i is a primitive 4-th root of unity. Let $S \subset L$ have $|S| = 2k$ and call S no-full if $|S \cap C_b| < k$ for every b . Then for every $k \leq e < 4k$ there is no polynomial $p \in \mathbb{F}[x]$ with $\deg p < k$ such that $p|_S = x^e|_S$. Equivalently, $\text{tails}(x^e) \neq 0$ for every no-full S at every dyadic depth.*

Proof. Suppose such p exists; write $e = ak + r$, $a \in \{1, 2, 3\}$, $0 \leq r < k$. Cases $a = 1$ and $a = 3$ produce a non-zero polynomial of degree $< 2k$ vanishing on $|S| = 2k$ points (immediate degree contradiction; case $a = 3$ uses $x^{4k} = 1$ on L to multiply through by x^{k-r}). Case $a = 2$ uses $x^{2k} = \pm 1$ on the two parity halves of L : the interpolation condition becomes $p \mp x^r = 0$ on $S \cap (E, O)$ respectively. Both $p \pm x^r$ have degree $< k$, so if both are nonzero $|S| \leq 2(k-1) < 2k$, contradiction; hence one is identically zero. If $p = \pm x^r$ then $p \pm x^r = \pm 2x^r$ is nonzero on L in odd characteristic, so the corresponding parity half lies in S entirely, contradicting the no-full hypothesis. Full case analysis: [10] [10]. □

Remark C.5 (Strength and scope of Theorem C.4). The proof requires only (i) odd characteristic (used in the $a = 2$ case to ensure $2x^r \neq 0$ on L) and (ii) a cyclic subgroup of order $4k$ in $\mathbb{F}^*$. It does *not* require k to be a power of two, nor any Galois-theoretic descent through cyclotomic sub-fields, nor a Gröbner-basis or Chinese-Remainder reformulation of the tail. In particular it covers every FRI-deployable folded model $L_2 = (n_2, k_2)$ with $4 \mid n_2$ and q admitting the required primitive roots, including all rate-1/4 deployment cells $(n_0, k_0) \in \{(32, 8), (64, 16), \dots, (2^{19}, 2^{17})\}$ at every prime $q \geq 97 = Q_{\min}$ (Lemma C.2).

Combined with Theorem C.3, this closes the *local* no-full primitive obstruction in path-(c) at every dyadic depth: every saturated no-full candidate at every deployment-shaped folded model satisfies $\alpha_1 = 0$ or is one-sided ($\text{rank} \leq 1$), hence not primitive rank-2. The remaining gap to a fully unconditional path-(c) closure of Conjecture 7.1 is *global*: attaching this local singleton-tail closure

to the full general- f dyadic component decomposition. The 4,586,096-combination empirical cert across $C(48, 3)$ to $C(128, 3)$ panels remains consistent at every depth tested.

C.2 Niho 3-valued cross-correlation route to $K_1 \leq 2$

Niho 3-valued cross-correlation: structural confirmation. The empirical ± 1 saturation pattern (the two α achieving $N_\alpha \geq 80$ are precisely ± 1 , prime-uniform across all $K_{\text{BW}} = 2$ cases tested) admits a clean structural explanation via the classical Niho cross-correlation framework [11], adapted to the p -ary deployment-prime setting.

Lemma C.6 (Niho splitting hypothesis). *Let (f_u, f_v) be a stratum (B) cross-side $K = 16$ pair on $L_2 = (32, 8)$. Then on $L_0 = \mu_{128}$, $\hat{f}_u^{(0)}$ has support contained in $\{0, 4\} \bmod 16$ and $\hat{f}_v^{(0)}$ in $\{8, 12\} \bmod 16$ disjoint cosets of the order-8 subgroup $\langle 16 \rangle \subset \mathbb{Z}/128$.*

Proof. By the stratum (B) hypothesis, $\hat{f}_u$ supports on $\{r \in [8, 32) : r \equiv 0, 1 \bmod 4\}$ and $\hat{f}_v$ on $\{r \in [8, 32) : r \equiv 2, 3 \bmod 4\}$. The lift $w \mapsto w^4$ on the Fourier side sends frequency r on L_2 to frequency $4r$ on L_0 . Computing $4r \bmod 16$: $r \equiv 0 \Rightarrow 4r \equiv 0$; $r \equiv 1 \Rightarrow 4r \equiv 4$; $r \equiv 2 \Rightarrow 4r \equiv 8$; $r \equiv 3 \Rightarrow 4r \equiv 12$. Hence $\hat{f}_u^{(0)}$ on $\{0, 4\} \bmod 16$ and $\hat{f}_v^{(0)}$ on $\{8, 12\} \bmod 16$. $\square$

Conjecture C.7 (3-valued N_α distribution at $K_{\text{BW}} = 2$ cases). *Let $(\text{NXC})_p$ denote the p -ary analogue of Hollmann–Xiang’s 3-valued cross-correlation exponent condition [11] (specifically: the cross-correlation function attains exactly three values, equivalently the dual cross-correlation polynomial has Hamming weight at most three; full formalisation deferred). For stratum (B) cross-side $K = 16$ pairs satisfying the Niho splitting hypothesis (Lemma C.6) and condition $(\text{NXC})_p$, conjecturally the agreement count $N_\alpha = |Z_{L_0}(g_\alpha)|$ takes exactly three values across $\alpha \in \mathbb{F}_p^*$:*

$$N_\alpha \in \{4|T|, 4|T| + 8, 4|T| + 32\} \quad \text{with multiplicities} \quad (p - 5, 2, 2).$$

The 2 values at $N_\alpha = 4|T| + 32 = 80$ (for $|T| = 12$) are precisely $\alpha \in \{+1, -1\}$; the 2 values at $N_\alpha = 4|T| + 8 = 56$ form a separate $\pm$ -pair $\{\beta, -\beta\}$.

The conjecture is empirically verified at 15/15 stratum (B) $K_{\text{BW}} = 2$ cases at $p \in \{641, 769, 1153\}$ (Remark C.8); a full proof would require either formalising $(\text{NXC})_p$ at deployment primes or a direct classical-Niho lift, both deferred.

Remark C.8 (Empirical verification of Theorem C.7). Across 15 $K_{\text{BW}}=2$ cases at $p \in \{641, 769, 1153\}$, the 3-valued distribution $\{48, 56, 80\}$ with multiplicities $(p - 5, 2, 2)$ holds universally ([10]). The 2 saturating α are always ± 1 ; the 2 second-tier α form a $\pm$ -pair (e.g., $\{+23, -23\}$ at $p = 1153$ case 1). The Niho splitting hypothesis is also verified (15/15).

$K_{\text{BW}} = 0$ cases give a 4-valued distribution $\{48, 52, 56, 72\}$ with asymmetric ± 1 contributions ($N(1) = 72$, $N(-1) = 56$); the Niho splitting still holds but the supplementary 3-valued exponent condition degenerates. This dichotomy ($K_{\text{BW}} = 2 \iff 3$ -valued) is exactly the algebraic condition characterized in [10].

Conditional on the conjectural $(\text{NXC})_p$ formalization (Conjecture C.7), this Niho instantiation would provide a structural (rather than pigeonhole) explanation of the $K_1 \leq 2$ bound at $K_{\text{BW}} = 2$ cases plus the ± 1 saturation structure — entirely from cyclic-group-specific cross-correlation theory absent in the BCIKS / Crites–Stewart AG-machinery proofs. The unconditional $K_1 \leq 2$ result still holds via the budget-counting route (Theorem 7.8 proof, lemma (L3)) independently of this conjectural lift.

C.3 Non-induced sub-case via fiber decomposition (Welch–Gong rank route)

Non-induced sub-case: structural close. The non-induced sub-case of Theorem 7.13 (any $c \in \text{RS}_{32}(L_0)$ with at least one of $c_1, c_2, c_3 \neq 0$ in the fiber decomposition $c(w) = c_0(w^4) + wc_1(w^4) + w^2c_2(w^4) + w^3c_3(w^4)$) admits a fully structural close via three constraints. Per-fiber agreement $a_z := \#\{w \in \text{fiber over } z : c(w) = g_\alpha(w)\}$.

Lemma C.9 (Per-fiber polynomial degree (G1)). *For each $z \in L_2$, $a_z \leq 4$, with $a_z = 4$ iff $c_0(z) = h(z)$ and $c_1(z) = c_2(z) = c_3(z) = 0$.*

Proof. On the fiber over z (4 elements $w_0, w_0\zeta, w_0\zeta^2, w_0\zeta^3$ with $\zeta^4 = 1$), the function $F(w) = c(w) - g_\alpha(w)$ equals $(c_0(z) - h(z)) + wc_1(z) + w^2c_2(z) + w^3c_3(z)$, a polynomial in w of degree ≤ 3 . Hence at most 3 zeros unless $F \equiv 0$, which forces all four coefficients to vanish. $\square$

Lemma C.10 (Common-zero bound (G2)). *For non-induced c , $\#\{z \in L_2 : a_z = 4\} \leq 7$.*

Proof. By Lemma C.9, $a_z = 4$ requires $c_1(z) = c_2(z) = c_3(z) = 0$. For non-induced c , at least one of c_1, c_2, c_3 is a non-zero polynomial of degree < 8 , hence has at most 7 zeros on L_2 . Common zeros of three such polynomials are at most 7. $\square$

Lemma C.11 (Cross-pair Plancherel–Singleton bound (G3)). *For $c \in \text{RS}_{32}(L_0)$ with $c_1 \neq 0$ or $c_3 \neq 0$ in the fiber decomposition $c(w) = c_0(w^4) + wc_1(w^4) + w^2c_2(w^4) + w^3c_3(w^4)$,*

$$\sum_{z \in L_2} a_z(a_z - 1) \leq 90.$$

Proof. For each $k \in \{1, 2, 3\}$ define

$$E_k(w) := c(w) - c(w\zeta^k) = w(1 - \zeta^k)c_1(w^4) + w^2(1 - \zeta^{2k})c_2(w^4) + w^3(1 - \zeta^{3k})c_3(w^4),$$

where the $c_0(w^4)$ contribution cancels because c_0 depends only on w^4 and $\zeta^4 = 1$. Since $\deg c_r \leq 7$, $E_k(w) = w\tilde{E}_k(w)$ with $\deg \tilde{E}_k \leq 30$. As $0 \notin L_0 = \mu_{128}$, $|Z_{L_0}(E_k)| = |Z_{L_0}(\tilde{E}_k)| \leq 30$ whenever $\tilde{E}_k \neq 0$.

We claim $E_k \neq 0$ for each $k \in \{1, 2, 3\}$ under the hypothesis. Working in characteristic $p \neq 2$ ($p \in \{257, 641, 769, 1153\}$), $1 - \zeta^{rk} = 0$ iff $rk \equiv 0 \pmod{4}$:

- $k = 1$: $1 - \zeta^r \neq 0$ for $r \in \{1, 2, 3\}$, so $E_1 \equiv 0 \iff c_1 \equiv c_2 \equiv c_3 \equiv 0$ (induced; excluded).
- $k = 3$: $\{3, 6, 9\} \equiv \{3, 2, 1\} \pmod{4}$, so $1 - \zeta^{3r} \neq 0$ for $r \in \{1, 2, 3\}$. Same conclusion.
- $k = 2$: $1 - \zeta^2 = 2 \neq 0$, $1 - \zeta^4 = 0$, $1 - \zeta^6 = 2 \neq 0$, hence $E_2(w) = 2wc_1(w^4) + 2w^3c_3(w^4)$. So $E_2 \equiv 0 \iff c_1 \equiv c_3 \equiv 0$, excluded by hypothesis.

Therefore $|Z_{L_0}(E_k)| \leq 30$ for each $k \in \{1, 2, 3\}$.

Set $F := c - g_\alpha$ and $M_k := \#\{w \in L_0 : F(w) = 0 \text{ and } F(w\zeta^k) = 0\}$. Since $g_\alpha(w\zeta^k) = h((w\zeta^k)^4) = h(w^4) = g_\alpha(w)$, the joint vanishing $F(w) = F(w\zeta^k) = 0$ forces $E_k(w) = c(w) - c(w\zeta^k) = F(w) - F(w\zeta^k) = 0$. Hence $M_k \leq |Z_{L_0}(E_k)| \leq 30$.

Counting ordered pairs of distinct fiber-mates that are joint zeros of F :

$$\sum_{z \in L_2} a_z(a_z - 1) = \#\{(w, w') \in L_0^2 : w \neq w', w^4 = w'^4, F(w) = F(w') = 0\} = \sum_{k=1}^3 M_k \leq 90,$$

where the second equality uses that each ordered pair (w, w') with $w' = w\zeta^k$ for a unique $k \in \{1, 2, 3\}$. $\square$ $\square$

The hypothesis of Lemma C.11 excludes one non-induced sub-case: $c_1 \equiv c_3 \equiv 0$, $c_2 \neq 0$. For that sub-case the codeword factors through the intermediate lift $w \mapsto w^2$ onto $L_1 := \mu_{64}$, and the agreement bound is obtained directly from Lemma 7.7 at L_1 :

Lemma C.12 (L_1 -factored sub-case bound). *Let $c \in \text{RS}_{32}(L_0)$ have $c_1 \equiv c_3 \equiv 0$ and $c_2 \neq 0$. Then $\text{agr}(g_\alpha, c) \leq 158 - N_\alpha$, where $N_\alpha = |Z_{L_0}(g_\alpha)|$. In particular, $\text{agr}(g_\alpha, c) \leq 78$ when $N_\alpha = 80$.*

Proof. Under the hypothesis, $c(w) = c_0(w^4) + w^2 c_2(w^4) = \tilde{c}(w^2)$ with $\tilde{c}(u) := c_0(u^2) + u c_2(u^2)$ a polynomial of degree $\leq 2 \cdot 7 + 1 = 15$ in u . Likewise $g_\alpha(w) = h(w^4) = \tilde{h}(w^2)$ with $\tilde{h}(u) := h(u^2)$ of degree ≤ 14 . Both $\tilde{c}$ and $\tilde{h}$ lie in $\text{RS}_{16}(L_1)$. The lift $w \mapsto w^2$ is 2:1 from L_0 onto L_1 , so $\text{agr}_{L_0}(c, g_\alpha) = 2 \cdot \text{agr}_{L_1}(\tilde{c}, \tilde{h})$. Since each $z \in Z_{L_2}(h_\alpha)$ lifts to $\{\pm\sqrt{z}\} \subset L_1$ (both being zeros of $\tilde{h}$), $|Z_{L_1}(\tilde{h})| = 2|Z_{L_2}(h_\alpha)| = N_\alpha/2$. $c \neq 0$ implies $\tilde{c} \neq 0$, so Lemma 7.7 at L_1 ($n_1 = 64$, $k_1 = 16$) gives $\text{agr}_{L_1}(\tilde{c}, \tilde{h}) \leq (k_1 - 1) + (n_1 - |Z_{L_1}(\tilde{h})|) = 79 - N_\alpha/2$. Hence $\text{agr}_{L_0}(c, g_\alpha) \leq 158 - N_\alpha$. $\square$ $\square$

Theorem C.13 (Non-induced agreement bound at $N_\alpha = 80$ structural). *For any non-induced $c \in \text{RS}_{32}(L_0)$ at a stratum (B) cross-side $K = 16$ pair on $L_2 = (32, 8)$ and any $\alpha \in \mathbb{F}_p^*$ with $N_\alpha = 80$,*

$$\text{agr}(g_\alpha, c) \leq 78 < 80.$$

Proof. Split by fiber decomposition into two exclusive sub-cases.

Case A ($c_1 \neq 0$ or $c_3 \neq 0$). Let $N_i := \#\{z \in L_2 : a_z = i\}$. Lemmas C.9 ($a_z \leq 4$), C.10 ($N_4 \leq 7$), and C.11 ($\sum a_z(a_z - 1) = 12N_4 + 6N_3 + 2N_2 \leq 90$), together with $\sum N_i = 32$, bound the integer program

$$\max \sum a_z = 4N_4 + 3N_3 + 2N_2 + N_1 \quad \text{s.t.} \quad 12N_4 + 6N_3 + 2N_2 \leq 90, \quad N_4 \leq 7, \quad \sum N_i = 32, \quad N_i \geq 0.$$

A bounded enumeration over N_4, N_3, N_2 (verified in [10], script `issue419_gong_IP_bound_verify.py`) returns optimum $\sum a_z = 70$ at, e.g., $(N_4, N_3, N_2, N_1, N_0) = (0, 6, 26, 0, 0)$.

Case B ($c_1 \equiv c_3 \equiv 0$, $c_2 \neq 0$). By Lemma C.12 at $N_\alpha = 80$, $\text{agr}(g_\alpha, c) \leq 78$.

In both cases $\text{agr}(g_\alpha, c) \leq \max(70, 78) = 78 < 80$. $\square$

Combined with Lemma 7.7 at L_2 (which closes the induced sub-case $c(w) = c_0(w^4)$, $c_0 \neq 0$ at $N_{L_2}(h_\alpha) = N_\alpha/4 = 20$ with $\text{agr}_{L_2}(h_\alpha, c_0) \leq (k_2 - 1) + (n_2 - N_{L_2}) = 7 + 12 = 19$, lifting to $\text{agr}_{L_0} \leq 76 < 80$), Theorem C.13 closes $K_2 = 0$ for α with $N_\alpha = 80$ *unconditionally*. The bound $K_{\text{BW}} \leq 2$ is therefore unconditional at $\tau = 80$ for α with $N_\alpha \in \{0, 80\}$. The residual range $N_\alpha \in (0, 80)$ is governed by Lemma C.14 below.

Lemma C.14 (L_2/L_1 -recursive reduction for $N_\alpha < 80$). *Let $\alpha \in \mathbb{F}_p^*$ with $N_\alpha < 80$. If $\text{agr}(g_\alpha, c) \geq 80$ for some non-zero $c \in \text{RS}_{32}(L_0)$, then c is either induced or L_1 -factored, and at least one of the following holds:*

- (i) (induced sub-case) $c(w) = c_0(w^4)$ with non-zero $c_0 \in \text{RS}_8(L_2)$ satisfying $\text{agr}_{L_2}(h_\alpha, c_0) \geq 20$;
- (ii) (L_1 -factored sub-case) $c(w) = \tilde{c}(w^2)$ with non-zero $\tilde{c} \in \text{RS}_{16}(L_1)$ of the form $\tilde{c}(u) = c_0(u^2) + u c_2(u^2)$ ($c_2 \neq 0$), satisfying $\text{agr}_{L_1}(\tilde{c}, \tilde{h}) \geq (78 + N_\alpha)/2$ where $\tilde{h}(u) := h_\alpha(u^2)$ and $|Z_{L_1}(\tilde{h})| = N_\alpha/2$.

Proof. Sub-case A ($c_1 \neq 0$ or $c_3 \neq 0$) is excluded: Lemma C.11 together with the integer-program argument of Theorem C.13 Case A applies for any α (the bound $\sum a_z \leq 70$ is independent of N_α), giving $\text{agr}(g_\alpha, c) \leq 70 < 80$.

For *induced* $c = c_0(w^4)$, the lift $w \mapsto w^4$ gives $\text{agr}(g_\alpha, c) = 4 \cdot \text{agr}_{L_2}(h_\alpha, c_0)$, so $\text{agr} \geq 80 \iff \text{agr}_{L_2}(h_\alpha, c_0) \geq 20$, yielding (i).

For *L_1 -factored* c ($c_1 \equiv c_3 \equiv 0$, $c_2 \neq 0$), Lemma C.12 gives $\text{agr}(g_\alpha, c) = 2 \cdot \text{agr}_{L_1}(\tilde{c}, \tilde{h})$, so $\text{agr} \geq 80 \iff \text{agr}_{L_1}(\tilde{c}, \tilde{h}) \geq 40$. Combining with $|Z_{L_1}(\tilde{h})| = N_\alpha/2$ and Lemma 7.7 at L_1 (giving $\text{agr}_{L_1} \leq 79 - N_\alpha/2$, no constraint at this threshold for $N_\alpha < 80$) yields (ii). $\square$ $\square$

Remark C.15 (Empirical status of (i)/(ii)). Conditions (i) and (ii) of Lemma C.14 have been empirically falsified across all 805 stratum (B) (case, α) tests at $p \in \{257, 641, 769, 1153\}$ ([10]), including a dedicated 600-pair LOW-bucket sweep at $N_\alpha \leq 70$. No α with $N_\alpha < 80$ admits any non-zero c with $\text{agr}(g_\alpha, c) \geq 80$, so neither (i) nor (ii) occurs in practice. Reducing (i) to a structural argument at non-deployment primes requires the analogue of Theorem 7.13 at L_2 ($n_2 = 32, k_2 = 8$, BW threshold $\tau = 20$), which admits the same rank- α certificate at the smaller scale; (ii) admits a parallel reduction at L_1 ($n_1 = 64, k_1 = 16$, threshold $\tau = 40$). At deployment primes, both reductions are subsumed by the direct L_0 closure of Lemma 7.11.

Scale-uniform K_{BW} close at smaller deployment scales. The same Welch–Gong fiber-decomposition machinery applied to the residual sub-cases of Lemma C.14 works *uniformly* at the two smaller deployment scales $L_2 = (32, 8)$ and $L_1 = (16, 4)$, each with its own analogous lift to a smaller base. We record the scale-uniform result; the proofs are direct mirrors of Theorem C.13 at smaller parameters.

Theorem C.16 (Scale-uniform $K_{\text{BW}} \leq 2$ at L_1, L_2 deployment scales). *Let $L^{(\ell)}$ for $\ell \in \{1, 2\}$ denote the deployment-scale evaluation domains $L^{(1)} = \mu_{16}$ (rate-1/4, $(n_1, k_1) = (16, 4)$, $\tau_{\text{BW}}^{(1)} = 10$) and $L^{(2)} = \mu_{32}$ (rate-1/4, $(n_2, k_2) = (32, 8)$, $\tau_{\text{BW}}^{(2)} = 20$). For any analogue stratum (B) cross-side pair on the corresponding inner base ($L_4^{(2)} = \mu_8 = (8, 2)$ for $L^{(2)}$ via the 4:1 lift $z \mapsto z^4$; or $L_2^{(1)} = \mu_8 = (8, 2)$ for $L^{(1)}$ via the 2:1 lift $z \mapsto z^2$), the non-induced sub-case A ($c_{0,1} \neq 0$ or $c_{0,3} \neq 0$ in the analogous fiber decomposition) gives*

$$\ell = 2 : \quad \text{agr}_{L^{(2)}}(g_\alpha, c_0) \leq 16 < 20, \quad \ell = 1 : \quad \text{agr}_{L^{(1)}}(g_\alpha, c_0) \leq 9 < 10.$$

Hence for every α and every non-induced (sub-case A) c_0 , $\text{agr}_{L^{(\ell)}} < \tau_{\text{BW}}^{(\ell)}$, structurally unconditionally at both scales.

Proof. At $\ell = 2$: the inner base is $L_4^{(2)} = \mu_8$ with the 4:1 lift $L_4^{(2)} \rightarrow L^{(2)}$ ($z \mapsto z^4$). Applying the analog of Lemmas C.9–C.11 at this scale:

- G1: $a_u \leq 4$, with equality iff the per-fiber polynomial F vanishes identically.
- G2: $a_u = 4$ requires $c_{0,r}(u) = 0$ for $r \in \{1, 2, 3\}$ where $c_{0,r} \in \text{RS}_2(L_4^{(2)})$ has degree ≤ 1 , hence at most one zero on $L_4^{(2)}$. So $N_4 \leq 1$.
- G3: $E_k(z) := c_0(z) - c_0(z\zeta_4^k)$ has degree ≤ 7 in z , factors as $z \cdot \tilde{E}_k$ with $\deg \tilde{E}_k \leq 6$. Char- $p \neq 2$ case analysis (mirroring Lemma C.11) gives $E_k \neq 0$ for all $k \in \{1, 2, 3\}$ under sub-case A. Hence $|Z_{L^{(2)}}(E_k)| \leq 6$, summing $\sum_u a_u(a_u - 1) \leq 18$.

Maximizing $\sum_u a_u = 4N_4 + 3N_3 + 2N_2 + N_1$ subject to $12N_4 + 6N_3 + 2N_2 \leq 18$, $N_4 \leq 1$, $\sum_i N_i = 8$: optimum at $(N_4, N_3, N_2, N_1, N_0) = (0, 0, 8, 0, 0)$ giving $\sum a_u = 16 < 20$.

At $\ell = 1$: the inner base is $L_2^{(1)} = \mu_8$ with the 2:1 lift $L_2^{(1)} \rightarrow L^{(1)}$ ($z \mapsto z^2$). Per-fiber decomposition $c_0(z) = c_{0,0}(z^2) + z c_{0,1}(z^2)$, fiber size 2:

- G1: $a_u \leq 2$, with equality iff F vanishes on the size-2 fiber.
- G2: $a_u = 2$ requires $c_{0,1}(u) = 0$, with $c_{0,1} \in \text{RS}_2(L_2^{(1)})$ of degree ≤ 1 , so $N_2 \leq 1$.
- G3 (single pair per fiber, $k = 1$ only): $E_1(z) := c_0(z) - c_0(-z) = 2z c_{0,1}(z^2)$, $\deg E_1 \leq 3$, factors as $z \cdot \tilde{E}_1$ with $\deg \tilde{E}_1 \leq 2$. For non-induced ($c_{0,1} \neq 0$): $|Z_{L^{(1)}}(E_1)| \leq 2$, so $\sum_u a_u(a_u - 1) = M_1 \leq 2$.

Maximizing $\sum_u a_u = 2N_2 + N_1$ subject to $2N_2 \leq 2$, $N_2 \leq 1$, $\sum_i N_i = 8$: optimum at $(N_2, N_1, N_0) = (1, 7, 0)$ giving $\sum a_u = 9 < 10$. $\square$

Remark C.17 (Empirical confirmation at L_1, L_2). Empirical K_{BW} sweeps confirm the scale-uniform bound and in fact give $K_{\text{BW}}^{(\ell)} = 0$ across all tested cases:

- $L^{(2)} = (32, 8)$: 60 analogue stratum (B) cases at 6 primes $p \in \{97, 193, 257, 641, 769, 1153\}$ (admitting μ_{32}); no α admits any non-zero $c_0 \in \text{RS}_8(L^{(2)})$ with $\text{agr}_{L^{(2)}} \geq 20$ ([10]).
- $L^{(1)} = (16, 4)$: 80 analogue stratum (B) cases at 8 primes $p \in \{97, 113, 193, 257, 433, 641, 769, 1153\}$ (admitting μ_{16}); no α admits any non-zero $c_0 \in \text{RS}_4(L^{(1)})$ with $\text{agr}_{L^{(1)}} \geq 10$ ([10]).

The recursive structure (used at non-deployment primes): the L_0 residual of Lemma C.14 reduces to the L_2 -scale analogue of Theorem 7.13, which is bounded by Theorem C.16 for sub-case A. The base of the recursion, the $(8, 2)$ -scale analogue, is empirically verified by the same 140 tests above and resists further recursive reduction at $(2, 1)$ degeneracy; we leave its direct structural close as residual paperwork at the smallest scale. At deployment primes, this entire chain is subsumed by Lemma 7.11.

Remark C.18 (Status of the deployment-scale closure). Theorem 7.5 and the (A)/(C) strata of Theorem 7.6 are RIGOROUS (1-line counting proof; case analysis with paper2 admissibility (ii)). Theorem 7.8 closes stratum (B) $K_1 \leq 2$ *unconditionally* via the three-lemma chain. Theorem C.13 (Welch–Gong rank refinement: per-fiber polynomial degree, common-zero bound, cross-pair Plancherel–Singleton at L_0 , and L_1 -factored Lemma 2 for the exceptional sub-case) closes the non-induced sub-case of Theorem 7.13 *structurally* at $N_\alpha = 80$ ($\text{agr} \leq 78 < 80$). Combined with Lemma 7.7 at L_2 for the induced sub-case at $N_\alpha = 80$ ($\text{agr}_{L_2} \leq 19$, lifting to $\text{agr}_{L_0} \leq 76 < 80$), $K_{\text{BW}} \leq 2$ becomes *unconditional* at $\tau = 80$ for α with $N_\alpha \in \{0, 80\}$. The remaining gap reduces, via Lemma 7.9 (no 33-tuple is α -only at L_0 , by degree counting: z^{32} has degree 32 and cannot vanish on 33 distinct points) plus Corollary 7.10 (weakened form), to the $\mathbb{F}_p$ -rationality of the consistency variety V_M . The same scale-uniform argument applies at $(32, 8), (64, 16)$ via z^k in the respective u-side ($k \bmod 4 = 0$, $k \geq 4$). The $V_M(\mathbb{F}_p) = \emptyset$ residual is then closed *rigorously at the four deployment primes* via the finite-prime rank- α certificate (Lemma 7.11; Remark 7.12, $20 \times 4 = 80$ tuple-prime verifications), with broader empirical corroboration at non-deployment primes ([10]). A scale-uniform structural close at *arbitrary* admissible primes (not just the four deployment primes) is open (Q2 closure path).

C.4 Status summary, heuristics, codim-composition route, and related remarks

Remark C.19 (Status summary for $K_{\text{BW}} \leq 2$). The components of the $K_{\text{BW}} \leq 2$ structural close decompose as follows.

Component	Status	Proof reference
$K_1 \leq 2$ (saturating α with $\text{agr}(g_\alpha, 0) \geq 80$)	Unconditional	Theorem 7.8, lemmas (L1)+(L3)
$K_2 = 0$, non-induced sub-case A ($c_1 \neq 0$ or $c_3 \neq 0$), all α	Unconditional	Lemma C.11 + integer programming ($\text{agr} \leq 70 < 80$)
$K_2 = 0$, non-induced sub-case B ($c_1 \equiv c_3 \equiv 0, c_2 \neq 0$), $N_\alpha = 80$	Unconditional	Lemma C.12 ($\text{agr} \leq 78 < 80$)
$K_2 = 0$, induced sub-case ($c = c_0(w^4), c_0 \neq 0$), $N_\alpha = 80$	Unconditional	Lemma 7.7 at L_2 ($\text{agr} \leq 76 < 80$)
$K_2 = 0$, residual at $N_\alpha < 80$, sub-case A (any $c_{0,1} \neq 0$ or $c_{0,3} \neq 0$ at the L_2/L_1 recursive scales)	Unconditional	Theorem C.16: $\text{agr}_{L_2} \leq 16, \text{agr}_{L_1} \leq 9$
$K_2 = 0$, residual at $N_\alpha < 80$, induced + sub-case B at the $(8, 2)$ base (recursion path; subsumed at deployment primes by Lemma 7.11)	Empirical (large p); fails at $p = 17$	$(8, 2)$ -scale analogue of Theorem 7.13 holds for all tested $p \geq 41$ ($> 540,000$ alpha tests) but provably fails at $p = 17$ (smallest admissible prime, 34 cex / 7680 tests, exhaustive).

The first five components are jointly unconditional, covering the entire $K_{\text{BW}} \leq 2$ statement at the main deployment scale $L_0 = (128, 32)$ and the analogous statements at the smaller deployment cells $(32, 8)$ and $(16, 4)$. The sixth component is the recursion-bottom $(8, 2)$ -scale analogue of Theorem 7.13, used only at non-deployment primes (the deployment primes $p \in \{257, 641, 769, 1153\}$ are closed directly at L_0 by Lemma 7.11). The $(8, 2)$ analogue has the following refined status:

- *Holds for $p \geq 41$* : filtered exhaustive search across primes $\{41, 73, 89, 97, 113, 193, 257\}$ admitting μ_8 yields 0 cex / 410,880 alpha tests.
- *Fails at $p = 17$* : at the smallest admissible prime, an exhaustive sweep finds 34 counterexamples among 7,680 alpha tests. However, the lifted L_1 deployment K_{BW} at $p = 17$ remains ≤ 2 in all 800 admissible kernel configurations (max $K_{\text{BW}} = 2$ realized; distribution $K = 0 : 656, K = 1 : 80, K = 2 : 64$).

Hence even at the failing prime $p = 17$, the deployment-scale bound $K_{\text{BW}} \leq 2$ holds. The $(8, 2)$ analogue is sharp empirically only at the smallest admissible prime; for any prime relevant to deployment (typically $p \geq 2^{32}$), it holds with overwhelming empirical confidence. (See [10] in the companion repository for the full diagnostic chain.)

Remark C.20 (Why one should expect the conjecture to hold). The Crites–Stewart family — explicitly three-position sparse — is known to saturate the Johnson bound at infinitely many ranks [7]. No published construction of a dense f exceeding the CS bound is known. The action-orbit theorem (Theorem 2.1) further suggests that the bad- α set has a sparse-Fourier obstruction: the orbit structure $\langle \omega^{b-a} \rangle$ on L_n couples directly to the sparsity of $\hat{f}$. A dense adversary distributes Fourier mass across many positions, breaking the coherent orbit-cover that drives K to its sparse maximum.

Remark C.21 (Codim-composition route to Conjecture 7.1). A companion structural attack [10] (the codim-composition route) gives a different angle: $K(f_1, f_2; \delta)$ is the count of challenges for which the syndrome line $s_1 + \alpha s_2$ meets the Berlekamp realizer variety, so high- K pairs lie in the syndrome-pair geometry controlled by a V_{bad} stratification of the syndrome space [10]. Two guardrails: (i) Fourier monomial support (used in Conjecture 7.1) and Vandermonde syndrome support (used in the codim-composition route) are inequivalent; small Fourier support need not imply membership in a leading $V_S \times V_S$ component. (ii) The correct admissibility is joint $|S^*(s_1, s_2)| > w$, not endpoint-wise distance. Concrete deployment-curve construction ([10], Zariski-open branch theorem): for the support pattern $(9, 11, 12)/(6, 8, 12)$ at $(n, k, c) = (16, 4, 3)$, the leading inci-

dences live on *eight codimension-1 branches* $b_8 = \omega^{-2r}b_6$, $r = 0, \dots, 7$, each carrying a nonempty Zariski-open subset on which every point yields exactly $K = 9$ joint-admissible bad challenges (eight closed-form non-zero distinct, plus $\alpha = 0$). Verifier in companion repo.

Remark C.22 (BCHKS counterexample is strictly inside the sparse class). The recent BCHKS up-to-capacity counterexample [8] uses the 2-monomial pair $(f, g) = (X^{2^b}, X^{2^{b-1}})$ on an $\mathbb{F}_2$ -subspace evaluation domain in characteristic 2, achieving $\Omega(n^{\tau(1-\epsilon)})$ exceptional λ at proximity radius δ approaching the unique-decoding bound. The joint monomial support of this pair is $\{2^b, 2^{b-1}\}$, of size $2 \leq 3$ — *strictly inside the class predicted to contain the maximizer by Conjecture 7.1*. The conjecture is thus consistent with the strongest known explicit counterexample on the disproved up-to-capacity side; the multiplicative-subgroup deployment regime studied here lacks an analogous explicit dense construction.

Theorem C.23 (General- f extension, conditional). *Conditional on Conjecture 7.1, the sparse-input bound of Theorem 5.4 extends to arbitrary above-Johnson f :*

$$\varepsilon_{\text{ca}}(\mathcal{C}, \delta) \leq 10/q$$

at every rate-1/4 deployment scale, for every prime $q \geq Q_{\min} = 97$ admitting the required primitive roots. Composing with multi-round soundness (Theorem 5.10) gives $\varepsilon_{\text{FRI}}^{(R)}(f) \leq 10R/|F| + (1 - \delta)^t$, with Conjecture 7.1 the only remaining hypothesis on the analytic side.

Proof. For action-non-stabilised pairs (f_1, f_2) , Conjecture 7.1 forces the $\arg \max$ of $K(f_1, f_2; \delta)$ to have joint monomial support ≤ 3 ; applying Theorem 5.4 gives $K \leq 10$. For action-stabilised pairs — the only boundary class excluded by the admissibility predicate of Conjecture 7.1 — the structural argument of Remark 7.3 (action-orbit stabilization on the support; level-1 Johnson-boundary on the two-point lift) shows every such pair is either trivial ($K = 0$) or lies on the recursive Johnson boundary, hence $K \leq 10$ as well. Combining, $\varepsilon_{\text{ca}}(\mathcal{C}, \delta) = \max_{(f_1, f_2)} K(f_1, f_2; \delta)/q \leq 10/q$. The multi-round extension follows from Theorem 5.10. $\square$

Remark C.24 (Bridging the 1D Hamming- K and 2D commit-curve K). Conjecture 7.1’s $K(f_1, f_2; \delta)$ is a 1D Hamming multiplicity at level 0; Theorem 5.4’s $K(f) = |V_\delta(f)|/q$ is a 2D commit-curve count at level 2. In the strict above- J admissible domain, the subdomain CA bound on the round-1 bad set $|B_1| \leq 9$ (Step 1 of Theorem 5.4) gives $K_{1D} \leq |B_1| + 1 \leq 10$ on every admissible adversary, so the $\varepsilon_{\text{ca}} \leq 10/q$ statement holds without overhead. Parity-aligned boundary saturations (Remark 7.2) live at $\delta = \delta_J$ exact and do not propagate into strict above- J at any fold level ([10]).

References

- [1] R. Chai and X. Fan. *FRI Soundness Above the Johnson Bound via Threshold Halving*. Cryptology ePrint Archive, Paper 2026/858, 2026. <https://eprint.iacr.org/2026/858>.
- [2] R. Chai and X. Fan. *Codim-2($c-1$) Reed–Solomon Proximity Bounds via the Berlekamp Re-alizer*. Companion paper, 2026. Manuscript and computational artifacts available at <https://github.com/iotexproject/rs-proximity-gaps>.
- [3] Ethereum Foundation, *The Proximity Prize: \$1M Reed–Solomon Challenge*. <https://proximityprize.org>, 2026.
- [4] Ethereum Foundation, *The Proximity Prize* (preliminary v2 framing: Grand MCA Challenge and Grand List Decoding Challenge with $\varepsilon^* = 2^{-128}$ at rates $\rho \in \{1/2, 1/4, 1/8, 1/16\}$), <https://proximityprize.org>, accessed May 2026.

- [5] G. Arnon, D. Boneh, and G. Fenzi. *Open Problems in List Decoding and Correlated Agreement*. Cryptology ePrint Archive, Paper 2026/680, 2026. <https://eprint.iacr.org/2026/680>.
- [6] E. Ben-Sasson, D. Carmon, Y. Ishai, S. Kopparty, and S. Saraf. *Proximity gaps for Reed–Solomon codes*. Journal of the ACM 70(5), Article 32, October 2023. <https://doi.org/10.1145/3614423>. Conference version: 61st IEEE FOCS, 2020.
- [7] Elizabeth Crites and Alistair Stewart. *On Reed–Solomon Proximity Gaps Conjectures*. Cryptology ePrint Archive, Paper 2025/2046, 2025. <https://eprint.iacr.org/2025/2046>.
- [8] E. Ben-Sasson, D. Carmon, U. Haböck, S. Kopparty, S. Saraf. *On proximity gaps for Reed–Solomon codes*. Cryptology ePrint Archive, Paper 2025/2055, 2025. <https://eprint.iacr.org/2025/2055>.
- [9] B. E. Diamond and A. Gruen. *On the Distribution of the Distances of Random Words*. Cryptology ePrint Archive, Paper 2025/2010, 2025. <https://eprint.iacr.org/2025/2010>.
- [10] R. Chai, X. Fan, *Computational companion: notes, scripts, and verification outputs for the action-orbit FRI soundness analysis*, <https://github.com/iotexproject/rs-proximity-gaps>, 2026.
- [11] H. D. L. Hollmann and Q. Xiang. *A proof of the Welch and Niho conjectures on cross-correlations of binary m -sequences*. Finite Fields and Their Applications, 7(2):253–286, 2001. <https://doi.org/10.1006/ffta.2000.0281>.